Graph Similarity Meets Optimal Transport

Jens Breitung[1] **Timo Gervens**[1] **Martin Grohe**[1]

[1] RWTH Aachen University, Germany
{lastname}@informatik.rwth-aachen.de

Abstract

A natural way to compare two graphs is to align their vertex sets by a bijection and to measure the difference of the aligned adjacency matrices in a matrix norm. Depending on the norm, this yields, for example, the edit distance, the cut distance, the spectral distance, and the local edit distance. To make these distances applicable to graphs of different order, one can consider blowups of the graphs or, equivalently, relax bijections to fractional alignments, that is, optimal transport plans between the uniform distributions on the vertex sets, in the spirit of Gromov–Wasserstein distances.

We investigate how much the optimal transport versions can differ from their bijective counterparts, that is, we quantify their integrality gaps. For the edit distance, we improve the best known upper bound on the integrality gap from 3 to 2 and the lower bound from 6/5 to 4/3. For the cut distance, we approach Lovász's conjecture that $\delta_\square \leq 2\delta_\square^{\mathsf{OT}}$ from both sides. We improve the previously best known upper bound $\delta_\square \leq 32 \cdot (\delta_\square^{\mathsf{OT}})^{1/67}$ (Borgs, Chayes, Lovász, Sós and Vesztergombi, Adv. Math. 2008) to $\delta_\square \leq C_\varepsilon \cdot (\delta_\square^{\mathsf{OT}})^{1-\varepsilon}$ for every $\varepsilon > 0$, and we construct graphs G, H with $\delta_\square(G,H) = 2\delta_\square^{\mathsf{OT}}(G,H)$, establishing a factor-2 lower bound that matches Lovász's conjecture. For the local edit and spectral distances, we obtain upper bounds of order $\sqrt{t\ln(1/t)}$ and $\sqrt{t}$, respectively, where t is the corresponding optimal transport distance, and we show that, unlike for the edit and cut distances, no linear upper bounds are possible.

Finally, we show that computing the optimal transport distances is NP-hard, and we give hardness of approximation results for the cut, spectral, and local edit distances.

Keywords: graph similarity, optimal transport, edit distance, cut distance, spectral distance

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1 Introduction

Metrics on graphs are of fundamental importance for many applications, for example in machine learning [BLH+23, WVN+26], and also for an analytical theory such as the theory of graph limits [Lov12]. Two widely used graph metrics are *graph edit distance* [GXTL10] and *cut distance* [Lov12].

A natural way to measure the distance between two graphs G, H with the same number n of vertices is to align their vertex sets via a bijection π and to quantify how much the adjacency matrices A_G, A_H differ under this alignment. For a matrix norm $\|\cdot\|_\Diamond$, we define

$$\mathsf{d}_\Diamond(G,H) \coloneqq \min_\pi \|A_G^\pi - A_H\|_\Diamond ,$$

minimising over all bijections $\pi : V(G) \to V(H)$. Here, A_G^π denotes the adjacency matrix of G with rows and columns permuted according to π. To obtain an isomorphism-invariant metric, we need to assume that the matrix norm is invariant under simultaneous permutations of rows and columns, which most matrix norms are (including all norms we consider here). Different matrix norms yield a variety of interesting graph metrics. First of all, the entrywise 1-norm $\|\cdot\|_{\mathsf{ed}}$ (that is, the ℓ_1-norm of the vector of length n^2 obtained by flattening the matrix) yields edit distance,[1] and the Frobenius norm yields the square root of graph edit distance. The cut norm $\|\cdot\|_\square$ [FK99] yields cut distance, which is the metric underlying the theory of graph limits. It has been observed in [GG22] that the operator 1-norm $\|\cdot\|_1$ yields *local edit distance*, which measures the maximum number of edge edits necessary at a single vertex to obtain isomorphic graphs. More generally, operator norms are quite interesting for defining graph metrics. For example, based on a result from [Nik09], we observe that the spectral norm (i.e., the 2-operator norm) yields a graph metric that is equivalent to cut distance.

The matrix norms and their graph metrics scale differently in terms of the order n of the matrices or graphs. To allow for comparisons between different metrics, and also to be able to compare graphs of different orders (discussed below), we need to normalise the metrics. The most natural scaling factor might be $1/s(n)$, where $s(n)$ is the maximum distance between graphs of order n, often the distance between the empty graph and the complete graph. However, more pragmatically, we just scale by the asymptotic growth rate of $s(n)$, that is, by $1/n^2$ for edit distance and cut distance and by $1/n$ for the metrics based on p-operator norms. So for every matrix norm $\|\cdot\|_\Diamond$ we obtain an unnormalised graph metric $\mathsf{d}_\Diamond$, which we normalise to the graph metric $\delta_\Diamond$.

So far, all these graph metrics have a serious limitation: they only allow for comparing graphs of the same order n. This renders them unsuitable for many theoretical considerations, where we may want to approximate large graphs by small ones using tools like the regularity lemma or consider limit objects for sequences of graphs of increasing order, and also for many applications. Lovász [Lov12, Chapter 8] proposed two different, but actually equivalent, approaches to generalise cut distance to graphs of different orders. Both can be generalised to arbitrary graph metrics based on matrix norms [Gro25], and it turns out they remain equivalent for the metrics considered here.

The Blowup Construction. The *k-blowup* $G^{\bullet k}$ of a graph G is obtained by replacing every vertex v by a *cloud* of k independent copies of v and each edge by a complete bipartite graph. Now if G, H are graphs of order m, n, respectively, the idea is to blow them up to the same order N and then compare them using a normalised metric $\delta_\Diamond$. For example, we can take N to be the least common multiple of m, n, or we can take $N = mn$. To make the resulting metric independent of any particular choice, we define the *blowup metric* associated with $\delta_\Diamond$ by

$$\delta_\Diamond^\bullet(G,H) \coloneqq \lim_{\ell\to\infty} \delta_\Diamond(G^{\bullet n\ell}, H^{\bullet m\ell}).$$

[1] We ignore the factor 2 the matrix norm induces by counting each undirected edge twice.

It is important here that we work with the *normalised* metric, as otherwise the limit would not exist. Even for the normalised metrics it is not clear a priori that the limit exists, but for all matrix norms considered here it does (see [Gro25]). Note, however, that for graphs G, H of the same order, it may happen that $\delta_{\Diamond}^{\bullet}(G,H) < \delta_{\Diamond}(G,H)$, because there are alignments between the blowups $G^{\bullet k}$ and $H^{\bullet k}$ that do not project to alignments between G and H. This leads to the idea of considering fractional alignments.

Fractional Alignments and Optimal Transport. Rather than blowing the graphs up to a common size and then aligning them by a bijection, we can use *fractional alignments*. Assume that the vertex sets of our graphs G, H are V, W, and let $m \coloneqq |V|$ and $n \coloneqq |W|$. A fractional alignment between G and H is a matrix $Q^* \in \mathbb{R}^{V\times W}$ with nonnegative entries and row sums $\sum_{w\in W} Q^*(v,w) = \frac{1}{m}$ for all $v \in V$ and column sums $\sum_{v\in V} Q^*(v,w) = \frac{1}{n}$ for all $w \in W$. Observe that such fractional alignments are precisely couplings between the uniform distributions on V, W. Moreover, note that if $m = n$, then for every permutation matrix $P \in \mathbb{R}^{V\times W}$ describing an alignment between G and H the matrix $\frac{1}{n}P$ is a fractional alignment. In this way, (integral) alignments, that is, bijections between the vertex sets, can also be viewed as special fractional alignments.

Let us now define the optimal transport version of graph edit distance. Recall that the matrix norm $\|\cdot\|_{\mathsf{ed}}$ underlying the definition of δ_{ed} is the entrywise 1-norm, that is, the sum of the absolute values of all entries of a matrix, and the normalisation factor is $\frac{1}{n^2}$. Thus for graphs G, H of the same order $m = n$ we have

$$\delta_{\mathsf{ed}}(G,H) = \frac{1}{n^2}\min_{P} \sum_{v,v'\in V}\sum_{w,w'\in W} \left|(A_G(v,v') - A_H(w,w'))P(v,w)P(v',w')\right|,$$

where P ranges over all permutation matrices $P \in \mathbb{R}^{V\times W}$. To understand this expression, note that $P(v,w)P(v',w') = 1$ if and only if the permutation underlying P maps v to w and v' to w'. Now for arbitrary G, H, we define the optimal transport version of edit distance by

$$\delta_{\mathsf{ed}}^{\mathsf{OT}}(G,H) = \min_{Q^*} \sum_{v,v'\in V}\sum_{w,w'\in W} \left|(A_G(v,v') - A_H(w,w'))Q^*(v,w)Q^*(v',w')\right|, \tag{1}$$

where Q^* ranges over (the compact set of) all fractional alignments between G and H. Letting $B(G,H,Q^*)$ be the $mn \times mn$ matrix with row and column indices from $V \times W$ and entries $B(G,H,Q^*)((v,w),(v',w')) = (A_G(v,v') - A_H(w,w'))Q^*(v,w)Q^*(v',w')$, we can write $\delta_{\mathsf{ed}}^{\mathsf{OT}}(G,H) = \min_{Q^*} \|B(G,H,Q^*)\|_{\mathsf{ed}}$. Observe that for graphs of the same order $m = n$, we have $\delta_{\mathsf{ed}}(G,H) = \min_{P^*} \|B(G,H,P^*)\|_{\mathsf{ed}}$, where P^* ranges over all matrices $\frac{1}{n}P$ for a permutation matrix P. This shows that $\delta_{\mathsf{ed}}^{\mathsf{OT}}(G,H) \le \delta_{\mathsf{ed}}(G,H)$.

A moment of reflection reveals that the definitions of $\delta_{\mathsf{ed}}^{\bullet}$ and $\delta_{\mathsf{ed}}^{\mathsf{OT}}$ are very similar. Unsurprisingly, it holds that $\delta_{\mathsf{ed}}^{\bullet} = \delta_{\mathsf{ed}}^{\mathsf{OT}}$ [Gro25].

An optimal-transport version of other graph metrics can be defined similarly. For cut distance, we use exactly the same matrix $B(G,H,Q^*)$ and define

$$\delta_{\square}^{\mathsf{OT}}(G,H) = \min_{Q^*} \|B(G,H,Q^*)\|_{\square}.$$

Again, we have $\delta_{\square}^{\bullet} = \delta_{\square}^{\mathsf{OT}}$ [Lov12].

It is less obvious how to define an optimal transport version of graph metrics based on operator norms. Recall that, for $1 \le p, q \le \infty$, the $p \to q$ operator norm of a matrix $A \in \mathbb{R}^{n\times n}$ is defined by $\|A\|_{p\to q} \coloneqq \max_{\boldsymbol{x}\in\mathbb{R}^n\setminus\{\mathbf{0}\}} \|A\boldsymbol{x}\|_q / \|\boldsymbol{x}\|_p$. In particular, the 1-operator norm is $\|\cdot\|_1 \coloneqq \|\cdot\|_{1\to 1}$ and the spectral norm is $\|\cdot\|_2 \coloneqq \|\cdot\|_{2\to 2}$. We propose the following definition of an optimal transport

version of the graph metric $\delta_{p\to q}$ based on $\|\cdot\|_{p\to q}$: for every fractional alignment Q^* between G and H, with the convention $0^0 = 0$, we let

$$B_{p\to q}(G,H,Q^*)((v,w),(v',w')) \coloneqq (A_G(v,v') - A_H(w,w'))Q^*(v,w)^{1/q}Q^*(v',w')^{1-1/p}$$

(and also write $B_\square(G,H,Q^*) \coloneqq B_{\mathsf{ed}}(G,H,Q^*) \coloneqq B(G,H,Q^*) = B_{\infty\to 1}(G,H,Q^*)$) and define

$$\delta^{\mathsf{OT}}_{p\to q}(G,H) \coloneqq \min_{Q^*} \|B_{p\to q}(G,H,Q^*)\|_{p\to q}\,, \tag{2}$$

where, as usual, the minimum ranges over all fractional alignments Q^*. Note that if the graphs have the same order and we restrict Q^* to be $\frac{1}{n}P$ for a permutation matrix P, this yields the graph metric $\delta_{p\to q}$, where $s(n) = n^{1/q+1-1/p}$. We show that with this definition, blowup distance and optimal transport distance coincide for all operator norms.

Theorem 1. *For all $1 \le p,q \le \infty$, it holds that $\delta_{p\to q} = \delta^{\mathsf{OT}}_{p\to q}$.*

To prove Theorem 1, we make the correspondence between alignments of blowups and fractional alignments precise. The main technical work consists in relating optimal test vectors for the operator norms of the aligned blowups to those of $B_{p\to q}(G,H,Q^*)$, and in approximating an optimal fractional alignment by one with rational entries. The proof is given in Appendix B.

Integrality Gaps. We have already remarked that, since every bijection is also a fractional alignment, we always have $\delta^{\mathsf{OT}}_\Diamond \le \delta_\Diamond$ for all pairs of graphs of the same order. It was known that this inequality can be strict for $\Diamond \in \{\mathsf{ed}, \square\}$ [Gro25, Example 3.6], [BCL$^+$08, Section 5.1]. We study how large the resulting *integrality gap* of relaxing permutation matrices to doubly stochastic matrices can be, i.e. whether $\delta_\Diamond$ is bounded by a function of $\delta^{\mathsf{OT}}_\Diamond$ independent of the input graphs and their order. This question appears in several places in the literature. In the theory of graph limits, the fractional distance is the one that extends to graphons, while the integral one is the natural distance between finite graphs; bounding the gap shows that both define the same notion of convergence [BCL$^+$08, Lov12]. It allows transferring statements about limits back to finite graphs, as in Pikhurko's analytic approach to stability [Pik10]. In property testing, Goldreich, Krivelevich, Newman and Rozenberg [GKNR08] asked how much blowing up two graphs can decrease their edit distance. In graph matching, the relaxation of edit distance is known as the *indefinite relaxation*.

Some bounds were previously known. For edit distance, Pikhurko [Pik10, Lemma 14] proved that for all graphs G, H of the same order it holds that $\delta_{\mathsf{ed}}(G,H) \le 3 \cdot \delta^{\mathsf{OT}}_{\mathsf{ed}}(G,H)$. Combined with an example of Matsliah [GKNR08, Appendix B], the optimal factor was known to lie in $[\frac{6}{5}, 3]$. For cut distance, Borgs, Chayes, Lovász, Sós and Vesztergombi [BCL$^+$08, Theorem 2.3] proved that for all graphs G, H of the same order it holds that $\delta_\square(G,H) \le 32 \cdot (\delta^{\mathsf{OT}}_\square(G,H))^{1/67}$. Lovász conjectured that the bound can be made linear, more precisely, that $\delta_\square(G,H) \le 2 \cdot \delta^{\mathsf{OT}}_\square(G,H)$ for all graphs G, H of the same order [Lov12, Conjecture 9.28].

We substantially improve both bounds. Furthermore, we obtain the first upper bounds for local edit and spectral distance. In the following we abbreviate $\delta_{p\to p}$ and $\delta^{\mathsf{OT}}_{p\to p}$ to δ_p and δ^{OT}_p, respectively.

Theorem 2 (Upper Bounds). *For all pairs of graphs of the same order and all $\varepsilon > 0$, we have*

$$\delta_{\mathsf{ed}} \le 2 \cdot \delta^{\mathsf{OT}}_{\mathsf{ed}}, \qquad \delta_1 \le 15\sqrt{\delta^{\mathsf{OT}}_1 \ln(4/\delta^{\mathsf{OT}}_1)}, \qquad \delta_\square \le C_\varepsilon \cdot (\delta^{\mathsf{OT}}_\square)^{1-\varepsilon}, \qquad \delta_2 \le (1+2\sqrt{2})\sqrt{\delta^{\mathsf{OT}}_2},$$

where $C_\varepsilon > 0$ is a constant depending only on ε. Here we just write $\delta^{\mathsf{OT}}_\Diamond$ instead of $\delta^{\mathsf{OT}}_\Diamond(G,H)$.

Theorem 2 follows from Theorems 6, 9, 11 and 14.

We complement the upper bounds by the following lower bounds.

Theorem 3 (Lower Bounds).

1. *For every $\varepsilon > 0$, there are graphs G, H such that $\delta_{\mathsf{ed}}(G,H) \geq (\frac{4}{3} - \varepsilon) \cdot \delta_{\mathsf{ed}}^{\mathsf{OT}}(G,H)$.*
2. *For every even $r \geq 2$, there are graphs G_r, H_r such that $\delta_1(G_r,H_r) = r \cdot \delta_1^{\mathsf{OT}}(G_r,H_r)$.*
3. *There exist graphs G, H such that $\delta_\square(G,H) = 2 \cdot \delta_\square^{\mathsf{OT}}(G,H)$.*
4. *For every $r \geq 1$, there are graphs G_r, H_r such that $\delta_2(G_r,H_r) \geq \sqrt{r} \cdot \delta_2^{\mathsf{OT}}(G_r,H_r)$.*

Theorem 3 follows from Propositions 5, 8, 10 and 13.

Computational Complexity. In addition to these bounds on the integrality gaps, we prove several hardness results for computing the optimal transport distances. As nonconvex quadratic programs over the transportation polytope, they are neither covered by the hardness proofs for $\mathsf{d}_\diamondsuit$ [GRW18, GG22] nor by algorithms for linear optimal transport, and their membership in NP is unclear.

Theorem 4 (NP-hardness and Hardness of Approximation).

1. *Computing $\mathsf{d}_{\mathsf{ed}}^{\mathsf{OT}}$ is NP-hard, even when restricted to collections of paths.*
2. *For every $1 \leq p \leq \infty$, computing $\mathsf{d}_p^{\mathsf{OT}}$ is NP-hard. Moreover, unless $\mathsf{P} = \mathsf{NP}$, there is no polynomial-time algorithm approximating $\mathsf{d}_p^{\mathsf{OT}}$ within a multiplicative factor of $2^{\max\{1/p,1-1/p\}} - \varepsilon$ for any $\varepsilon > 0$. For $p = 1$, this holds even when restricted to graphs of maximum degree two.*
3. *Computing $\mathsf{d}_\square^{\mathsf{OT}}$ is NP-hard. For every $\varepsilon > 0$, unless $\mathsf{P} = \mathsf{NP}$, there is no polynomial-time algorithm approximating $\mathsf{d}_\square^{\mathsf{OT}}$ within multiplicative factor $\frac{17}{16} - \varepsilon$.*

Theorem 4 follows from Theorems 15 to 18. For the first two assertions, we encode the partitioning constraints of Three-Partition in the optimisation over fractional alignments, while the cut distance hardness is based on approximately fixing the identity bijection between two suitably defined graphs and using that the cut norm cannot be approximated in polynomial time within a multiplicative factor of $\frac{16}{17} + \varepsilon$ [Hås01] unless $\mathsf{P} = \mathsf{NP}$.

We supply Lean certificates for all our proofs, available at `https://github.com/jensbreitung/graph-similarity-meets-optimal-transport`.

1.1 Related Work

Graph metrics are of great practical importance. Comparing graphs by aligning their vertex sets, known as *graph matching*, has a long history in pattern recognition and computer vision [CFSV04, FPV14], and it is used, for example, to align biological networks and to compare chemical structures [ESDS16]; graph edit distance in particular has become a standard tool [GXTL10]. In machine learning, graph similarity underlies graph kernels [KJM20] and informs the analysis of graph neural networks [BLH+23, WVN+26]. In contrast, graph metrics have received limited attention in theory. The computational complexity of edit distance has been studied under the names approximate and robust graph isomorphism [AKKV12, OWWZ14, GRW18], and Gervens and Grohe [GG22] extended this study to metrics based on other matrix norms. Lyzinski et al. [LFF+16] analysed convex relaxations of graph matching. On the analytic side, cut distance is the foundation of the theory of graph limits [BCL+08, Lov12].

An important inspiration for our "optimal transport" metrics is the Gromov–Wasserstein distance on finite metric spaces [Mém11]. Of course we can view graphs as metric spaces on their vertex sets with the shortest path distance. We always assume that the probability distribution on both spaces (i.e. vertex sets) is uniform. In this setting, the Gromov–Wasserstein distance with respect to the ℓ_r

norm, where $1 \le r \le \infty$, can be expressed as

$$\delta_{\mathsf{GW}}(G,H) \coloneqq \frac{1}{2}\min_{Q^*}\Big(\sum_{v,v'\in V}\sum_{w,w'\in W}|D_G(v,v')-D_H(w,w')|^r Q^*(v,w)Q^*(v',w')\Big)^{1/r},$$

where D_G, D_H are the distance matrices of our graphs and Q^* ranges over all fractional alignments. Compare this with Eq. (1), in particular for $r=1$: all we do is replace the adjacency matrices by the distance matrices. Note that, in contrast to classical optimal transport, where one minimises a cost that is linear in Q^* [Vil09, PC19], the objective of our distances is nonlinear in Q^*: whether it is beneficial to transport mass from v to w depends on where the neighbours of v are transported. The complexity of δ_{GW} and other metrics based on the distance matrices is investigated in [BGP26].

2 Preliminaries

The set of real numbers is denoted $\mathbb{R}$. We write log and ln for the binary and natural logarithm, respectively. For a positive integer n, we let $[n] \coloneqq \{1,\dots,n\}$. We denote vectors by bold symbols and write $x_1,\dots,x_n$ for the entries of $\boldsymbol{x} \in \mathbb{R}^n$.

For finite sets V, W of the same size, we write $\mathrm{Bij}(V,W)$ and $\mathrm{Perm}(V,W)$ for the bijections between V and W and their permutation matrices, where $P_\pi(v,w)=1$ if and only if $\pi(v)=w$. $\mathrm{DS}(V,W)$ denotes the set of doubly stochastic matrices. Hence, if $Q \in \mathrm{DS}(V,W)$, then $\frac{1}{n}Q \in \mathrm{DS}^*(V,W)$. The *support* of $Q \in \mathbb{R}^{V\times W}$ is $\mathrm{supp}(Q) \coloneqq \{(v,w) \mid Q(v,w)>0\}$.

The vertex set of a graph G is denoted $V(G)$, its edge set $E(G)$. Unless explicitly stated otherwise, graphs are finite, simple and undirected, so $E(G) \subseteq \binom{V(G)}{2}$. We denote edges by vw instead of $\{v,w\}$. The *order of* G is $|V(G)|$. The *adjacency matrix* $A_G \in \{0,1\}^{V(G)\times V(G)}$ is defined in the usual way. For a graph G and a vertex $v \in V(G)$, we denote the *open neighbourhood* of v in G by $N_G(v) \coloneqq \{w \in V(G) \mid vw \in E(G)\}$. Unless stated otherwise, graphs G and H have vertex sets V and W, respectively. If the graphs are of the same order, for $Q \in \mathrm{DS}(V,W)$, we define the *(unnormalised) cost* $\mathcal{C}_\Diamond(Q) \coloneqq \|B_\Diamond(G,H,Q)\|_\Diamond$, corresponding to $\mathsf{d}_\Diamond$ instead of $\delta_\Diamond$. For $\pi \in \mathrm{Bij}(V,W)$, we let $\mathcal{C}_\Diamond(\pi) \coloneqq \mathcal{C}_\Diamond(P_\pi) = \|A_G^\pi - A_H\|_\Diamond$. Thus $\mathsf{d}_\Diamond(G,H)$ and $\mathsf{d}_\Diamond^{\mathsf{OT}}(G,H)$ are the minima of $\mathcal{C}_\Diamond$ over $\mathrm{Bij}(V,W)$ and $\mathrm{DS}(V,W)$, respectively. The *local cost* of a pair $(v,w) \in V\times W$ is

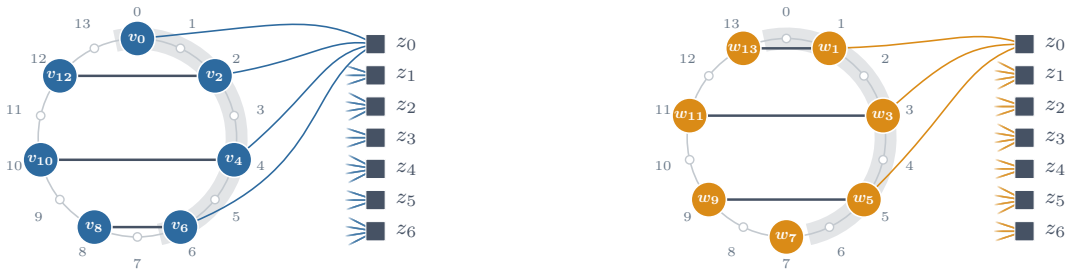
$$\mathcal{L}_Q(v,w) \coloneqq \sum_{(v',w')\in V\times W}|A_G(v,v')-A_H(w,w')|Q(v',w') \qquad \text{and} \qquad \mathcal{C}_1(Q) = \max_{(v,w)\in\mathrm{supp}(Q)}\mathcal{L}_Q(v,w),$$

the latter identity is also shown in Eq. (7) (compare Appendix A). For $\pi \in \mathrm{Bij}(V,W)$, we let $\mathcal{L}_\pi \coloneqq \mathcal{L}_{P_\pi}$ and note that $\mathcal{L}_\pi(v,\pi(v)) = |\pi(N_G(v)) \mathbin{\triangle} N_H(\pi(v))|$ counts the pairs $\{v,v'\}$ *mismatched* by π, i.e. with exactly one of $vv' \in E(G)$ and $\pi(v)\pi(v') \in E(H)$, so $\mathcal{C}_{\mathsf{ed}}(\pi)$ is twice the number of mismatched pairs.

By the Birkhoff–von Neumann theorem [Sch86], every $Q \in \mathrm{DS}(V,W)$ can be written as a convex combination of permutation matrices, that is, there are $\pi_1,\dots,\pi_m \in \mathrm{Bij}(V,W)$ such that

$$Q = \sum_{i=1}^m \lambda_i P_{\pi_i} \qquad \text{with } \lambda_i > 0 \text{ for } i \in [m] \text{ and } \sum_{i=1}^m \lambda_i = 1. \tag{3}$$

We call $\{\pi_1,\dots,\pi_m\}$ the *support* of the decomposition.



(a) The graph G with core vertices $\{v_0, v_2, \ldots, v_{12}\}$. **(b)** The graph H with core vertices $\{w_1, w_3, \ldots, w_{13}\}$.

Figure 1 The graphs G and H of Proposition 5 for $N = 7$, without the clique added by Lemma 22. According to their positions in $\mathbb{Z}_{14}$, the core vertices are drawn on top of the cycle graph C_{14}, which is drawn in light grey (although it is neither part of G nor H). In both graphs, distinct core vertices are adjacent if their positions sum to 0 modulo $2N$. The core-marker edges are drawn in full only for z_0; for the other markers they are indicated by stubs and can be seen in Figure 4. The semicircle corresponding to z_0 is drawn shaded.

3 The Integrality Gaps of Optimal Transport Formulations

In this section, we analyse how far $\delta_\Diamond$ and $\delta_\Diamond^{\mathsf{OT}}$ may deviate from each other. Clearly, $\delta_\Diamond^{\mathsf{OT}} \leq \delta_\Diamond$, so we are interested in the converse direction, that is, in functions f such that $\delta_\Diamond \leq f(\delta_\Diamond^{\mathsf{OT}})$.

For the upper bounds (Theorem 2), we start with an optimal fractional alignment $Q \in \mathrm{DS}(V, W)$ which is subsequently "rounded" to a bijection whose cost remains bounded in terms of $\mathcal{C}_\Diamond(Q)$. Since $\mathcal{C}_\Diamond$ is not linear in Q, a random bijection $\boldsymbol{\pi}$ from a Birkhoff decomposition Eq. (3), generally, need not satisfy $\mathbb{E}[\mathcal{C}_\Diamond(\boldsymbol{\pi})] = \mathcal{C}_\Diamond(Q)$. We use three increasingly sophisticated rounding schemes to deal with this. For edit distance, even though the cost is not linear in Q, there always exists a bijection within the Birkhoff decomposition with at most twice the cost. For the local edit distance, we use the dependent rounding scheme of Chekuri, Vondrák and Zenklusen [CVZ11] to obtain a large matching $M \subseteq V \times W$. We complete this into a bijection by breaking up long "paths" into short cycles. For the cut distance, we combine the thin-perfect-matching theorem of Haqi and Oveis Gharan [HOG26] with an iterated rounding scheme that merges two weighted bijections into a single new one in each step. The new bijection is obtained by breaking long permutation cycles into short ones and filling the remaining gaps with an arbitrary bijection.

For the lower bounds (Theorem 3), we construct graph pairs witnessing gaps between $\delta_\Diamond$ and $\delta_\Diamond^{\mathsf{OT}}$. Their vertices comprise *core vertices* and *markers*, which encode structural restrictions on the core vertices. With markers fixed, suitable fractional alignments of the core outperform every bijection. To force low-cost bijections to fix the markers, we attach them to successively longer prefixes of a large clique, separating their degrees from one another and from those of core and clique vertices. The formal statement (Lemma 22) and its proof are in Appendix C.

3.1 Edit Distance

Our first result improves on the previously-known lower bound of $\frac{6}{5}$ on the integrality gap of Matsliah [GKNR08, Appendix B] to $\frac{4}{3}$.

Proposition 5. *For every odd $N \geq 3$, there are graphs G and H with $\mathsf{d}_{\mathsf{ed}}(G, H) \geq 4N - 2$ and $\mathsf{d}_{\mathsf{ed}}^{\mathsf{OT}}(G, H) \leq 3N + 1$. Thus, for every $\varepsilon > 0$, there is a sufficiently large N, such that*

$$\delta_{\mathsf{ed}}(G, H) \geq \left(\frac{4}{3} - \varepsilon\right) \delta_{\mathsf{ed}}^{\mathsf{OT}}(G, H).$$

Proof Idea. Fix an odd $N \geq 3$. Both graphs G and H consist of N core vertices and N markers

$\{z_0, \dots, z_{N-1}\}$, which are fixed by Lemma 22. We assign each core vertex a *position* on the cycle $\mathbb{Z}_{2N}$ so that G occupies all even positions and H the odd ones. In both graphs, for each $r \in \{0, \dots, N-1\}$, the marker z_r is adjacent to all core vertices positioned on the semicircle $\{r, \dots, r+N-1\}$ (see Figure 1). Hence, mapping a core vertex at position i in G to a vertex at position j in H incurs a change of adjacency to Δ markers, where Δ equals the cyclic distance between i and j in $\mathbb{Z}_{2N}$ (Lemma 23). Since even and odd positions have distance at least one, the core-marker pairs contribute at least $2N$ to the cost of any alignment (considering both pair orientations).

This lower bound is attained for example by mapping each core vertex one position "forward", from i to $i+1$, or "backward", from i to $i-1$ in $\mathbb{Z}_{2N}$. However, we place $(N-1)/2$ edges between core vertices in each graph in a way so that none of them is preserved by either of these mappings, yielding a total cost of $4N-2$ for them. Furthermore, preserving edges requires any bijection π to move core vertices so far that the core-marker pairs keep $\mathcal{C}_{\mathsf{ed}}(\pi) \geq 4N-2$ (Lemma 25).

A fractional alignment Q can map core vertices one position forward with weight $1/2$ and one position backward with weight $1/2$, which preserves the core edges (except for one) in the two out of four cases in which one endpoint is moved forward while the other is moved backward (see Figure 5), yielding $\mathcal{C}_{\mathsf{ed}}(Q) \leq 3N+1$ (Lemma 24). The formal proof is given in Appendix C.1. ◀

For the upper bound, we show that the relaxation of the edit distance can only differ from the bijective variant by a factor of at most two. Remarkably, this does not require a clever rounding argument: every bijection in the support of a decomposition is already good *on average*. This improves on the factor 3 of Pikhurko [Pik10, Lemma 14].

Theorem 6. *Let $Q \in \mathrm{DS}(V, W)$ with a decomposition as in Eq. (3). Then*

$$\sum_{i=1}^{m} \lambda_i \mathcal{C}_{\mathsf{ed}}(\pi_i) \leq 2 \cdot \mathcal{C}_{\mathsf{ed}}(Q).$$

In particular, $\mathsf{d}_{\mathsf{ed}}(G, H) \leq 2 \cdot \mathsf{d}_{\mathsf{ed}}^{\mathsf{OT}}(G, H)$ for all graphs G, H of the same order.

The following example shows that any further improvement must either exploit additional properties of Q, such as optimality, or consider bijections outside the support of its decomposition.

Example 7. Let $V = W = \mathbb{Z}_6$, let G be the cycle with edges $\{v, v \pm 1\}$, and let H consist of two triangles with edges $\{v, v \pm 2\}$. Set $\pi(v) \coloneqq v+3$ and $Q \coloneqq \frac{1}{2}(P_{\mathsf{id}} + P_\pi)$. Since G and H are edge-disjoint and π is an automorphism of H, we have $\mathcal{C}_{\mathsf{ed}}(\mathsf{id}) = \mathcal{C}_{\mathsf{ed}}(\pi) = 24$. For each pair of vertices (v, v') in G, the fractional cost compares their adjacency with that of four equally weighted pairs (v, v'), $(v+3, v')$, $(v, v'+3)$, and $(v+3, v'+3)$ in H. The first and last comparisons give the costs of id and π, respectively. The other two produce no disagreement, since shifting exactly one of v, v' by 3 turns adjacency in G into adjacency in H. Thus $\mathcal{C}_{\mathsf{ed}}(Q) = \frac{1}{4}(24+0+0+24) = 12$, whereas both bijections in the prescribed decomposition have cost 24. However, this does not show tightness of the integrality gap since $\mathsf{d}_{\mathsf{ed}}(G, H) = 8$, attained by mapping two consecutive triples of cycle vertices to the two triangles.

3.2 Local Edit Distance

In contrast to the edit distance, we cannot bound the gap for the local edit distance by any constant.

Proposition 8. *For every even $r \geq 2$, there are graphs G_r, H_r on $\Theta(r^3)$ vertices such that*

$$\mathsf{d}_1(G_r, H_r) = r \qquad \text{and} \qquad \mathsf{d}_1^{\mathsf{OT}}(G_r, H_r) = 1.$$

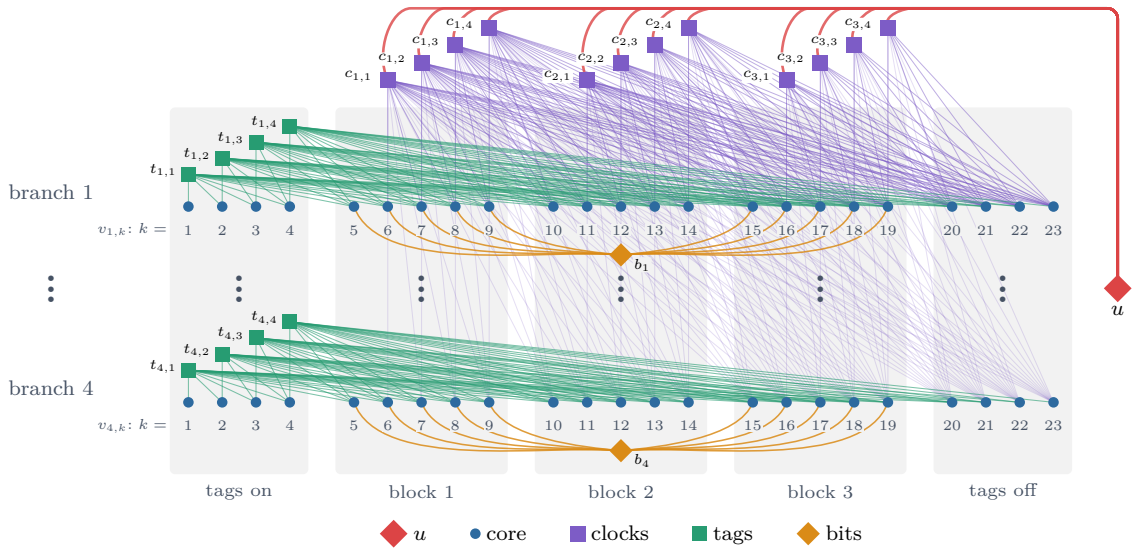


Figure 2 The graph H_4 of Proposition 8, with the clique fixing markers omitted. Of the four branches, only the first and last are shown. Each branch is arranged horizontally and split into blocks based on connectivity to the tags, clocks and bits. Edges from clocks to core vertices in the fourth branch are faded to improve readability. The graph G_4 is obtained by removing all edges incident to u.

Proof Idea. The graphs G_r and H_r are defined on a common vertex set V and share a set of *core vertices*, consisting of a special vertex u and r *branches* $v_{i,1}, \dots, v_{i,\ell-1}$ for $i \in [r]$ and $\ell := r^2 + 2r$, and a set of *markers* (Figure 2). Every edge joins a core vertex to a marker. This is done in such a way that every core vertex is described by its *pattern* (a binary vector) indicating its adjacencies to the markers. There are three types of markers: $r(r-1)$ *clock* markers, r *tags* for each branch, and one *bit* for each branch. In G_r, the vertex u is isolated, while in H_r it is adjacent to all clock markers. The construction is finished by fixing the markers. For this, we apply Lemma 22 with stepsize $r+1$. The number of vertices is $\Theta(r^3)$.

In each "step" $v_{i,k} \to v_{i,k+1}$ along the branch i, only one entry is changed in either graph: Along a fixed branch i, first the tags of branch i are switched on one by one, then the clock markers are switched on one by one in $r-1$ blocks of size r where the entry for b_i is toggled on and off between these blocks. Finally, the tags are switched off again one by one. Hence, patterns corresponding to consecutive vertices along a branch have Hamming distance one (compare Figure 6).

For the upper bound on $\mathsf{d}_1^{\mathsf{OT}}$, for $i \in [r]$, let σ_i be the bijection from G to H that fixes marker vertices and shifts the core vertices of the i-th branch along one step, that is $\sigma_i(u) = v_{i,1}$, $\sigma_i(v_{i,k}) = v_{i,k+1}$ for $k \in [\ell-2]$ and $\sigma_i(v_{i,\ell-1}) = u$. Let $Q := \frac{1}{r}\sum_{i=1}^{r} P_{\sigma_i}$. Every pair of core vertices in $\operatorname{supp}(Q)$ differs in adjacency to at most one marker, and every marker is changed with total mass at most one: each clock marker changes once on each of the r branches, which receive mass $\frac{1}{r}$ each, and the bit of branch i flips r times, but again only with mass $\frac{1}{r}$. Hence $\mathcal{C}_1(Q) \le 1$ with equality holding by Observation 21 since u is isolated in G_r but H_r has no isolated vertex.

For the lower bound on d_1, the degrees force every bijection π with $\mathcal{C}_1(\pi) < r$ to fix all markers. Start at u in G_r, map it along π to H_r, continue from the same vertex in G_r, and repeat until reaching u in H_r. The patterns along this sequence lead from the G_r-pattern to the H_r-pattern of u, and consecutive patterns differ in fewer than r markers. Thus $\pi(u) \neq u$ and this sequence is forced to traverse a single branch i due to the r tags that branch vertices are connected to. Then, the clock markers enforce that an assignment $x \mapsto \pi(x)$ only "jumps" between consecutive blocks within the branch i because non-adjacent blocks differ in connectivity to at least r clock vertices.

However, each assignment $x \mapsto \pi(x)$ that moves one block forward incurs an error of one relative to b_i because $A_G(x, b_i) \neq A_H(\pi(x), b_i)$. By construction, this must occur at least r times, so $\mathcal{C}_1(\pi) \geq r$. Intuitively, averaging over r branches can spread the cost by a factor of r across the bits of all branches, while a bijection aggregates all errors at the bit of a single branch. ◀

While Proposition 8 rules out any bound of the form $\delta_1 \leq c \cdot \delta_1^{\mathsf{OT}}$, the distances are still related by a uniform, albeit non-linear, bound. In particular, $\delta_1^{\mathsf{OT}}(G_k, H_k) \to 0$ implies $\delta_1(G_k, H_k) \to 0$.

Theorem 9. *For all graphs G, H of the same order, we have*

$$\delta_1(G, H) \leq 15\sqrt{\delta_1^{\mathsf{OT}}(G, H) \cdot \ln \frac{4}{\delta_1^{\mathsf{OT}}(G, H)}},$$

where the right-hand side is defined as 0 if $\delta_1^{\mathsf{OT}}(G, H) = 0$.

Proof Idea. Let $L \coloneqq \mathsf{d}_1^{\mathsf{OT}}(G, H)$, fix an optimal Q, and choose a reference bijection $\sigma \in \mathrm{Bij}(V, W)$ contained in $\mathrm{supp}(Q)$ by Eq. (3). By Lemma 29, whenever $(v, w) \in \mathrm{supp}(Q)$, the neighbourhood of w must be similar to $\sigma(v)$, that is $|N_H(w) \triangle N_H(\sigma(v))| \leq 2L$. The same lemma shows that $L < 1$ implies that σ is an isomorphism, so we may assume $L \geq 1$. Viewing Q as a fractional perfect matching in the graph $\mathrm{supp}(Q)$, we apply the dependent rounding scheme of Chekuri, Vondrák and Zenklusen [CVZ11] with $\gamma \in (0, \frac{1}{2}]$. The resulting random matching has edge marginals $(1-\gamma)Q$, and the local error $\Phi_v(M) \coloneqq \sum_{(v',w') \in M} |A_G(v, v') - A_H(\sigma(v), w')|$ of each v (relative to σ and M) has expectation at most $\mathcal{L}_Q(v, \sigma(v)) \leq L$. Concentration and a union bound then yield a matching that leaves at most $2\gamma n$ vertices unmatched on each side and satisfies $\Phi_v(M) \leq 2L + \mathrm{O}(\ln(n)/\gamma)$ simultaneously for all v [CVZ11, Theorem 1.1]. To complete M to a bijection, we identify W with V via σ, turning each $(v, w) \in M$ into a directed edge $v \to \sigma^{-1}(w)$. The resulting directed graph consists of cycles and $\mathrm{O}(\gamma n)$ paths. We keep all cycles, and split the paths into blocks of at most ℓ vertices (closing each block into a cycle) to obtain a bijection π. Only $\mathrm{O}(\gamma n + n/\ell)$ closing assignments are needed, each adding at most one to any local error relative to σ. The neighbourhood bound above, applied along each block, also gives $|N_H(\sigma(v)) \triangle N_H(\pi(v))| \leq 2\ell L$. Together, these bounds control the actual local costs $\mathcal{L}_\pi$. Balancing the block length ℓ and rounding error based on γ gives $\mathsf{d}_1(G, H) = \mathrm{O}(L + \sqrt{nL} + \sqrt{n \ln(4n)})$. Finally, we use the fact that $L \geq 1$ to bound the remaining logarithmic term by a function of $\delta_1^{\mathsf{OT}}(G, H)$. ◀

For the graphs G_r, H_r of Proposition 8, we have $\delta_1^{\mathsf{OT}}(G_r, H_r) = \Theta(r^{-3})$ and $\delta_1(G_r, H_r) = \Theta(r^{-2}) = \Theta(\delta_1^{\mathsf{OT}}(G_r, H_r)^{2/3})$. Hence there is no uniform bound of the form $\delta_1 \leq C \cdot (\delta_1^{\mathsf{OT}})^\alpha$ with $\alpha > \frac{2}{3}$, while Theorem 9 yields such a bound for every $\alpha < \frac{1}{2}$.

3.3 Cut Distance

We first show that if Lovász's conjecture $\delta_\square \leq 2 \cdot \delta_\square^{\mathsf{OT}}$ [Lov12, Conjecture 9.28] holds, then the factor 2 is tight.

Proposition 10. *There are graphs G, H on 145 vertices with $\mathsf{d}_\square(G, H) = 8$ and $\mathsf{d}_\square^{\mathsf{OT}}(G, H) \leq 4$.*

The graphs are shown in Figure 3. Markers ensure that bijections may only swap vertices within the same pair. As adjacency between pairs of the pentagons is exactly flipped between G and H and all pairs are connected to each other, any bijection yields a cut norm of 8. A fractional alignment can reduce the value to 4 by spreading mass uniformly across each pair. This essentially decomposes the twisted-pair pentagons into two individual copies of weight $\frac{1}{2}$ with cut norms scaled by $\frac{1}{4}$ each.

Note that one of our Lean certificates[2] verifies that the $\mathsf{d}_{\square}^{\mathsf{OT}}$-bound is tight, so $\mathsf{d}_{\square}^{\mathsf{OT}}(G, H) = 4$.



(a) The graph G. **(b)** The graph H.

Figure 3 The graphs G and H of Proposition 10. The five pairs $\{v_i^+, v_i^-\}$, $i \in \mathbb{Z}_5$, sit at the corners of a pentagon, and each pair has its own set L_i of $9(i+1)$ markers (grey). Blue core edges join pairs i and $i \pm 1$, orange core edges join pairs i and $i \pm 2$.

While we were unable to confirm or refute Lovász's conjecture, we improve the known upper bound significantly and show that arbitrarily good uniform non-linear bounds are possible.

Theorem 11. *There is a universal constant $C > 0$ such that for all graphs G, H of the same order, $\delta_{\square} := \delta_{\square}(G, H)$, and $\delta_{\square}^{\mathsf{OT}} := \delta_{\square}^{\mathsf{OT}}(G, H)$ we have*

$$\delta_{\square}^{\mathsf{OT}} \leq \delta_{\square} \leq C\delta_{\square}^{\mathsf{OT}} \ln \frac{e}{\delta_{\square}^{\mathsf{OT}}} \ln \left(e \ln \frac{e}{\delta_{\square}^{\mathsf{OT}}} \right),$$

where $e \approx 2.718$ is the Euler number and the right-hand side is defined as 0 if $\delta_{\square}^{\mathsf{OT}} = 0$. Consequently, for every $\varepsilon > 0$ we have $\delta_{\square} = \mathrm{O}((\delta_{\square}^{\mathsf{OT}})^{1-\varepsilon})$.

Proof Overview.

As usual, the lower bound is trivial, and we only need to prove the upper bound. Let $Q \in \mathrm{DS}(V, W)$ be an optimal fractional alignment of normalised cost $\zeta_Q := n^{-2}\mathcal{C}_{\square}(Q) = \delta_{\square}^{\mathsf{OT}}$. It is our goal to "round" Q to a permutation matrix P whose cost ζ_P is sufficiently close to ζ_Q. We follow two completely different strategies. The first yields a multiplicative upper bound on the cost of an optimal permutation matrix in terms of ζ_Q, and the second an additive upper bound. The upper bound of the theorem is then obtained by applying the two estimates in complementary regimes, using the multiplicative bound for small ζ_Q and the additive bound for larger ζ_Q.

The Multiplicative Bound. We prove that we can find a permutation matrix P with $\zeta_P = \mathrm{O}(\zeta_Q \log n \log\log n)$ (Theorem 38). To obtain this upper bound, we use a recent result due to Haqi and Oveis Gharan [HOG26] stating that for each fractional perfect matching of a bipartite graph there is a "thin matching". In our context, for the fractional matching $Q \in \mathrm{DS}(V, W)$ this yields a perfect matching, described by a permutation matrix $P \in \mathrm{Perm}(V, W)$, such that for every $U \subseteq V \cup W$, the number of matching edges from U to its complement is at most α times the total weight the fractional matching Q assigns to the edges from U to its complement, for some $\alpha \in \mathrm{O}(\log n \log\log n)$. Such a matching P is called *α-thin* with respect to Q.

We then prove that if P is α-thin with respect to Q, then $\zeta_P \leq (1+4\alpha)\zeta_Q$. The proof turns thinness into a transport inequality $\sum_v |x_v - y_{\pi(v)}| \leq \alpha \sum_{v,w} Q_{vw}|x_v - y_w|$ via a level-set decomposition

[2] `https://github.com/jensbreitung/graph-similarity-meets-optimal-transport/GsimOT/Proposition12_CutLowerBound/OptimalAlignment.lean`

(Lemma 36), and combines it with the observation that the fractional alignment Q only matches vertices with similar neighbourhoods (Lemma 37).

The Additive Bound. We prove that we can find a permutation matrix P with $\zeta_P \leq \zeta_Q + \mathrm{O}(n^{-1/6}\sqrt{\log n})$ (Theorem 39). Technically, this part of the proof is significantly harder. The high-level argument goes as follows. We describe ζ_Q in terms of the $n^2 \times n^2$ *discrepancy matrix* D with entries $D_{vw,v'w'} \coloneqq A_{vv'} - B_{ww'}$, where A and B are the adjacency matrices of the graphs G and H. For every $R \in \mathrm{DS}(V,W)$ we can rewrite the normalised cost $n^{-2}\mathcal{C}_\square(R)$ as

$$\zeta_R = \max_{S,T \subseteq V \times W} \frac{1}{n^2} \left| \sum_{(v,w)\in S,(v',w')\in T} R_{vw} R_{v'w'} D_{vw,v'w'} \right|. \tag{4}$$

The first step uses standard arguments to construct a spectral approximation of D. We obtain a family of n^2 vectors $\boldsymbol{z}_{vw} \in \mathbb{R}^{O(n)}$ and a matrix T of bounded spectral norm such that if we replace D in Eq. (4) by the matrix D' with entries $\langle \boldsymbol{z}_{vw}, T\boldsymbol{z}_{v'w'} \rangle$ then the resulting value ζ'_R approximates ζ_R sufficiently closely.

Next, for each $R \in \mathrm{DS}(V,W)$ we consider the zonotope Z_R generated by the line segments $[0, \frac{1}{n} R_{vw}\boldsymbol{z}_{vw}]$. The Hausdorff distance between these zonotopes yields a pseudometric ρ on $\mathrm{DS}(V,W)$. We show that matrices R that are ρ-close have approximately the same ζ'_R. This motivates the main result (Lemma 47): if $R \in \mathrm{DS}(V,W)$ is a convex combination $R = (1-\theta)P_1 + \theta P_2$ of permutation matrices P_1, P_2, then there is a permutation matrix P with $\rho(P,R) = \mathrm{O}(n^{-1/3})$. Viewing P_1, P_2 as perfect matchings in $K_{V,W}$, the complete bipartite graph between V and W, their union decomposes into alternating cycles, which we cut into blocks of roughly $n^{1/3}$ consecutive vertices. Independently for each block, all its vertices follow P_2 with probability θ and P_1 otherwise. The resulting random map has expected support function h_R, deviates from it by $\mathrm{O}(n^{-1/3})$ in expectation uniformly over all directions (by symmetrisation and Rademacher contraction), and fails to be a bijection only at block boundaries, so it can be repaired to a permutation matrix P at an additional cost of $\mathrm{O}(n^{-1/3})$.

Since every doubly stochastic matrix is a convex combination of permutation matrices, we can use this to prove that every $R \in \mathrm{DS}(V,W)$ is ρ-close to a permutation matrix.

3.4 Spectral Distance

The spectral distance δ_2 is closely related to the cut distance. The following proposition shows that the two bound each other uniformly, so in particular $\delta_2(G_k, H_k) \to 0$ if and only if $\delta_\square(G_k, H_k) \to 0$. We note that a similar relation holds for all δ_p with $1 < p < \infty$, but not for δ_1 or δ_{ed}.

Proposition 12. *For all graphs G, H of the same order, it holds that*

$$\delta_\square(G,H) \leq \delta_2(G,H) \leq 2\sqrt{\delta_\square(G,H)}.$$

The second inequality uses a result of Nikiforov [Nik09]. Despite their close relation, our following results illustrate a fundamentally different behaviour between their optimal transport formulations.

Proposition 13. *For every $r \geq 1$, there are graphs G_r, H_r on $n = \Theta(r^2)$ vertices such that*

$$\mathsf{d}_2(G_r,H_r) = \sqrt{r} \qquad \text{and} \qquad \frac{1}{2\sqrt{2}} \leq \mathsf{d}_2^{\mathsf{OT}}(G_r,H_r) \leq 1.$$

Since $\delta_2(G_r,H_r) = \Theta(n^{-3/4})$ and $\delta_2^{\mathsf{OT}}(G_r,H_r) = \Theta(n^{-1})$, it follows that there is no uniform bound of the form $\delta_2 \leq C_\alpha(\delta_2^{\mathsf{OT}})^\alpha$ for any $\alpha > \frac{3}{4}$, whereas for the cut distance, bounds arbitrarily close to linear are possible.

For the upper bound, we closely follow the proof of the multiplicative estimate for the cut distance. However, here a transport inequality similar to Lemma 36 does not hold. We replace this step with a probabilistic argument combined with spectral truncation as for the additive cut estimate.

Theorem 14. *For all graphs G, H of the same order, it holds that*

$$\delta_2(G,H) \le \delta_2^{\mathsf{OT}}(G,H) + 2\sqrt{2\,\delta_2^{\mathsf{OT}}(G,H)}.$$

As $\delta_2^{\mathsf{OT}} \le 1$, this implies $\delta_2 \le (1+2\sqrt{2})\sqrt{\delta_2^{\mathsf{OT}}}$ of Theorem 2. Together with the previous observation, the best exponent α in a uniform bound of the form $\delta_2 \le C_\alpha(\delta_2^{\mathsf{OT}})^\alpha$ therefore lies in $[\frac{1}{2}, \frac{3}{4}]$.

4 Computational Hardness of Optimal Transport Formulations

In this section, we give an overview of our NP-hardness results for the optimal transport formulations of the edit distance, the operator p-distances, and the cut distance. The full proofs are given in Appendix G. Since our reductions always produce instances with pairs of graphs of equal order, we work with $\mathsf{d}_\square^{\mathsf{OT}}$ rather than $\delta_\square^{\mathsf{OT}}$ for simplicity. For $\mathsf{d}_\square^{\mathsf{OT}}$, we denote by $\textsc{GSim}(\mathsf{d}_\square^{\mathsf{OT}})$ the problem of deciding whether $\mathsf{d}_\square^{\mathsf{OT}}(G,H) \le q$ for a given pair of graphs G, H and a rational threshold $q \in \mathbb{Q}$.

We first establish hardness for $\textsc{GSim}(\mathsf{d}_{\mathsf{ed}}^{\mathsf{OT}})$ using the reduction of Grohe, Rattan and Woeginger [GRW18, Theorem 8] from the NP-hard problem Three-Partition [GJ79] to $\textsc{GSim}(\mathsf{d}_{\mathsf{ed}})$, with a modified correctness argument. In Three-Partition, the input consists of integers A and $a_1, \dots, a_{3m}$ in unary representation satisfying $\sum_{i=1}^{3m} a_i = mA$ and $A/4 < a_i < A/2$ for $1 \le i \le 3m$. The question then is to decide whether $a_1, \dots, a_{3m}$ can be partitioned into m triples such that the sum of each triple is exactly A. For such an instance, they construct two graphs G, H where G consists of $3m$ disjoint paths on a_i vertices each and H is the disjoint collection of m paths on A vertices. In case of a YES-instance, three paths of G can be mapped into one path of H such that only the two edges joining the three segments in H are not preserved. For NO-instances, for any bijection π, at least one path of G must be split across multiple paths in H, which incurs additional edge mismatches.

Theorem 15. $\textsc{GSim}(\mathsf{d}_{\mathsf{ed}}^{\mathsf{OT}})$ *is* NP*-hard, even when restricted to collections of paths.*

For the local edit distance, these additional edge mismatches of a NO-instance need not increase the cost, as they do not necessarily increase the maximum number of mismatches occurring at one vertex. A path split across several target paths may have its split points mapped to target endpoints, keeping the local error at one. Replacing the target paths by cycles prevents this: a split then forces at least two mismatches at the same vertex, whereas a valid partition gives at most one per vertex.

Theorem 16. *For any $\varepsilon > 0$, unless* P = NP*, there is no polynomial-time approximation algorithm computing $\mathsf{d}_1^{\mathsf{OT}}$ with multiplicative error guarantee $2-\varepsilon$, even for graphs of maximum degree two.*

Extending this argument to all operator p-norms requires additional control over fractional alignments. For $p = 1$, the identity in Eq. (7) expresses the objective as a maximum over $\operatorname{supp}(Q)$. Every supported pair is considered fully, regardless of how little mass Q assigns to it. For $p > 1$, averaging can reduce the contribution of such pairs. To control this effect, we augment the construction by attaching sufficiently many leaves to each vertex. The resulting approximation threshold depends on p. We let $b(p) \coloneqq \max\{\frac{1}{p}, 1-\frac{1}{p}\}$, where $b(\infty) \coloneqq 1$.

Theorem 17. *For $1 \le p \le \infty$ and any $\varepsilon > 0$, unless* P = NP*, there is no polynomial-time approximation algorithm computing $\mathsf{d}_p^{\mathsf{OT}}$ with multiplicative error guarantee $2^{b(p)} - \varepsilon$.*

Our proof makes use of an auxiliary matrix $A^Q_{G,H}$ to obtain a lower bound on $\mathsf{d}_p^{\mathsf{OT}}$ (compare Lemma 53). Due to its simplified structure, for each NO-instance, we find a column with sufficiently many large entries, separating it from all YES-instances.

While Theorem 17 shows that the operator p-distances are NP-hard, it does not settle their membership in NP, since we do not know whether an optimal fractional alignment can always be encoded with polynomially many bits. For $p = 1$ (and by symmetry $p = \infty$) and the edit distance, we can resolve this because of their particularly simple structure. In particular, for any two input graphs there is a blowup size, exponential in their order, at which $\delta_1^{\bullet}$ and $\delta_{\mathsf{ed}}^{\bullet}$ are attained. See Proposition 57. Finally, we show hardness for $\mathsf{d}_{\square}^{\mathsf{OT}}$.

Theorem 18. *For every $\varepsilon > 0$, unless $\mathsf{P} = \mathsf{NP}$, there is no polynomial-time approximation algorithm computing $\mathsf{d}_{\square}^{\mathsf{OT}}$ with multiplicative error guarantee $\frac{17}{16} - \varepsilon$.*

In the proof, we reduce from MaxCut, adapting the construction of [GG22, Theorem 23], which encodes the maximum cut of an input graph as the cost $\mathcal{C}_{\square}(\pi)$ of a canonical bijection π between two auxiliary graphs. Attaching sufficiently many leaves forces even fractional alignments to approximately respect π: We show that any substantial deviation already incurs a larger cost than $\mathcal{C}_{\square}(\pi)$. As a result, for a fractional alignment Q that is close to π, the cut (S, T) witnessing the original distance can be lifted to a cut in the matrix $B_{\square}(G, H, Q)$ with their corresponding cut values remaining close. The resulting bounds transfer Håstad's inapproximability result for MaxCut [Hås01] to $\textsc{GSim}(\mathsf{d}_{\square}^{\mathsf{OT}})$.

5 Conclusion

We study optimal transport versions of matrix-based graph metrics and establish a fairly general correspondence between the fractional alignment and the blowup approach. We prove upper and lower bounds on the integrality gaps of these metrics, significantly improving the known bounds for edit and cut distance and establishing the first bounds for metrics based on operator norms. The latter separate cut and spectral distance: both induce the same notion of convergence, yet only the former admits almost linear bounds. Our proof techniques, including spectral methods, are new and fundamentally different from previous approaches.

Determining the exact integrality gaps remains open: the optimal factor for edit distance, the optimal exponents for local edit and spectral distance, and in particular Lovász's conjectured factor 2 for cut distance.

Statement on AI Use. We used ChatGPT 6 Astra to obtain improved bounds relating the integral and fractional versions of the distances studied in this paper. AI-generated results and arguments contributed substantially to the results in Section 3 and their proofs in Appendices C to F. We subsequently simplified these proofs and improved their presentation significantly. We take full responsibility for the correctness of all results.

A Extended Preliminaries

We denote the natural numbers by $\mathbb{N}$. We denote the set of all doubly stochastic $n \times n$ matrices by $\mathrm{DS}(n) \coloneqq \mathrm{DS}([n],[n])$ and the subset of $n \times n$ permutation matrices by $\mathrm{Perm}(n)$.

Graphs The *degree* of a vertex $v \in V(G)$ is $\deg_G(v) \coloneqq |N_G(v)|$. We write $N_G[v] \coloneqq N_G(v) \cup \{v\}$ for the closed neighborhood of v.

Matrix Norms For finite sets V, W, a matrix $A \in \mathbb{R}^{V \times W}$ and $S \subseteq V$, $T \subseteq W$, we let $A(S,T) \coloneqq \sum_{s \in S, t \in T} A(s,t)$. We consider the following norms on $\mathbb{R}^{V \times W}$, see [GG22] for a systematic treatment. The *Frobenius norm* is $\|A\|_F \coloneqq (\sum_{v \in V, w \in W} |A(v,w)|^2)^{1/2}$, the *entrywise 1-norm* is $\|A\|_{\mathsf{ed}} \coloneqq \|A\|_{(1)} \coloneqq \sum_{v \in V, w \in W} |A(v,w)|$ (i.e. the squared Frobenius norm), for $1 \leq p, q \leq \infty$, the *operator $p \to q$-norm* is

$$\|A\|_{p \to q} \coloneqq \max_{\boldsymbol{x} \in \mathbb{R}^W,\, \|\boldsymbol{x}\|_p = 1} \|A\boldsymbol{x}\|_q .$$

We write $\|A\|_p \coloneqq \|A\|_{p \to p}$. For instance, $\|A\|_1$ is the maximum absolute column sum of A. The *cut norm* is $\|A\|_\square \coloneqq \max_{S \subseteq V, T \subseteq W} |A(S,T)|$.

For $1 \leq q \leq \infty$, we denote by q^* the *Hölder conjugate* of q, i.e. the unique value with $1/q + 1/q^* = 1$; thus $1/q^* = 1 - 1/q$, $q^* = 1$ if $q = \infty$, $q^* = \infty$ if $q = 1$, and $(q^*)^* = q$. By *Hölder's inequality*, $\sum_{v \in V} |\boldsymbol{x}(v)\boldsymbol{y}(v)| \leq \|\boldsymbol{x}\|_q \|\boldsymbol{y}\|_{q^*}$ for every finite set V and all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^V$. Choosing $\boldsymbol{y}$ to be the all-ones vector, we obtain

$$\sum_{v \in V} |\boldsymbol{x}(v)| \leq \|\boldsymbol{x}\|_q \cdot |V|^{1-1/q}. \tag{5}$$

The $p \to q$-norms have the following well-known characterisation in terms of the Hölder conjugate.

Lemma 19 (see, e.g., [HJ13, Section 5.5]). *For $1 \leq p, q \leq \infty$ and all $A \in \mathbb{R}^{V \times W}$ it holds that*

$$\|A\|_{p \to q} = \max_{\|\boldsymbol{y}\|_{q^*} = 1, \|\boldsymbol{x}\|_p = 1} \left| \boldsymbol{y}^\top A \boldsymbol{x} \right| .$$

Indeed, by the duality of the ℓ_q- and ℓ_{q^*}-norms, $\|\boldsymbol{z}\|_q = \max_{\|\boldsymbol{y}\|_{q^*}=1} |\boldsymbol{y}^\top \boldsymbol{z}|$ for all $\boldsymbol{z} \in \mathbb{R}^V$, and it suffices to apply this to $\boldsymbol{z} = A\boldsymbol{x}$. As $\boldsymbol{y}^\top A \boldsymbol{x} = \boldsymbol{x}^\top A^\top \boldsymbol{y}$, it follows that $\|A\|_{p \to q} = \|A^\top\|_{q^* \to p^*}$. In particular, $\|A\|_p = \|A^\top\|_{p^*}$.

Optimal Transport Distances For finite sets V, W, we denote by $\mathrm{DS}^*(V,W)$ the set of nonnegative matrices $\mathbb{R}^{V \times W}$ with row sums $1/|V|$ and column sums $1/|W|$. Note that if $|V| = |W| = n$, then $\mathrm{DS}(V,W) = n\mathrm{DS}^*(V,W)$. For matrices $A \in \mathbb{R}^{V \times V}$, $B \in \mathbb{R}^{W \times W}$, the *discrepancy matrix* $D_{A,B} \in \mathbb{R}^{(V \times W) \times (V \times W)}$ has entries

$$D_{A,B}(v,w,v',w') \coloneqq D_{A,B}\big((v,w),(v',w')\big) \coloneqq A(v,v') - B(w,w'). \tag{6}$$

For graphs G, H, we let $D_{G,H} \coloneqq D_{A_G, A_H}$, and we drop the subscripts if G and H are clear from the context. In this notation, $B_{p \to q}(G,H,Q^*)$ has entries $Q^*(v,w)^{1/q}\, D_{G,H}(v,w,v',w')\, Q^*(v',w')^{1-1/p}$. For $p = q = 1$, the convention $0^0 = 0$ ensures that column (v',w') of $B_{1 \to 1}(G,H,Q^*)$ vanishes whenever $Q^*(v',w') = 0$, so

$$\delta_1^{\mathsf{OT}}(G,H) = \min_{Q^* \in \mathrm{DS}^*(V,W)} \max_{(v',w') \in \mathrm{supp}(Q^*)} \sum_{(v,w) \in V \times W} |D_{G,H}(v,w,v',w')| Q^*(v,w). \tag{7}$$

All minima in the definitions of the graph distances are attained. For minimisation over $\mathrm{Bij}(V,W)$, this follows from finiteness. For optimal transport, the coupling polytopes $\mathrm{DS}^*(V,W)$ and $\mathrm{DS}(V,W)$

are compact, and the objectives are continuous for the entrywise 1-norm and the cut norm and lower semicontinuous for all operator $p \to q$-norms. For the operator norms, let (Q_k^*) be a convergent sequence in $\mathrm{DS}^*(V,W)$. Note that every pair in the support of $\lim_{k\to\infty} Q_k^*$ lies in $\mathrm{supp}(Q_k^*)$ for almost all k; restricting the weighted discrepancy matrices to the rows and columns indexed by this support gives convergent matrices and cannot increase the operator norm. The same argument applies to $\mathrm{DS}(V,W)$.

B Proof of Theorem 1

Let G, H be graphs with vertex sets V, W of orders m and n, respectively. To simplify the notation below, we let $\kappa := \kappa_{p\to q} = 1/q + 1 - 1/p$ and $D := D_{G,H}$. Note that $s(n) = n^{\kappa_{p\to q}}$ for the function $s(n)$ corresponding to the $p \to q$-operator norm. Thus, the entries of $B_{p\to q}(G,H,Q^*)$ are $Q^*(v,w)^{1/q}\, D(v,w,v',w')\, Q^*(v',w')^{1-1/p}$.

We first prove

$$\delta^{\bullet}_{p\to q}(G,H) \geq \min_{Q^*\in \mathrm{DS}^*(V,W)} \|B_{p\to q}(G,H,Q^*)\|_{p\to q}\,. \tag{8}$$

Let $\ell \in \mathbb{N}_{>0}$ and $G^{\bullet} := G^{\bullet \ell n}$, $H^{\bullet} := H^{\bullet \ell m}$. Recall that the vertex sets of $G^{\bullet}$ and $H^{\bullet}$ are $V^{\bullet} := V \times [\ell n]$ and $W^{\bullet} := W \times [\ell m]$, respectively, and let $N := \ell m n = |V^{\bullet}| = |W^{\bullet}|$. We prove that

$$\mathsf{d}_{p\to q}(G^{\bullet}, H^{\bullet}) \geq N^{\kappa} \cdot \min_{Q^*\in \mathrm{DS}^*(V,W)} \|B_{p\to q}(G,H,Q^*)\|_{p\to q} \tag{9}$$

which implies Eq. (8).

Choose $\pi \in \mathrm{Bij}(V^{\bullet}, W^{\bullet})$ such that $\mathsf{d}_{p\to q}(G^{\bullet}, H^{\bullet}) = \|M^{\pi}\|_{p\to q}$, where $M^{\pi} := A^{\pi}_{G^{\bullet}} - A_{H^{\bullet}}$. For all $v \in V$, $w \in W$, let

$$\begin{aligned} I(v,w) &:= \{i \in [\ell n] \mid \pi(v,i) \in \{w\} \times [\ell m]\}\,, \\ J(v,w) &:= \{j \in [\ell m] \mid \pi^{-1}(w,j) \in \{v\} \times [\ell n]\}. \end{aligned}$$

Then π maps $\{v\} \times I(v,w)$ bijectively to $\{w\} \times J(v,w)$, and thus $|I(v,w)| = |J(v,w)|$. Note that for $j \in J(v,w)$, $j' \in J(v',w')$ it holds that

$$\begin{aligned} M^{\pi}((w,j),(w',j')) &= A_{G^{\bullet}}(\pi^{-1}(w,j), \pi^{-1}(w',j')) - A_{H^{\bullet}}((w,j),(w',j')) \\ &= D(v,w,v',w'). \end{aligned} \tag{10}$$

We define a matrix $Q^* \in \mathbb{R}^{V\times W}$ by

$$Q^*(v,w) := \frac{|I(v,w)|}{N}, \quad \text{that is,} \quad Q^*(v,w) = \frac{|J(v,w)|}{N}.$$

Observe that for all $v \in V$, $w \in W$ we have $\bigcup_{w'\in W} I(v,w') = [\ell n]$ and $\bigcup_{v'\in V} J(v',w) = [\ell m]$. Thus

$$\sum_{w'\in W} Q^*(v,w') = \frac{\ell n}{N} = \frac{1}{m} \qquad \text{and} \qquad \sum_{v'\in V} Q^*(v',w) = \frac{\ell m}{N} = \frac{1}{n},$$

which implies that $Q^* \in \mathrm{DS}^*(V,W)$.

Let $B := B_{p\to q}(G,H,Q^*)$. We prove that

$$\|M^{\pi}\|_{p\to q} \geq N^{\kappa} \cdot \|B\|_{p\to q}\,. \tag{11}$$

This will imply Eq. (9). Using Lemma 19, let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{V \times W}$ with $\|\boldsymbol{y}\|_{q^*} = 1$ and $\|\boldsymbol{x}\|_p = 1$ such that $\left|\boldsymbol{y}^\top B \boldsymbol{x}\right|$ is maximised. Note that we regard $\boldsymbol{x}$ and $\boldsymbol{y}$ as vectors of length mn and not as matrices.

For $w \in W$ and $j \in [\ell m]$ let $v(w,j)$ be the unique $v \in V$ such that $j \in J(v,w)$. We define the vectors $\tilde{\boldsymbol{x}}, \tilde{\boldsymbol{y}} \in \mathbb{R}^{W^\bullet}$ with entries

$$\tilde{\boldsymbol{y}}(w,j) \coloneqq \begin{cases} \frac{\boldsymbol{y}(v(w,j),w)}{|J(v(w,j),w)|^{1-1/q}} & \text{if } |J(v(w,j),w)| > 0, \\ 0 & \text{otherwise,} \end{cases}$$

and

$$\tilde{\boldsymbol{x}}(w,j) \coloneqq \begin{cases} \frac{\boldsymbol{x}(v(w,j),w)}{|J(v(w,j),w)|^{1/p}} & \text{if } |J(v(w,j),w)| > 0, \\ 0 & \text{otherwise.} \end{cases}$$

In the following, we simply write v instead of $v(w,j)$ whenever the choice is clear from context. Let $(\boldsymbol{z}, r) \in \{(\boldsymbol{x}, p), (\boldsymbol{y}, q^*)\}$ and note that $1/q^* = 1 - 1/q$. If $r < \infty$, it holds that

$$\begin{aligned}
\|\tilde{\boldsymbol{z}}\|_r &= \left(\sum_{(v,w) \in V \times W} \sum_{j \in J(v,w)} |\tilde{\boldsymbol{z}}(w,j)|^r \right)^{1/r} \\
&= \left(\sum_{\substack{(v,w) \in V \times W \\ |J(v,w)|>0}} \sum_{j \in J(v,w)} |\tilde{\boldsymbol{z}}(w,j)|^r \right)^{1/r} \\
&= \left(\sum_{\substack{(v,w) \in V \times W \\ |J(v,w)|>0}} \sum_{j \in J(v,w)} \left| \frac{\boldsymbol{z}(v,w)}{|J(v,w)|^{1/r}} \right|^r \right)^{1/r} \\
&= \left(\sum_{\substack{(v,w) \in V \times W \\ |J(v,w)|>0}} \sum_{j \in J(v,w)} \frac{|\boldsymbol{z}(v,w)|^r}{|J(v,w)|} \right)^{1/r} \\
&= \left(\sum_{\substack{(v,w) \in V \times W \\ |J(v,w)|>0}} \frac{1}{|J(v,w)|} \cdot \sum_{j \in J(v,w)} |\boldsymbol{z}(v,w)|^r \right)^{1/r} \\
&= \left(\sum_{\substack{(v,w) \in V \times W \\ |J(v,w)|>0}} |\boldsymbol{z}(v,w)|^r \right)^{1/r} \\
&\leq \left(\sum_{(v,w) \in V \times W} |\boldsymbol{z}(v,w)|^r \right)^{1/r} = \|\boldsymbol{z}\|_r = 1.
\end{aligned}$$

Note that the second equality holds because the inner sum is empty whenever $|J(v,w)| = 0$. If $r = \infty$, then $\tilde{\boldsymbol{z}}(w,j) = \boldsymbol{z}(v,w)$ for all $j \in J(v,w)$, and thus $\|\tilde{\boldsymbol{z}}\|_\infty \leq \|\boldsymbol{z}\|_\infty = 1$.

Then we have

$$
\begin{aligned}
&\|M^\pi\|_{p\to q} \\
&\geq \left|\tilde{\boldsymbol{y}}^\top M^\pi \tilde{\boldsymbol{x}}\right| \\
&= \left|\sum_{(w,j),(w',j')\in W^\bullet} \tilde{\boldsymbol{y}}(w,j)\, M^\pi\big((w,j),(w',j')\big)\, \tilde{\boldsymbol{x}}(w',j')\right| \\
&= \left|\sum_{(v,w),(v',w')\in V\times W} \sum_{\substack{j\in J(v,w)\\ j'\in J(v',w')}} \frac{\boldsymbol{y}(v,w)}{|J(v,w)|^{1-1/q}}\, D(v,w,v',w')\, \frac{\boldsymbol{x}(v',w')}{|J(v',w')|^{1/p}}\right| \\
&= \left|\sum_{(v,w),(v',w')\in V\times W} D(v,w,v',w')\, |J(v,w)|^{1/q}\, |J(v',w')|^{1-1/p}\, \boldsymbol{y}(v,w)\, \boldsymbol{x}(v',w')\right| \\
&= N^\kappa \cdot \left|\sum_{(v,w),(v',w')\in V\times W} D(v,w,v',w')\, Q^*(v,w)^{1/q}\, Q^*(v',w')^{1-1/p}\, \boldsymbol{y}(v,w)\, \boldsymbol{x}(v',w')\right| \\
&= N^\kappa \cdot \left|\sum_{(v,w),(v',w')\in V\times W} \boldsymbol{y}(v,w)\, D(v,w,v',w')\, Q^*(v,w)^{1/q}\, Q^*(v',w')^{1-1/p}\, \boldsymbol{x}(v',w')\right| \\
&= N^\kappa \cdot \left|\boldsymbol{y}^\top B \boldsymbol{x}\right| \\
&= N^\kappa \cdot \|B\|_{p\to q}\,.
\end{aligned}
$$

Here, the first inequality uses Lemma 19 and $\|\tilde{\boldsymbol{y}}\|_{q^*}, \|\tilde{\boldsymbol{x}}\|_p \leq 1$, the third equality uses Eq. (10), and the fifth equality uses that terms with $|J(v,w)| = 0$ or $|J(v',w')| = 0$ vanish (recall the convention $0^0 = 0$). The last equality holds because $\boldsymbol{x}$ and $\boldsymbol{y}$ were chosen such that they maximise the quantity $\left|\boldsymbol{y}^\top B \boldsymbol{x}\right|$, which is $\|B\|_{p\to q}$.

For the converse inequality

$$
\delta^\bullet_{p\to q}(G,H) \leq \min_{Q^*\in \mathrm{DS}^*(V,W)} \|B_{p\to q}(G,H,Q^*)\|_{p\to q}\,, \tag{12}
$$

let $\varepsilon > 0$ and $Q^* \in \mathrm{DS}^*(V,W)$ be a rational matrix such that

$$
\|B_{p\to q}(G,H,Q^*)\|_{p\to q} \leq \min_{Q_0^*\in \mathrm{DS}^*(V,W)} \|B_{p\to q}(G,H,Q_0^*)\|_{p\to q} + \varepsilon. \tag{13}
$$

Such a matrix exists, this can be shown similarly to [Gro25, Corollary 3.12].

Claim 20. *The rational matrix $Q^* \in \mathrm{DS}^*(V,W)$ fulfilling Eq. (13) exists.*

Proof. Let $Q_0^* \in \mathrm{DS}^*(V,W)$ be a minimiser and put $S \coloneqq \mathrm{supp}(Q_0^*)$. The couplings in $\mathrm{DS}^*(V,W)$ whose entries outside S are zero form a rational polytope. Hence there is a sequence of rational matrices $Q_k^* \in \mathrm{DS}^*(V,W)$ with $\mathrm{supp}(Q_k^*) = S$ and $Q_k^* \to Q_0^*$. On this fixed support, the entries of $B_{p\to q}(G,H,Q_k^*)$ converge to those of $B_{p\to q}(G,H,Q_0^*)$, also when $p = 1$ or $q = \infty$: entries outside $S \times S$ remain zero by the convention $0^0 = 0$, and the remaining entries depend continuously on the positive entries of Q_k^*. By continuity of the norm,

$$
\|B_{p\to q}(G,H,Q_k^*)\|_{p\to q} \to \|B_{p\to q}(G,H,Q_0^*)\|_{p\to q}\,.
$$

Taking $Q^* \coloneqq Q^*_k$ for sufficiently large k therefore gives the required inequality. $\triangleleft$

Let $B \coloneqq B_{p\to q}(G, H, Q^*)$. We shall prove

$$\delta^{\bullet}_{p\to q}(G, H) \leq \|B\|_{p\to q}\,. \tag{14}$$

As $\varepsilon > 0$ was arbitrary, this will imply Eq. (12) and hence the assertion. Let k be the least common multiple of the denominators of all entries $Q^*(v, w)$ of Q^*. We shall prove that for every $s \in \mathbb{N}_{>0}$, it holds that

$$\delta_{p\to q}(G^{\bullet nks}, H^{\bullet mks}) \leq \|B\|_{p\to q}\,. \tag{15}$$

Then

$$\begin{aligned}\delta^{\bullet}_{p\to q}(G, H) &= \lim_{\ell\to\infty} \delta_{p\to q}(G^{\bullet n\ell}, H^{\bullet m\ell})\\ &= \lim_{s\to\infty} \delta_{p\to q}(G^{\bullet nks}, H^{\bullet mks}) \leq \|B\|_{p\to q}\,,\end{aligned}$$

which proves Eq. (14). Thus, it remains to prove Eq. (15).

Let $s \in \mathbb{N}_{>0}$ and $\ell \coloneqq ks$, and $G^{\bullet} \coloneqq G^{\bullet n\ell}$, $H^{\bullet} \coloneqq H^{\bullet m\ell}$. Then the vertex sets of $G^{\bullet}$ and $H^{\bullet}$ are $V^{\bullet} \coloneqq V \times [\ell n]$ and $W^{\bullet} \coloneqq W \times [\ell m]$, respectively, of size $N \coloneqq \ell mn = |V^{\bullet}| = |W^{\bullet}|$. For all $v \in V$, $w \in W$ we let $q(v, w) \coloneqq N \cdot Q^*(v, w)$. As N is a multiple of k, we have $q(v, w) \in \mathbb{N}$. For all $v \in V$ we have

$$\sum_{w\in W} q(v, w) = N \cdot \sum_{w'\in W} Q^*(v, w') = \frac{N}{m} = \ell n.$$

We choose an arbitrary partition of $[\ell n]$ into sets $I(v, w)$ of size $|I(v, w)| \coloneqq q(v, w)$ for $w \in W$. Similarly, for $w \in W$ we have $\sum_{v\in V} q(v, w) = \ell m$, and we choose an arbitrary partition of $[\ell m]$ into sets $J(v, w)$ of size $|J(v, w)| \coloneqq q(v, w)$, for $v \in V$. For all $v \in V$, $w \in W$ we let π_{vw} be an arbitrary bijection from $I(v, w)$ to $J(v, w)$. Then we define a bijection $\pi : V^{\bullet} \to W^{\bullet}$ by letting $\pi(v, i) \coloneqq (w, \pi_{vw}(i))$ for the unique w such that $i \in I(v, w)$. Let $M^{\pi} \coloneqq A^{\pi}_{G^{\bullet}} - A_{H^{\bullet}}$. Observe that Eq. (10) holds for this π as well, that is, for $j \in J(v, w)$, $j' \in J(v', w')$ it holds that

$$M^{\pi}((w, j), (w', j')) = D(v, w, v', w').$$

Let $\tilde{\boldsymbol{x}}, \tilde{\boldsymbol{y}} \in \mathbb{R}^{W^{\bullet}}$ with $\|\tilde{\boldsymbol{y}}\|_{q^*} = 1$ and $\|\tilde{\boldsymbol{x}}\|_p = 1$ such that $\left|\tilde{\boldsymbol{y}}^{\top} M^{\pi} \tilde{\boldsymbol{x}}\right|$ is maximised. We define vectors $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{(V\times W)}$ by letting

$$\boldsymbol{y}(v, w) \coloneqq \begin{cases} \frac{\sum_{j\in J(v,w)} \tilde{\boldsymbol{y}}(w,j)}{|J(v,w)|^{1/q}} & \text{if } |J(v, w)| > 0,\\ 0 & \text{otherwise,}\end{cases}$$

and

$$\boldsymbol{x}(v, w) \coloneqq \begin{cases} \frac{\sum_{j\in J(v,w)} \tilde{\boldsymbol{x}}(w,j)}{|J(v,w)|^{1-1/p}} & \text{if } |J(v, w)| > 0,\\ 0 & \text{otherwise.}\end{cases}$$

Let $(\boldsymbol{z}, r) \in \{(\boldsymbol{x}, p), (\boldsymbol{y}, q^*)\}$ and note that $1 - 1/q^* = 1/q$. If $r < \infty$, it holds that

$$\|\boldsymbol{z}\|_r = \left(\sum_{(v,w)\in V\times W} |\boldsymbol{z}(v, w)|^r\right)^{1/r}$$

$$
\begin{aligned}
&= \left(\sum_{\substack{(v,w)\in V\times W \\ |J(v,w)|>0}} \left| \frac{\sum_{j\in J(v,w)} \tilde{\boldsymbol{z}}(w,j)}{|J(v,w)|^{1-1/r}} \right|^r \right)^{1/r} \\
&= \left(\sum_{\substack{(v,w)\in V\times W \\ |J(v,w)|>0}} \frac{1}{|J(v,w)|^{r-1}} \cdot \left| \sum_{j\in J(v,w)} \tilde{\boldsymbol{z}}(w,j) \right|^r \right)^{1/r} \\
&\overset{(1)}{\leq} \left(\sum_{\substack{(v,w)\in V\times W \\ |J(v,w)|>0}} \frac{1}{|J(v,w)|^{r-1}} \cdot \left(\sum_{j\in J(v,w)} |\tilde{\boldsymbol{z}}(w,j)| \right)^r \right)^{1/r} \\
&\overset{(2)}{\leq} \left(\sum_{\substack{(v,w)\in V\times W \\ |J(v,w)|>0}} \frac{1}{|J(v,w)|^{r-1}} \cdot \left(\left(\sum_{j\in J(v,w)} |\tilde{\boldsymbol{z}}(w,j)|^r \right)^{1/r} \cdot |J(v,w)|^{\frac{r-1}{r}} \right)^r \right)^{1/r} \\
&= \left(\sum_{\substack{(v,w)\in V\times W \\ |J(v,w)|>0}} \frac{1}{|J(v,w)|^{r-1}} \cdot \left(\sum_{j\in J(v,w)} |\tilde{\boldsymbol{z}}(w,j)|^r \right) \cdot |J(v,w)|^{r-1} \right)^{1/r} \\
&= \left(\sum_{(v,w)\in V\times W} \sum_{\substack{j\in J(v,w) \\ |J(v,w)|>0}} |\tilde{\boldsymbol{z}}(w,j)|^r \right)^{1/r} \\
&= \left(\sum_{(v,w)\in V\times W} \sum_{j\in J(v,w)} |\tilde{\boldsymbol{z}}(w,j)|^r \right)^{1/r} = \|\tilde{\boldsymbol{z}}\|_r = 1,
\end{aligned}
$$

where (1) holds because taking the absolute value of each individual summand can only increase the value of the sum and (2) is an application of Hölder's inequality in the form of Eq. (5). If $r = \infty$, then $\boldsymbol{z}(v,w)$ is the average of the values $\tilde{\boldsymbol{z}}(w,j)$, $j \in J(v,w)$, and thus $\|\boldsymbol{z}\|_\infty \leq \|\tilde{\boldsymbol{z}}\|_\infty = 1$.

Putting things together we obtain:

$$
\begin{aligned}
&\mathsf{d}_{p\to q}(G^{\bullet}, H^{\bullet}) \\
&\leq \|M^\pi\|_{p\to q} \\
&= \left| \tilde{\boldsymbol{y}}^\top M^\pi \tilde{\boldsymbol{x}} \right| \\
&= \left| \sum_{(w,j),(w',j')\in W^{\bullet}} \tilde{\boldsymbol{y}}(w,j)\, M^\pi\big((w,j),(w',j')\big)\, \tilde{\boldsymbol{x}}(w',j') \right| \\
&= \left| \sum_{(v,w),(v',w')\in V\times W} \sum_{\substack{j\in J(v,w) \\ j'\in J(v',w')}} \tilde{\boldsymbol{y}}(w,j)\, M^\pi\big((w,j),(w',j')\big)\, \tilde{\boldsymbol{x}}(w',j') \right|
\end{aligned}
$$

$$
\begin{aligned}
&= \left| \sum_{(v,w),(v',w') \in V \times W} D(v,w,v',w') \cdot \left(\sum_{\substack{j \in J(v,w) \\ j' \in J(v',w')}} \tilde{\boldsymbol{y}}(w,j)\, \tilde{\boldsymbol{x}}(w',j') \right) \right| \\
&= \left| \sum_{(v,w),(v',w') \in V \times W} D(v,w,v',w') \cdot \left(\sum_{j \in J(v,w)} \tilde{\boldsymbol{y}}(w,j) \right) \left(\sum_{j' \in J(v',w')} \tilde{\boldsymbol{x}}(w',j') \right) \right| \\
&= \left| \sum_{(v,w),(v',w') \in V \times W} D(v,w,v',w') \cdot |J(v,w)|^{1/q}\, \boldsymbol{y}(v,w) \cdot |J(v',w')|^{1-1/p}\, \boldsymbol{x}(v',w') \right| \\
&= N^{\kappa} \cdot \left| \sum_{(v,w),(v',w') \in V \times W} D(v,w,v',w') \cdot Q^*(v,w)^{1/q}\, \boldsymbol{y}(v,w) \cdot Q^*(v',w')^{1-1/p}\, \boldsymbol{x}(v',w') \right| \\
&= N^{\kappa} \cdot \left| \sum_{(v,w),(v',w') \in V \times W} \boldsymbol{y}(v,w) \underbrace{D(v,w,v',w')\, Q^*(v,w)^{1/q}\, Q^*(v',w')^{1-1/p}}_{= B((v,w),(v',w'))} \boldsymbol{x}(v',w') \right| \\
&= N^{\kappa} \cdot \left| \boldsymbol{y}^\top B \boldsymbol{x} \right| \\
&\leq N^{\kappa} \cdot \max_{\|\boldsymbol{y}'\|_{q^*} \leq 1, \|\boldsymbol{x}'\|_p \leq 1} \left| \boldsymbol{y}'^\top B \boldsymbol{x}' \right| \\
&= N^{\kappa} \cdot \max_{\|\boldsymbol{y}'\|_{q^*} = 1, \|\boldsymbol{x}'\|_p = 1} \left| \boldsymbol{y}'^\top B \boldsymbol{x}' \right| \\
&= N^{\kappa} \cdot \|B\|_{p \to q}\,.
\end{aligned}
$$

Here, the fourth equality uses Eq. (10), and the second to last equality holds because the maximum of the bilinear form over the unit balls is attained on the unit spheres. Since $\delta_{p\to q}(G^{\bullet}, H^{\bullet}) = N^{-\kappa} \cdot \mathsf{d}_{p\to q}(G^{\bullet}, H^{\bullet})$, this proves Eq. (15). This concludes the proof.

C The Integrality Gap of the Edit Distance

The following two results are used throughout the lower bound constructions for the edit and local edit distances. We first prove a simple degree bound for $\mathcal{L}_Q$.

Observation 21. *For all graphs G, H of the same order, all $Q \in \mathrm{DS}(V,W)$ and $(v,w) \in V \times W$, we have $\mathcal{L}_Q(v,w) \geq |\deg_G(v) - \deg_H(w)|$.*

Proof. By the triangle inequality and since all row and column sums of Q are 1,

$$
\begin{aligned}
\mathcal{L}_Q(v,w) &\geq \left| \sum_{v' \in V} \sum_{w' \in W} D(v,w,v',w') Q(v',w') \right| \\
&= \left| \sum_{v' \in V} A_G(v,v') - \sum_{w' \in W} A_H(w,w') \right| = |\deg_G(v) - \deg_H(w)|.
\end{aligned}
$$

◀

Based on this observation, the following lemma allows us to fix the marker vertices in our constructions. This is achieved by attaching the markers, ordered by degree, to a large clique in a staircase whose stepsize is independent of the marker degrees in the original graph.

Lemma 22 (Locking Markers). *Let G, H be graphs on a common vertex set V with maximum degree*

at most Δ, and let $z_1, \dots, z_k \in V$ be markers*, ordered such that $\deg_G(z_1) \leq \cdots \leq \deg_G(z_k)$. Let $\tau \geq 1$ and $\delta \geq 0$ be integers with $|\deg_G(z_j) - \deg_H(z_j)| \leq \delta$ for all $j \in [k]$. Obtain G' and H' (with vertex set V') from G and H, respectively, by adding the same clique $\mathcal{K}$ on $m \coloneqq 2\Delta + 2\tau + k(\tau+\delta) + 1$ new vertices $y_1, \dots, y_m$ and joining every marker z_j to $y_1, \dots, y_{\Delta+\tau+j(\tau+\delta)}$. Then the following hold for $\Diamond \in \{\mathsf{ed}, 1\}$.*

1. *Every bijection $\pi \in \mathrm{Bij}(V', V')$ with $\mathcal{C}_\Diamond(\pi) < \tau$ with respect to (G', H') fixes every marker and satisfies $\pi(V) = V$.*
2. *Let $Q \in \mathrm{DS}(V, V)$ with $Q(z_j, z_j) = 1$ for all $j \in [k]$, and extend Q by the identity on $\mathcal{K}$ to Q'. Then $\mathcal{C}_\Diamond(Q')$ with respect to (G', H') equals $\mathcal{C}_\Diamond(Q)$ with respect to (G, H).*

Proof. Let $Z \coloneqq \{z_1, \dots, z_k\}$ and $s \coloneqq \tau + \delta$. Both costs are determined by the local costs, namely

$$\mathcal{C}_{\mathsf{ed}}(Q) = \sum_{(v,w) \in V \times W} Q(v,w)\mathcal{L}_Q(v,w) \qquad \text{and} \qquad \mathcal{C}_1(Q) = \max_{(v,w) \in \mathrm{supp}(Q)} \mathcal{L}_Q(v,w). \tag{16}$$

So by Observation 21, a bijection π with $\mathcal{C}_\Diamond(\pi) < \tau$ maps every vertex v to a vertex $\pi(v)$ with $|\deg_{G'}(v) - \deg_{H'}(\pi(v))| < \tau$.

In G' and H', vertices in $V \setminus Z$ have degree at most Δ, clique vertices have degree at least $m - 1$, and the degree of z_j lies in $[\Delta + \tau, 2\Delta + \tau + ks] = [\Delta + \tau, m - 1 - \tau]$. Moreover, for $i < j$, the order of the markers yields

$$\deg_{G'}(z_j) - \deg_{H'}(z_i) = (j-i)s + \deg_G(z_j) - \deg_H(z_i) \geq s + \deg_G(z_j) - \deg_G(z_i) - \delta \geq \tau,$$

and symmetrically $\deg_{H'}(z_j) - \deg_{G'}(z_i) \geq \tau$. Hence π maps every vertex of $V \setminus Z$ into $V \setminus Z$ and every marker z_j to z_j, so $\pi(V) = V$. This proves the first item.

For the second item, $Q(z_j, z_j) = 1$ implies that every $(v, w) \in \mathrm{supp}(Q')$ satisfies $v = w \in Z \cup \mathcal{K}$ or $v, w \in V \setminus Z$. In both cases, v in G' and w in H' have the same neighbours in $\mathcal{K}$. So for $(v, w), (v', w') \in \mathrm{supp}(Q')$, the entry $D_{G',H'}(v, w, v', w')$ is zero if one of the pairs lies in $\mathcal{K} \times \mathcal{K}$, and $D_{G,H}(v, w, v', w')$ otherwise. Thus $\mathcal{L}_{Q'}$ agrees with $\mathcal{L}_Q$ on $\mathrm{supp}(Q)$ and is zero on $\mathrm{supp}(Q') \setminus \mathrm{supp}(Q)$. The claim then follows from Eq. (16). ◀

C.1 The Lower Bound

In this section, we prove Proposition 5, following the proof overview in Section 3.1.

Fix an odd $N \geq 3$. We start with the formal construction of the graphs G and H. To this end, we first define graphs G' and H' which are defined on a common set of marker vertices $M \coloneqq \{z_0, \dots, z_{N-1}\}$ and a set of core vertices $S_G \coloneqq \{v_0, v_2, \dots, v_{2N-2}\}$ and $S_H \coloneqq \{w_1, w_3, \dots, w_{2N-1}\}$, respectively. In each graph, distinct core vertices $v_i, v_{i'}$ (respectively $w_i, w_{i'}$) are connected by an edge if $i + i' \equiv 0 \pmod{2N}$ (compare Figure 1). To describe the connections between core and marker vertices, for every $r \in \{0, \dots, N-1\}$, we let

$$I_r \coloneqq \{(r + s) \bmod (2N) \mid s \in \{0, \dots, N-1\}\} \subseteq \mathbb{Z}_{2N}$$

be the set of positions along a semicircle in the cycle $\mathbb{Z}_{2N}$. A *core-marker pair* is connected by an edge $v_i z_r$ (or $w_i z_r$) in either graph if the index of the core vertex is contained in the semicircle corresponding to the marker, that is $i \in I_r$. This is visualised for z_0 in Figure 1 and for all $z \in M$ in Figure 4.

Finally, we obtain G and H by applying Lemma 22 to G' and H' (where we identify v_i of G' with w_{i+1} of H'), with the markers M ordered nondecreasingly by their degree in G' and parameters $\Delta \coloneqq N + 1$, $\tau \coloneqq 4N - 1$ and $\delta \coloneqq 1$. These parameters are valid because every vertex has degree at

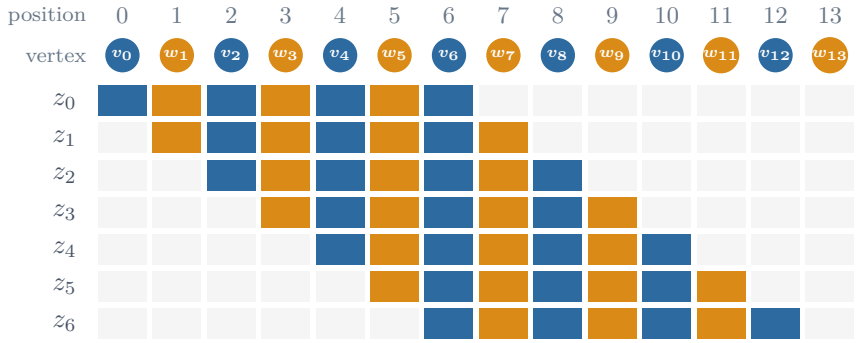


Figure 4 Visualisation of the adjacencies between core vertices and marker vertices in G and H of Proposition 5 for $N = 7$. A coloured cell indicates adjacency. Two columns differ in exactly as many rows as the cyclic distance of their positions, so mapping v_i to w_j changes the adjacency to $\ell(v_i, w_j)$ markers (Lemma 23).

most $N + 1$ in G' and H', and the degree of z_r in the two graphs differs by exactly one for each r.

All in all, the subgraph induced by the core vertices consists of $(N-1)/2$ disjoint edges and a single isolated vertex in both G and H. However, the positions assigned to these vertices (and thus their connections to marker vertices) are shifted by one step along the cycle $\mathbb{Z}_{2N}$.

Write $V \coloneqq V(G)$, $W \coloneqq V(H)$ and call a fractional alignment $Q \in \mathrm{DS}(V, W)$ *compliant* if Q acts as the identity on the marker vertices and the clique added by Lemma 22. For $v_i \in S_G$ and $w_j \in S_H$, we define $\ell(v_i, w_j) \coloneqq \min_{k\in\mathbb{Z}} |i - j + 2kN|$. Intuitively, this corresponds to the shortest path distance between v_i and w_j when placed at their assigned position in a common cycle graph C_{2N}.

Lemma 23. *For every compliant $Q \in \mathrm{DS}(V, W)$, the core-marker pairs contribute exactly*

$$2 \sum_{v_i \in S_G,\, w_j \in S_H} Q(v_i, w_j)\, \ell(v_i, w_j)$$

to its edit cost $\mathcal{C}_{\mathsf{ed}}(Q)$.

Proof. Mapping v_i to w_j moves its position on the cycle C_{2N} by $\ell(v_i, w_j) \leq N$ (compare Figure 1), incurring a change of adjacency to exactly $\ell(v_i, w_j)$ markers (compare Figure 4). Summing these differences weighted by $Q(v_i, w_j)$ and counting both orientations of each core-marker pair gives the formula. ◀

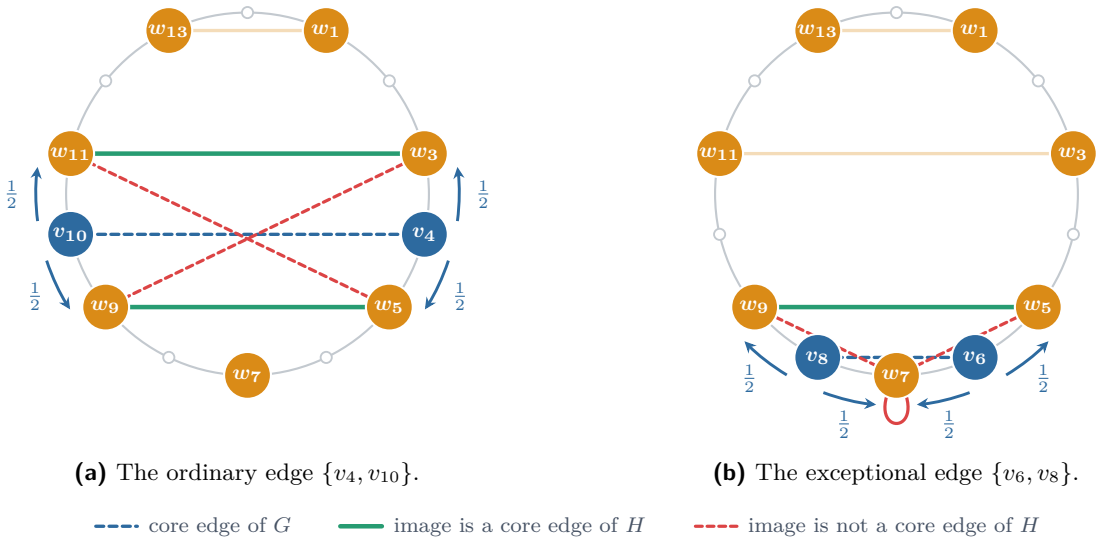
Lemma 24. $\mathsf{d}_{\mathsf{ed}}^{\mathsf{OT}}(G, H) \leq 3N + 1$.

Proof. Let $\pi_-, \pi_+ \in \mathrm{Bij}(V, W)$ be the compliant bijections with

$$\pi_-(v_i) \coloneqq w_{(i-1) \bmod 2N},$$
$$\pi_+(v_i) \coloneqq w_{(i+1) \bmod 2N}$$

for each $i \in \{0, 2, \ldots, 2N-2\}$ and let $Q \coloneqq \frac{1}{2}(P_{\pi_-} + P_{\pi_+})$. Hence Q sends half of the mass of v_i *backward* to $\pi_-(v_i)$ and half of it *forward* to $\pi_+(v_i)$. By Lemma 22, we may compute the cost on G' and H'. Pairs of markers contribute nothing to $\mathcal{C}_{\mathsf{ed}}(Q)$, and since $\ell(v_i, \pi_-(v_i)) = \ell(v_i, \pi_+(v_i)) = 1$, Lemma 23 gives a contribution of $2N$ from the core-marker pairs.

Each edge of G' is preserved under an alignment if one endvertex is moved forward and the other is moved backward. As vertices are mapped forward or backward by Q independently with



(a) The ordinary edge $\{v_4, v_{10}\}$.

(b) The exceptional edge $\{v_6, v_8\}$.

Figure 5 The fractional alignment $Q = \frac{1}{2}(P_{\pi_-} + P_{\pi_+})$ from Lemma 24 on a core edge of G for $N = 7$. (a) The edge is preserved, with image $\{w_5, w_9\}$ or $\{w_3, w_{11}\}$, in the two cases in which one endpoint is moved forward and the other backward, so with probability $\frac{1}{2}$; otherwise its image is $\{w_5, w_{11}\}$ or $\{w_3, w_9\}$. (b) Near the isolated core vertex w_7, only moving v_6 backward and v_8 forward preserves the edge; the other cases give $\{w_7, w_9\}$, $\{w_5, w_7\}$, and the pair (w_7, w_7), drawn as a loop, which is a nonedge of H. Symmetrically, the isolated core vertex v_0 of G sends mass to both endpoints of the core edge $\{w_{13}, w_1\}$ of H.

probability $\frac{1}{2}$, this happens in two of the four equally probable cases, so with probability $\frac{1}{2}$. Hence, each edge is not preserved with probability $\frac{1}{2}$. This is visualised in Figure 5a.

There are a total of $\frac{N-1}{2}$ core edges in G and considering both orientations for each thus yields a total cost of $2 \cdot \frac{N-1}{2} \cdot \frac{1}{2} = \frac{N-1}{2}$. Applying the same reasoning for the $\frac{N-1}{2}$ core edges of H (with the inverses of π_- and π_+) yields the same quantity again. The only imprecision of this argument occurs when moving forward or backward involves one of the vertices v_0, w_N since these are isolated in the two graphs and hence cannot preserve edges. This is visualised in Figure 5b. The cost incurred by this is at most one for each of these two vertices.

In total, the cost is at most $2N + (N-1) + 2 = 3N + 1$, which concludes the proof. ◀

Lemma 25. $\mathsf{d_{ed}}(G, H) \geq 4N - 2$.

Proof. Let $\pi \in \mathrm{Bij}(V, W)$ be arbitrary. If $\mathcal{C}_{\mathsf{ed}}(\pi) \geq \tau = 4N - 1$, there is nothing to prove. Otherwise, Lemma 22 implies that π fixes every marker and maps S_G onto S_H. We may assume that π is compliant, as replacing its action on the added clique by the identity cannot increase the cost. For every $v_i \in S_G$, let $d_i \in \{-N+2, -N+4, \ldots, N\}$ be the signed *displacement* of v_i under π, so that

$$\pi(v_i) = w_{(i+d_i) \bmod 2N}$$

for each $v_i \in S_G$. For example, moving v_i to the opposite end of the cycle corresponds to $d_i = N$.

Note that $\ell(v_i, \pi(v_i)) = |d_i|$ for each $v_i \in S_G$, so Lemma 23 gives a core-marker contribution of $2U$, where

$$U \coloneqq \sum_{v_i \in S_G} |d_i|.$$

Let k be the number of core edges of G' that π maps onto core edges of H'.

Claim 26. *We have $U \geq N + 2k$.*

Proof. If $d_i = N$ for some $v_i \in S_G$, then $U \geq N + (N-1) = 2N - 1 \geq N + 2k$, since every other displacement has absolute value at least 1 and $\frac{N-1}{2} \geq k$. So we can assume that every d_i lies in the interval $(-N, N)$. Since π maps the even positions bijectively onto the odd ones, we have

$$\sum_{v_i \in S_G} d_i \equiv \sum_{r=0}^{N-1} (2r+1) - \sum_{r=0}^{N-1} 2r = N \pmod{2N},$$

so the absolute value of this sum is at least N.

For a core edge $\{v_i, v_{i'}\}$ preserved under π, both the source positions and the target positions sum to 0 modulo $2N$. Since $d_i, d_{i'} \in (-N, N)$, their sum (as integers) is zero as well, while $|d_i| + |d_{i'}| \geq 2$ since all displacements are nonzero.

Let $S \subseteq S_G$ be the set of endpoints of the k preserved edges, so $|S| = 2k$. Then

$$U \geq \sum_{v_i \in S} |d_i| + \sum_{v_i \in S_G \setminus S} |d_i| \geq 2k + \left| \sum_{v_i \in S_G \setminus S} d_i \right| = 2k + \left| \sum_{v_i \in S_G} d_i - \underbrace{\sum_{v_i \in S} d_i}_{=0} \right| \geq 2k + N$$

where the second sum in the last expression is zero by pairing the endvertices of the k preserved edges. ◁

Both cores have $(N-1)/2$ edges, of which exactly k are preserved. Their ordered pairs therefore contribute $2(N - 1 - 2k)$ to the cost. Together with the marker contribution and Claim 26, this gives

$$\mathcal{C}_{\mathsf{ed}}(\pi) \geq 2(N - 1 - 2k) + 2U \geq 4N - 2. \qquad \blacktriangleleft$$

Proof of Proposition 5. By Lemmas 24 and 25,

$$\begin{aligned} \mathsf{d}_{\mathsf{ed}}(G, H) \geq 4N - 2 &= \left(\frac{4}{3} - \frac{10}{3(3N+1)}\right)(3N+1) \\ &\geq \left(\frac{4}{3} - \frac{10}{3(3N+1)}\right) \mathsf{d}_{\mathsf{ed}}^{\mathsf{OT}}(G, H). \end{aligned}$$

The two distances have the same normalisation factor. For any $\varepsilon > 0$, choosing an odd N sufficiently large so that $\frac{10}{3(3N+1)} \leq \varepsilon$ proves the claimed bound. ◀

C.2 The Upper Bound

We denote the Frobenius inner product by $\langle \cdot, \cdot \rangle_F$.

Proof of Theorem 6. Let $\boldsymbol{\pi}$ be the random bijection with $\Pr(\boldsymbol{\pi} = \pi_i) = \lambda_i$ and write $\boldsymbol{P} \coloneqq P_{\boldsymbol{\pi}}$, so that $\mathbb{E}[\boldsymbol{P}] = Q$ for a Birkhoff decomposition guaranteed by Eq. (3). Since all entries of A_G and A_H are in $\{0, 1\}$, we have $|a - b| = a^2 + b^2 - 2ab$ for all entries a of A_G and b of A_H. Expanding $\mathcal{C}_{\mathsf{ed}}$ accordingly, the row and column sums of Q reduce the first two terms to $\|A_G\|_F^2 + \|A_H\|_F^2$, so that

$$\mathcal{C}_{\mathsf{ed}}(Q) = \|A_G\|_F^2 + \|A_H\|_F^2 - 2\langle A_G Q, Q A_H \rangle_F. \tag{17}$$

We therefore have

$$\mathbb{E}[\mathcal{C}_{\mathsf{ed}}(\boldsymbol{\pi})] = \mathbb{E}\left[\left\| A_G - \boldsymbol{P} A_H \boldsymbol{P}^\top \right\|_F^2 \right]$$

$$
\begin{aligned}
&= \mathbb{E}\Big[\|A_G\boldsymbol{P} - \boldsymbol{P}A_H\|_F^2\Big] && \text{(a)}\\
&= \mathbb{E}\Big[\|(A_G\boldsymbol{P} - QA_H) + (QA_H - \boldsymbol{P}A_H)\|_F^2\Big]\\
&\le 2\,\mathbb{E}\Big[\|A_G\boldsymbol{P} - QA_H\|_F^2 + \|QA_H - \boldsymbol{P}A_H\|_F^2\Big] && \text{(b)}\\
&= 2\,\mathbb{E}\Big[\|A_G\boldsymbol{P} - QA_H\|_F^2 + \|QA_H\|_F^2 + \|\boldsymbol{P}A_H\|_F^2 - 2\langle QA_H, \boldsymbol{P}A_H\rangle_F\Big] && \text{(c)}\\
&= 2\,\mathbb{E}\Big[\|A_G\boldsymbol{P} - QA_H\|_F^2 + \|QA_H\|_F^2 + \|\boldsymbol{P}A_H\|_F^2 - 2\langle QA_H, QA_H\rangle_F\Big] && \text{(d)}\\
&= 2\,\mathbb{E}\Big[\|A_G\boldsymbol{P} - QA_H\|_F^2 - \|QA_H\|_F^2 + \|\boldsymbol{P}A_H\|_F^2\Big] && \text{(e)}\\
&= 2\,\mathbb{E}\Big[\|A_G\boldsymbol{P}\|_F^2 + \|QA_H\|_F^2 - 2\langle A_G\boldsymbol{P}, QA_H\rangle_F - \|QA_H\|_F^2 + \|\boldsymbol{P}A_H\|_F^2\Big] && \text{(f)}\\
&= 2\,\mathbb{E}\Big[\|A_G\boldsymbol{P}\|_F^2 + \|\boldsymbol{P}A_H\|_F^2 - 2\langle A_G\boldsymbol{P}, QA_H\rangle_F\Big]\\
&= 2\Big(\|A_G\|_F^2 + \|A_H\|_F^2 - 2\langle A_GQ, QA_H\rangle_F\Big) && \text{(g)}\\
&= 2\mathcal{C}_{\mathsf{ed}}(Q). && \text{Eq. (17)}
\end{aligned}
$$

Step (a) uses permutation invariance of the Frobenius norm. Step (b) is the squared triangle inequality $\|X+Y\|_F^2 \le 2\|X\|_F^2 + 2\|Y\|_F^2$. Step (c) expands the second square. Step (d) uses $\mathbb{E}[\boldsymbol{P}] = Q$, linearity of the inner product and linearity of expectation, while step (e) simplifies $\langle QA_H, QA_H\rangle_F = \|QA_H\|_F^2$. Step (f) expands the first square. In step (g), we use $\mathbb{E}[\boldsymbol{P}] = Q$ and permutation invariance of $\|\cdot\|_F$.

For the second assertion, let Q be a minimiser of $\mathcal{C}_{\mathsf{ed}}$ over $\mathrm{DS}(V,W)$ and sample $\boldsymbol{\pi}$ from a Birkhoff decomposition Eq. (3) of Q. Then

$$\mathsf{d}_{\mathsf{ed}}(G,H) \le \mathbb{E}[\mathcal{C}_{\mathsf{ed}}(\boldsymbol{\pi})] \le 2\mathcal{C}_{\mathsf{ed}}(Q) = 2\mathsf{d}_{\mathsf{ed}}^{\mathsf{OT}}(G,H). \qquad \blacktriangleleft$$

Note that Theorem 6 also holds for arbitrary matrices A, B since the argument does not rely on any properties of the input matrices. Hence, applying it to the threshold matrices $\mathbb{1}[A > \tau]$ and $\mathbb{1}[B > \tau]$ for a fixed decomposition Q and every $\tau \in \mathbb{R}$ and integrating over τ yields the claim since $|a-b| = \int_{\mathbb{R}} |\mathbb{1}[a>\tau] - \mathbb{1}[b>\tau]|\, d\tau$.

D The Integrality Gap of the Local Edit Distance

D.1 The Lower Bound

We start by giving the formal construction of the graphs G_r and H_r. (See Figure 2 for a visualisation of the case $r = 4$.) Fix an even integer $r \ge 2$ and let

$$\ell := r^2 + 2r, \qquad n_X := 1 + r(\ell - 1), \qquad n_M := 2r^2.$$

The n_X *core vertices* are

$$X := \{u\} \cup \{v_{i,k} \mid i \in [r], k \in [\ell - 1]\},$$

and the n_M *markers* are the disjoint union $M := C \cup T \cup B$ of the *clock markers* $C := \{c_{j,h} \mid j \in [r-1], h \in [r]\}$, the *tags* $T := \{t_{i,h} \mid i, h \in [r]\}$, and the *bits* $B := \{b_i \mid i \in [r]\}$. We first define a bipartite graph G_r^0 on $X \cup M$ by the following rules:

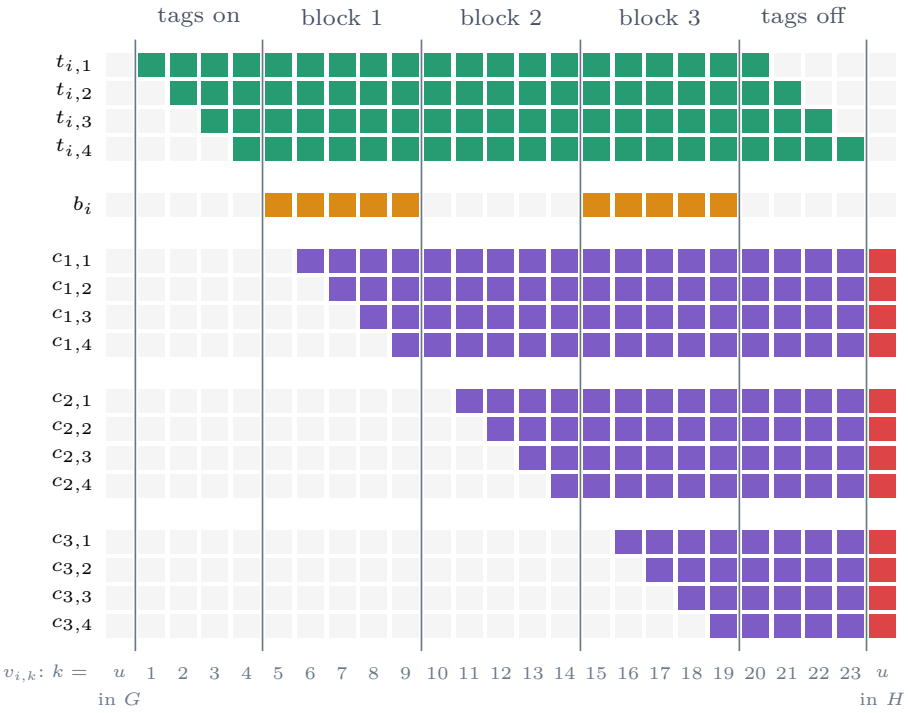


Figure 6 Visualisation of the adjacency patterns $\chi_{G_4}(u), \chi(v_{i,1}), \ldots, \chi(v_{i,23}), \chi_{H_4}(u)$ obtained by following the cyclic permutation σ_i of Lemma 27 in G_4 and H_4.

Core vertex	Marker	Condition for adjacency
$v_{i,k}$	$c_{j,h}$	$k \geq j(r+1)+h$
$v_{i,k}$	$t_{i',h}$	$i = i'$ and $h \leq k < \ell - r + h$
$v_{i,k}$	$b_{i'}$	$i = i'$ and $\lfloor k/(r+1) \rfloor$ is odd

In particular, u is an isolated vertex of G_r^0. Let H_r^0 be obtained from G_r^0 by adding all edges between u and the clock markers C. Hence $\deg_{H_r^0}(z) = \deg_{G_r^0}(z) + \mathbb{1}[z \in C]$ for every marker z.

To force every bijection of small cost to fix markers and preserve core vertices, let G_r and H_r be obtained from G_r^0 and H_r^0, respectively, by applying Lemma 22 with their markers enumerated $z_1, \ldots, z_{n_M}$ (ordered non-decreasingly in order of their degrees in G_r^0), $\Delta := n_X$, $\tau := r$ and $\delta := 1$. (Formally, Lemma 22 requires both graphs to be defined on the same vertex set, which can be achieved by relabelling vertices accordingly.) Here we use that all vertices of G_r^0 and H_r^0 have degree at most n_X and that core vertices are only adjacent to markers. Moreover, only the degree of a clock marker differs (by one) between the two graphs because of u.

For a core vertex $x \in X$, we let $\chi_{G_r}(x), \chi_{H_r}(x) \in \{0,1\}^M$ be its *patterns*, i.e. the indicator vectors of its neighbours in M in G_r and H_r, respectively. By construction, we have $\chi_{G_r}(v_{i,k}) = \chi_{H_r}(v_{i,k})$ for all i, k, so in the following we only write $\chi(v_{i,k})$. Moreover, $\chi_{G_r}(u) = \mathbf{0}$ and $\chi_{H_r}(u) = \mathbb{1}_C$. Fix a branch $i \in [r]$ and consider the sequence $\chi_{G_r}(u), \chi(v_{i,1}), \ldots, \chi(v_{i,\ell-1}), \chi_{H_r}(u)$ of patterns at *times* $0, 1, \ldots, \ell$. Consecutive patterns differ in exactly one coordinate: at time $h \in [r]$, the tag $t_{i,h}$ is switched on; at time $j(r+1)$ for $j \in [r]$, the bit b_i is flipped; at time $j(r+1)+h$ for $j \in [r-1]$ and $h \in [r]$, the clock marker $c_{j,h}$ is switched on; and at time $\ell - r + h$ for $h \in [r]$, the tag $t_{i,h}$ is switched off. These times partition $[\ell]$, and since r is even, the bit b_i is 0 at time ℓ, consistent with $\chi_{H_r}(u) = \mathbb{1}_C$. A visualisation of this is given in Figure 6.

The following two lemmas show the desired gap.

Lemma 27. $\mathsf{d}_1^{\mathsf{OT}}(G_r, H_r) = 1$.

Proof. For $i \in [r]$, let σ_i be the bijection from G_r to H_r that acts as the identity on the marker vertices and shifts the core vertices of the i-th branch along one step, that is $\sigma_i(u) = v_{i,1}$, $\sigma_i(v_{i,k}) = v_{i,k+1}$ for $k \in [\ell - 2]$ and $\sigma_i(v_{i,\ell-1}) = u$. Define

$$Q_r \coloneqq \frac{1}{r} \sum_{i=1}^{r} P_{\sigma_i}.$$

Intuitively, Q_r spreads u (in G_r) uniformly across the r start vertices of the r branches. Each branch vertex maps a $1/r$ fraction of its mass to the next vertex along the branch; the r endpoints then map mass $1/r$ each into u (in H).

Since Q_r fixes all markers and all vertices of $\mathcal{K}$, the second item of Lemma 22 shows that $\mathcal{C}_1(Q_r)$ is the same with respect to (G_r, H_r) and with respect to (G_r^0, H_r^0), so we may compute it in the latter pair. Since core vertices are only adjacent to markers, the local cost of a supported pair (x, y) of core vertices is the Hamming distance of $\chi_{G_r}(x)$ and $\chi_{H_r}(y)$. The supported pairs of core vertices are either of the form $(v_{i,k}, v_{i,k})$, of cost 0, or consecutive vertices along a branch, of cost 1. For a marker z, the local cost $\mathcal{L}_{Q_r}(z, z)$ is the total Q_r-mass of supported core pairs whose patterns differ at z, where every pair of consecutive vertices along a branch has mass $\frac{1}{r}$. Each clock marker changes once on each of the r branches, each bit changes r times on its own branch, and each tag changes twice on its own branch. Hence the local costs at clock markers, tags, and bits are 1, $\frac{2}{r}$, and 1, respectively. Thus $\mathcal{C}_1(Q_r) \leq 1$. Equality follows from Lemma 29 below since $G_r \ncong H_r$. ◀

Lemma 28. $\mathsf{d}_1(G_r, H_r) \geq r$.

Proof. Suppose that π is a bijection with $\mathcal{C}_1(\pi) < r$. By construction and Lemma 22 with $\tau \coloneqq r$, π fixes all markers and permutes X. Hence, for every core vertex $x \in X$, $\mathcal{L}_\pi(x, \pi(x))$ is the Hamming distance of the patterns $\chi_{G_r}(x)$ and $\chi_{H_r}(\pi(x))$, which is less than r.

Starting at u in G_r, follow π to $\pi(u)$ in H_r, continue from $\pi(u)$ in G_r, and so on, until $\pi^s(u) = u$ is reached in H_r. Consider the resulting sequence of patterns $x_0 \coloneqq \chi_{G_r}(u)$, $x_\iota \coloneqq \chi(\pi^\iota(u))$ for $0 < \iota < s$, and $x_s \coloneqq \chi_{H_r}(u)$. By assumption, consecutive patterns in this sequence have Hamming distance less than r. For a pattern x, let $c(x)$ be the number of clock markers in x, its *clock level*. The clock markers of every pattern form a prefix of the order $c_{1,1}, \ldots, c_{1,r}, c_{2,1}, \ldots, c_{r-1,r}$, so consecutive clock levels in the sequence differ by less than r, and we have $c(x_0) = 0$ and $c(x_s) = r(r-1)$.

Let β be the first index with $c(x_\beta) = r(r-1)$ and let $\alpha < \beta$ be the last index before β with $c(x_\alpha) = 0$. Every pattern x_ι with $\alpha < \iota < \beta$ has clock level strictly between 0 and $r(r-1)$, so it is the pattern of a vertex $v_{i,k}$ with $r + 2 \leq k \leq r^2 + r - 2$, at which all r tags of branch i are switched on and all other tags are switched off. Thus two such patterns on different branches have Hamming distance at least $2r$, and hence all patterns x_ι with $\alpha < \iota < \beta$ belong to the same branch i.

For $j \in [r-1]$, let ι_j be the first index in $(\alpha, \beta]$ with $c(x_{\iota_j}) \geq jr$. As consecutive clock levels differ by at most $r - 1$, we have $c(x_{\iota_j - 1}) \in ((j-1)r, jr)$, so $x_{\iota_j - 1}$ is the pattern of a vertex $v_{i,k}$ with $k = j(r+1) + h'$ for some $h' \in [r-1]$, and thus its value at the bit b_i is $j \bmod 2$. Since $\iota_1 < \iota_2 < \cdots < \iota_{r-1}$, and since x_0 and x_s both have value 0 at b_i, the sequence of values at b_i contains the subsequence $(0, 1, 0, 1, \ldots, 1, 0)$ of length $r + 1$ and r alternations. Thus at least r distinct core vertices x satisfy $A_{G_r}(b_i, x) \neq A_{H_r}(b_i, \pi(x))$. Hence, with $b_i = \pi(b_i)$, we have $\mathcal{L}_\pi(b_i, \pi(b_i)) \geq r$, a contradiction. ◀

Proof of Proposition 8. By Lemma 27, $\mathsf{d}_1^{\mathsf{OT}}(G_r, H_r) = 1$. The bijection σ_1 fixes all markers and all vertices of $\mathcal{K}$, so by the second item of Lemma 22 its local costs can be computed in (G_r^0, H_r^0) as

in the proof of Lemma 27 yielding $\mathsf{d}_1(G_r, H_r) = r$ by Lemma 28 and since $\mathcal{L}_{\sigma_1}(b_1, b_1) = r$. Finally, $|G_r^0| = \Theta(r^3)$, and Lemma 22 only adds $\mathrm{O}(r^3)$ vertices, so $|V(G_r)| = |V(H_r)| = \Theta(r^3)$. ◀

D.2 The Upper Bound

Throughout this subsection, let $L \coloneqq \mathsf{d}_1^{\mathsf{OT}}(G, H)$ and fix an optimal $Q \in \mathrm{DS}(V, W)$, i.e. $\mathcal{L}_Q(v, w) \leq L$ for all $(v, w) \in \mathrm{supp}(Q)$. The pseudometric $\rho_H(w, w') \coloneqq |N_H(w) \mathbin{\triangle} N_H(w')|$ on W measures how much the neighbourhood changes if w is replaced by w'. By Eq. (3), $\mathrm{supp}(Q)$ contains the graph of a bijection $\sigma \in \mathrm{Bij}(V, W)$, that is,

$$Q(v, \sigma(v)) > 0 \quad \text{and} \quad \mathcal{L}_Q(v, \sigma(v)) \leq L \qquad \text{for all } v \in V. \tag{18}$$

We fix σ and use it as a reference: $\sigma(v)$ is the vertex of H whose neighbourhood we compare with the neighbourhood of v in G.

Lemma 29. *If $Q(v, w) > 0$, then $\rho_H(\sigma(v), w) \leq 2L$. Moreover, if $L < 1$, then σ is an isomorphism from G to H. In particular, if G and H are not isomorphic, then $L \geq 1$.*

Proof. Let $v \in V$ and $w, w' \in W$. Using the column sums of Q and the triangle inequality, we obtain

$$\begin{aligned}
\rho_H(w, w') &= \sum_{v' \in V} \sum_{w'' \in W} Q(v', w'') |A_H(w, w'') - A_H(w', w'')| \\
&\leq \sum_{v' \in V} \sum_{w'' \in W} Q(v', w'') \Big(|A_G(v, v') - A_H(w, w'')| + |A_G(v, v') - A_H(w', w'')| \Big) \\
&= \mathcal{L}_Q(v, w) + \mathcal{L}_Q(v, w').
\end{aligned}$$

For $w \coloneqq \sigma(v)$ and $(v, w') \in \mathrm{supp}(Q)$, this is at most $2L$ by Eq. (18).

Now suppose that $L < 1$. By Observation 21, $\deg_G(v) = \deg_H(w)$ for every $(v, w) \in \mathrm{supp}(Q)$. Applying this to $(v, \sigma(v))$ and (v, w) shows that $\deg_H(\sigma(v)) = \deg_H(w)$ for all $(v, w) \in \mathrm{supp}(Q)$, hence $\rho_H(\sigma(v), w)$ is even, and as $\rho_H(\sigma(v), w) \leq 2L < 2$, it is 0. Thus $A_H(w', w) = A_H(w', \sigma(v))$ for all $w' \in W$ and $(v, w) \in \mathrm{supp}(Q)$. Hence, for every $v' \in V$, using the row sums of Q,

$$\mathcal{L}_Q(v', \sigma(v')) = \sum_{v \in V} \sum_{w \in W} Q(v, w) |A_G(v', v) - A_H(\sigma(v'), w)| = \sum_{v \in V} |A_G(v', v) - A_H(\sigma(v'), \sigma(v))|.$$

Since the right-hand side is an integer and strictly less than one, it is 0. It follows that $A_G(v', v) = A_H(\sigma(v'), \sigma(v))$ for all $v, v' \in V$, i.e. σ is an isomorphism from G to H. ◀

We view $\mathrm{supp}(Q) \subseteq V \times W$ as a bipartite graph with sides V and W, and Q as a fractional perfect matching in it. For a matching $M \subseteq \mathrm{supp}(Q)$ and $v \in V$, let

$$\Phi_v(M) \coloneqq \sum_{(v', w') \in M} |A_G(v, v') - A_H(\sigma(v), w')|.$$

be the local cost of v under σ relative to M. Note that $\Phi_v(M)$ is a linear statistic of M with coefficients in $\{0, 1\}$, and that

$$\sum_{v' \in V} \sum_{w' \in W} |A_G(v, v') - A_H(\sigma(v), w')| Q(v', w') = \mathcal{L}_Q(v, \sigma(v)) \leq L.$$

We use the following special case of the dependent rounding scheme of Chekuri, Vondrák and

Zenklusen.

Lemma 30 ([CVZ11, Theorem 1.1]). *Let $\boldsymbol{q}$ be a point in the bipartite matching polytope and $\gamma \in (0, \frac{1}{2}]$. Then there is a random matching $\boldsymbol{M}$ with $\mathbb{E}[\mathbb{1}_{\boldsymbol{M}}] = (1-\gamma)\boldsymbol{q}$ such that, for every linear statistic $Z \coloneqq \sum_e \alpha_e \mathbb{1}[e \in \boldsymbol{M}]$ with $\alpha_e \in [0,1]$, every $\mu \geq \mathbb{E}[Z]$ with $\mu > 0$, and every $\tau \geq 2$,*

$$\Pr\left(Z \geq \tau\mu\right) \leq \exp\left(-\frac{\mu\gamma(2\tau-3)}{20}\right). \tag{19}$$

Lemma 31. *Suppose that $L > 0$. For every $\gamma \in (0, \frac{1}{2}]$, there is a matching $M \subseteq \operatorname{supp}(Q)$ such that*

$$n - |M| \leq 2\gamma n \qquad \text{and} \qquad \Phi_v(M) \leq 2L + \frac{10\ln(4n)}{\gamma} \quad \text{for all } v \in V.$$

Proof. We apply Lemma 30 to $\boldsymbol{q} \coloneqq Q$, which is a convex combination of perfect matchings in $\operatorname{supp}(Q)$, and let $\boldsymbol{M}$ be the resulting random matching. For every $v \in V$, we have $\mathbb{E}[\Phi_v(\boldsymbol{M})] = (1-\gamma)\mathcal{L}_Q(v, \sigma(v)) \leq L$. For $Z \coloneqq \Phi_v(\boldsymbol{M})$, $\mu \coloneqq L$, and $\tau \coloneqq 2 + 10\ln(4n)/(\gamma L)$, Eq. (19) yields

$$\Pr\left(\Phi_v(\boldsymbol{M}) \geq 2L + \frac{10\ln(4n)}{\gamma}\right) \leq \frac{\exp(-\gamma L/20)}{4n} \leq \frac{1}{4n}.$$

By the union bound, the probability that this condition fails for some $v \in V$ is at most $\frac{1}{4}$. Moreover, $\mathbb{E}[|\boldsymbol{M}|] = (1-\gamma)n$, so Markov's inequality applied to $n - |\boldsymbol{M}|$ yields $\Pr\left(n - |\boldsymbol{M}| > 2\gamma n\right) \leq \frac{1}{2}$. Hence, with positive probability, neither event occurs. This yields the desired matching. ◀

Proposition 32. *Suppose that $L > 0$. For all $\gamma \in (0, \frac{1}{2}]$ and $\ell \in \mathbb{N}_{>0}$,*

$$\mathsf{d}_1(G, H) \leq 2L + \frac{10\ln(4n)}{\gamma} + 2\gamma n + \frac{n}{\ell} + 2\ell L.$$

Proof. Let M be the matching of Lemma 31 and $m \coloneqq n - |M|$, so that exactly m vertices of V and m vertices of W are unmatched. To turn M into a bijection, we use σ to go back from W to V: let μ be the partial injective map on V given by $\mu(v) \coloneqq \sigma^{-1}(w)$ for every $(v, w) \in M$. In the directed graph on V with the edges $v \to \mu(v)$, every vertex has in- and outdegree at most 1. Thus it decomposes into directed cycles (including loops) and exactly m directed paths (including isolated vertices), since every path ends in exactly one vertex on which μ is undefined. We keep all cycles, cut every path into consecutive blocks of ℓ vertices (plus a final shorter block), discard all edges between blocks, and close every block $v_1 \to \cdots \to v_k$ by adding the *closing edge* $v_k \to v_1$, where a single vertex is closed by a loop. The resulting directed cycles define a permutation τ of V, and we let $\pi \coloneqq \sigma \circ \tau \in \operatorname{Bij}(V, W)$. Hence $\pi(v) = w$ if $(v, w) \in M$, so all edges of the form $v \to \mu(v)$ are retained, and $\pi(v_k) = \sigma(v_1)$ for a closing edge $v_k \to v_1$. The number of closing edges is at most $\sum_j \lceil p_j/\ell \rceil \leq m + n/\ell$, where $p_1, \ldots, p_m$ are the numbers of vertices of the paths.

We claim that $\rho_H(\sigma(v), \pi(v)) \leq 2\ell L$ for all $v \in V$. If the edge $v \to \mu(v)$ is retained, then $(v, \pi(v)) \in M \subseteq \operatorname{supp}(Q)$, and Lemma 29 yields $\rho_H(\sigma(v), \pi(v)) \leq 2L$. For a closing edge $v_k \to v_1$ of a block $v_1 \to \cdots \to v_k$, we have $(v_j, \sigma(v_{j+1})) \in M$ for all $j < k$, so Lemma 29 yields $\rho_H(\sigma(v_j), \sigma(v_{j+1})) \leq 2L$, and the triangle inequality then gives

$$\rho_H(\sigma(v_k), \pi(v_k)) = \rho_H(\sigma(v_k), \sigma(v_1)) \leq \sum_{j=1}^{k-1} \rho_H(\sigma(v_j), \sigma(v_{j+1})) \leq 2(k-1)L.$$

Finally, for every $v \in V$, inserting $A_H(\sigma(v), \pi(v'))$ and using the triangle inequality, we obtain

$$\begin{aligned}\mathcal{L}_{P_\pi}(v, \pi(v)) &= \sum_{v' \in V} \left|A_G(v, v') - A_H(\pi(v), \pi(v'))\right| \\ &\leq \sum_{v' \in V} \left|A_G(v, v') - A_H(\sigma(v), \pi(v'))\right| + \sum_{v' \in V} \left|A_H(\sigma(v), \pi(v')) - A_H(\pi(v), \pi(v'))\right|.\end{aligned}$$

where the second sum equals $\rho_H(\sigma(v), \pi(v)) \leq 2\ell L$ since π is a bijection. In the first sum, the summands for the v' whose edge $v' \to \mu(v')$ is retained satisfy $(v', \pi(v')) \in M$, so their sum is at most $\Phi_v(M)$. All other v' are the starting points of closing edges, hence there are at most $m + n/\ell$ such vertices, with their respective summands being at most 1. This yields $\mathcal{L}_{P_\pi}(v, \pi(v)) \leq \Phi_v(M) + m + n/\ell + 2\ell L$ for all $v \in V$, and therefore $\mathsf{d}_1(G, H) \leq \mathcal{C}_1(\pi) = \max_{v \in V} \mathcal{L}_{P_\pi}(v, \pi(v))$ is bounded by the same quantity. Lemma 31 completes the proof. ◀

Proof of Theorem 9. Let $f \coloneqq \delta_1^{\mathsf{QT}}(G, H) = L/n$ and $b \coloneqq \delta_1(G, H)$. If $L = 0$, then G and H are isomorphic by Lemma 29, so $b = 0$. Hence, suppose that $L > 0$ and let $\ell \coloneqq \lceil \sqrt{n/(2L)} \rceil$ and $\gamma^* \coloneqq \sqrt{5 \ln(4n)/n}$. Then $\frac{n}{\ell} + 2\ell L \leq 2\sqrt{2nL} + 2L$. If $\gamma^* \leq \frac{1}{2}$, then Proposition 32 with $\gamma \coloneqq \gamma^*$ yields

$$\mathsf{d}_1(G, H) \leq 4L + 2\sqrt{2nL} + 4\sqrt{5n \ln(4n)}. \tag{20}$$

If $\gamma^* > \frac{1}{2}$, then $4\sqrt{5n \ln(4n)} = 4n\gamma^* > n \geq \mathsf{d}_1(G, H)$, so Eq. (20) holds as well. As $L > 0$, the graphs are not isomorphic, so $L \geq 1$ by Lemma 29, i.e. $n \geq 1/f \geq 1$. Since $x \mapsto \ln(4x)/x$ is decreasing for $x \geq 1$, we obtain $\ln(4n)/n \leq \frac{\ln(4/f)}{1/f} = f \ln(4/f)$. Dividing Eq. (20) by n yields

$$b \leq 4f + 2\sqrt{2f} + 4\sqrt{5}\sqrt{f \ln(4/f)}.$$

For $0 < f \leq 1$ and $h \coloneqq \ln(4/f) \geq \ln 4$, we have

$$\frac{b}{\sqrt{fh}} \leq \frac{4f + 2\sqrt{2f} + 4\sqrt{5fh}}{\sqrt{fh}} = 4\sqrt{\frac{f}{h}} + \frac{2\sqrt{2}}{\sqrt{h}} + 4\sqrt{5} \leq \frac{4 + 2\sqrt{2}}{\sqrt{\ln 4}} + 4\sqrt{5} < 15.$$

Multiplying both sides by $\sqrt{fh}$ proves the theorem. ◀

E The Integrality Gap of the Cut Distance

E.1 The Lower Bound

We give a formal construction of the graphs depicted in Figure 3. The graph G has vertex set V and the graph H has vertex set W, each consisting of a core of 10 vertices in five pairs, and 135 markers. We denote the core vertices of G by v_i^σ and the core vertices of H by w_i^σ with $i \in \mathbb{Z}_5$ and $\sigma \in \{+1, -1\}$, and write $C_V \subseteq V$ and $C_W \subseteq W$ for the two cores. Core vertices are connected by edges based on the following matrix

$$Z \coloneqq \begin{pmatrix} 0 & +1 & -1 & -1 & +1 \\ +1 & 0 & +1 & -1 & -1 \\ -1 & +1 & 0 & +1 & -1 \\ -1 & -1 & +1 & 0 & +1 \\ +1 & -1 & -1 & +1 & 0 \end{pmatrix} \in \{-1, 0, +1\}^{\mathbb{Z}_5 \times \mathbb{Z}_5},$$

that is, $Z_{ij} = 1$ for neighbouring corners of a pentagon ($j = i \pm 1$) and $Z_{ij} = -1$ for corners two apart ($j = i \pm 2$) where addition is modulo 5. For $i \neq j$, we let

$$v_i^\sigma v_j^\tau \in E(G) \iff \sigma\tau = Z_{ij}, \qquad w_i^\sigma w_j^\tau \in E(H) \iff \sigma\tau = -Z_{ij}.$$

So in G, along a side of the pentagon, + joins + and − joins −, while along a diagonal, + joins −, while in H it is exactly opposite (compare Figure 3). Thus, each core vertex has exactly four core neighbours. For each $i \in \mathbb{Z}_5$, where we identify $\mathbb{Z}_5$ with $\{0, \dots, 4\}$, the vertices v_i^+ and v_i^- are connected to a set $L_i \subseteq V$ of $9(i+1)$ markers, and likewise w_i^+ and w_i^- are connected to a set $L_i' \subseteq W$ of $9(i+1)$ markers. As a result, the pairs have degrees $13, 22, 31, 40, 49$ respectively and hence moving a core vertex to a different pair yields a degree mismatch of at least 9. Moreover, as markers are connected identically to core vertices of the same pair, mixing within a pair of core vertices incurs no cost (relative to the marker edges).

Lemma 33. $\mathsf{d}_\square(G, H) = 8$.

Proof. Let $\pi \in \mathrm{Bij}(V, W)$ and $A := A_G^\pi - A_H$, so that $\mathcal{C}_\square(\pi) = \|A\|_\square$. Taking $S := \{\pi(v)\}$ and $T := W$ shows $\mathcal{C}_\square(\pi) \geq |\deg_G(v) - \deg_H(\pi(v))|$. Since the pairs have degrees $13, 22, 31, 40, 49$ and the markers have degree 2, we get $\mathcal{C}_\square(\pi) \geq 9$ unless π maps every pair $\{v_i^+, v_i^-\}$ onto $\{w_i^+, w_i^-\}$. We may therefore assume that $\pi(v_i^\sigma) = w_i^{r_i\sigma}$ for signs $r_i \in \{-1, +1\}$. Hence, each pair is either kept as-is or swapped under π. Let $K := \{i \in \mathbb{Z}_5 \mid r_i = +1\}$ be the set of indices of pairs that are kept.

For $i \neq j$ we have $Z_{ij} \in \{-1, +1\}$, so $A_G(v_i^\sigma, v_j^\tau) = \frac{1}{2}(1 + Z_{ij}\sigma\tau)$ and $A_H(w_i^\sigma, w_j^\tau) = \frac{1}{2}(1 - Z_{ij}\sigma\tau)$. As $\pi^{-1}(w_i^\sigma) = v_i^{r_i\sigma}$, it follows that

$$A(w_i^\sigma, w_j^\tau) = \tfrac{1}{2} Z_{ij}\, \sigma\tau\, (1 + r_i r_j) = M_{ij}\, \sigma\tau,$$

where $M := Z[K] \oplus Z[\overline{K}]$. This also holds for $i = j$, since there are no edges within a pair and $Z_{ii} = 0$. Now let $S, T \subseteq C_W$, and define vectors $\boldsymbol{x}, \boldsymbol{y} \in \{-1, 0, +1\}^{\mathbb{Z}_5}$ with entries $x_i := \sum_\sigma \sigma\, \mathbb{1}[w_i^\sigma \in S]$ and $y_j := \sum_\tau \tau \mathbb{1}[w_j^\tau \in T]$ for $i, j \in \mathbb{Z}_5$. Then

$$\sum_{s \in S,\, t \in T} A(s, t) = \sum_{i,j \in \mathbb{Z}_5} M_{ij} \Big(\sum_{\sigma\colon w_i^\sigma \in S} \sigma \Big) \Big(\sum_{\tau\colon w_j^\tau \in T} \tau \Big) = \boldsymbol{x}^\top M \boldsymbol{y},$$

and every pair $\boldsymbol{x}, \boldsymbol{y} \in \{-1, 0, +1\}^{\mathbb{Z}_5}$ arises in this way. In particular, S and T realise all pairs of sign vectors. Hence $\mathcal{C}_\square(\pi) \geq \|M\|_{\infty\to 1}$ and since M is block diagonal we can compute its norm as the sum of the norms of its blocks. The four remaining cases (up to symmetries) are $|K| = 0$, $|K| = 1$ and $|K| = 2$ (once with two neighbouring pairs in K and once two non-adjacent pairs) and one easily checks that in each case $\|M\|_{\infty\to 1} \geq 8$. To show that equality holds, let $\mathsf{id} \in \mathrm{Bij}(V, W)$ denote the natural bijection between the two graphs with $v_i^\sigma \mapsto w_i^\sigma$ and $L_i \to L_i'$ for each $i \in \mathbb{Z}_5$. For this alignment, we have $M = Z$ and A is zero outside $C_W \times C_W$ because G and H agree outside their cores. Every cut value is thus exactly of the form $\boldsymbol{x}^\top Z \boldsymbol{y}$ with $\boldsymbol{x}, \boldsymbol{y} \in \{-1, 0, +1\}^{\mathbb{Z}_5}$. The maximum of this bilinear expression is attained at sign vectors, and one can show that $\mathcal{C}_\square(\mathsf{id}) = \|Z\|_{\infty\to 1} = 8$. ◀

Lemma 34. $\mathsf{d}_\square^{\mathsf{OT}}(G, H) \leq 4$.

Proof. Let $Q \in \mathrm{DS}(V, W)$ map each marker set L_i bijectively onto L_i' and match each pair $\{v_i^+, v_i^-\}$ uniformly to $\{w_i^+, w_i^-\}$, that is

$$Q(v_i^\sigma, w_i^\tau) \coloneqq \tfrac{1}{2} \qquad \text{for all } i \in \mathbb{Z}_5 \text{ and } \sigma, \tau \in \{+1, -1\}.$$

Since each marker is adjacent to both vertices of its pair, every discrepancy involving a marker is zero. The remaining 20 atoms (v_i^σ, w_i^τ) in $\mathrm{supp}(Q)$ each carry mass $\frac{1}{2}$, with discrepancy

$$D(v_i^\sigma, w_i^\tau, v_j^{\sigma'}, w_j^{\tau'}) = \tfrac{1}{2} Z_{ij}(\sigma\sigma' + \tau\tau').$$

Call an atom *straight* if $\sigma = \tau$ and *crossed* if $\sigma = -\tau$. Between a straight and a crossed atom, the discrepancy is 0, while between two atoms of the same kind it is $Z_{ij}\sigma\sigma'$. As a result, the discrepancy decomposes into two disjoint copies of the pentagon pattern, each with weight of $\frac{1}{2}$. In each copy, a cut has value $\frac{1}{4}\boldsymbol{x}^\top Z \boldsymbol{y}$ with $\boldsymbol{x}, \boldsymbol{y} \in \{-1, 0, +1\}^{\mathbb{Z}_5}$, hence at most $\frac{1}{4} \cdot 8 = 2$ in absolute value since $\mathcal{C}_\square(\mathsf{id}) = 8$. Combining both copies then shows $\mathsf{d}_\square^{\mathsf{OT}}(G, H) \le \mathcal{C}_\square(Q) \le 4$. ◀

Proof of Proposition 10. This follows from Lemmas 33 and 34. ◀

E.2 The Upper Bound

Throughout this section, G and H are graphs of the same order $n \ge 2$. For simplicity, we assume that $V(G) = V(H) = [n]$. The adjacency matrices of G, H are $A, B \in \{0,1\}^{n \times n}$. For $Q \in \mathrm{DS}(n)$, we let

$$\zeta_Q \coloneqq \max_{S,T \subseteq [n]^2} \frac{1}{n^2} \Big| \sum_{\substack{(i,j) \in S \\ (k,\ell) \in T}} Q_{ij} Q_{k\ell} (A_{ik} - B_{j\ell}) \Big| \tag{21}$$

$$= \max_{f,g:[n]^2 \to [0,1]} \frac{1}{n^2} \Big| \sum_{i,j,k,\ell \in [n]} f(i,j)\, Q_{ij} Q_{k\ell} (A_{ik} - B_{j\ell})\, g(k,\ell) \Big|, \tag{22}$$

where equality holds because the expression is bilinear in (f, g), so its absolute value is maximised at vertices of the cube $[0,1]^{[n]^2}$, i.e. at indicator functions of sets $S, T \subseteq [n]^2$. Note that

$$\delta_\square^{\mathsf{OT}}(G, H) = \min_{Q \in \mathrm{DS}(n)} \zeta_Q \qquad \text{and} \qquad \delta_\square(G, H) = \min_{P \in \mathrm{Perm}(n)} \zeta_P.$$

As explained in the proof sketch, we prove two upper bounds, a multiplicative and an additive one, using completely different techniques. We start with the multiplicative upper bound.

E.2.1 Multiplicative Rounding via Thin Matchings

Let $\alpha > 0$. A permutation matrix $P \in \mathrm{Perm}(n)$ is *α-thin* with respect to $Q \in \mathrm{DS}(n)$ if

$$\sum_{(i,j) \in (X \times \overline{Y}) \cup (\overline{X} \times Y)} P_{ij} \le \alpha \sum_{(i,j) \in (X \times \overline{Y}) \cup (\overline{X} \times Y)} Q_{ij} \qquad \text{for all } X, Y \subseteq [n]. \tag{23}$$

Here we write $\overline{X}, \overline{Y}$ to denote the complements $[n] \setminus X, [n] \setminus Y$ of the sets. If we view P as describing a matching of the complete bipartite graph $K_{n,n}$ and Q as a fractional matching, then for all cuts $(X \cup Y, \overline{X} \cup \overline{Y})$ the number of matching edges across the cut is bounded from above by α times the total weight of the fractional edges across the cut. Haqi and Oveis Gharan showed that thin permutations exist where α only grows polylogarithmically in n.

Theorem 35 ([HOG26, Theorem 1.4]). *There is an absolute constant $C_{\mathrm{thin}} > 0$ such that every $Q \in \mathrm{DS}(n)$ admits an α-thin permutation matrix with respect to Q with $\alpha \le C_{\mathrm{thin}} \ln(2n) \ln\ln(4n)$.*

The following lemma turns thinness (Eq. (23)) into a cost bound for line metrics based on a level-set decomposition. A similar but different argument was used in the thin-tree approach to

ATSP [AGM+17].

Lemma 36. *Let π be a permutation of $[n]$ whose permutation matrix $P = P_\pi$ is α-thin with respect to $Q \in \mathrm{DS}(n)$. Then, for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$, we have*

$$\sum_{i\in[n]} |x_i - y_{\pi(i)}| \le \alpha \sum_{i,j\in[n]} Q_{ij}|x_i - y_j|.$$

Proof. For $s \in \mathbb{R}$, consider the level sets $X_s \coloneqq \{i \in [n] : x_i > s\}$ and $Y_s \coloneqq \{j \in [n] : y_j > s\}$. For $i, j \in [n]$, we have $\big|\mathbb{1}[x_i > s] - \mathbb{1}[y_j > s]\big| = 1$ if $(i, j) \in (X_s \times \overline{Y_s}) \cup (\overline{X_s} \times Y_s)$, and the expression is 0 otherwise. Hence Eq. (23), applied to $X \coloneqq X_s$ and $Y \coloneqq Y_s$, yields

$$\sum_{i\in[n]} \big|\mathbb{1}[x_i > s] - \mathbb{1}[y_{\pi(i)} > s]\big| = \sum_{i,j\in[n]} P_{ij}\big|\mathbb{1}[x_i > s] - \mathbb{1}[y_j > s]\big| \le \alpha \sum_{i,j\in[n]} Q_{ij}\big|\mathbb{1}[x_i > s] - \mathbb{1}[y_j > s]\big|.$$

For $x, y \in \mathbb{R}$, the function $s \mapsto |\mathbb{1}[x > s] - \mathbb{1}[y > s]|$ is the indicator of the interval between x and y, so $\int_{\mathbb{R}} |\mathbb{1}[x > s] - \mathbb{1}[y > s]|\, ds = |x - y|$. Integrating the inequality over $s \in \mathbb{R}$ and exchanging the integral with the finite sums therefore proves the claim. ◀

The above lemma allows us to transfer guarantees about the pairs of vertices that are matched by Q to the pairs that are matched by π.

Lemma 37 (Local Agreement). *Let $Q \in \mathrm{DS}(n)$ and $\boldsymbol{y} \in [0, 1]^n$, and let $\boldsymbol{x} \coloneqq A\boldsymbol{y} \in \mathbb{R}^n$. Then*

$$\frac{1}{n^2} \sum_{i,j\in[n]} Q_{ij}|x_i - (Q^\top \boldsymbol{x})_j| \le 4\zeta_Q.$$

To understand the intuition behind this lemma, consider the case where $\boldsymbol{y} = \mathbb{1}_Y$ for a subset $Y \subseteq [n]$. Then the entry x_i of $\boldsymbol{x}$ is exactly the number of neighbours vertex i has in the set Y in the graph G. For $j \in [n]$, the entry $(Q^\top \boldsymbol{x})_j = \sum_i Q_{ij}x_i$ is a convex combination of the entries x_i of the vertices i that Q matches to j. Intuitively, x_i and $(Q^\top \boldsymbol{x})_j$ are close if on average the vertices i' that Q matches to j have about the same number of neighbours in Y as i. The lemma states that x_i and $(Q^\top \boldsymbol{x})_j$ are close for most pairs (i, j) in the support of Q: the sum of $Q_{ij}|x_i - (Q^\top \boldsymbol{x})_j|$ over all pairs (i, j) is small. With a bit of a stretch, this means that Q only matches different vertices i, i' of G to the same vertex j of H if i and i' have similar neighbourhoods.

Proof. For $i, j \in [n]$, let

$$F_{ij} \coloneqq \sum_{k,\ell\in[n]} Q_{k\ell}(A_{ik} - B_{j\ell})y_k = x_i - r_j, \qquad \text{where } \boldsymbol{r} \coloneqq BQ^\top \boldsymbol{y},$$

and the identity uses that the rows of Q sum to one. We first show that

$$\frac{1}{n^2} \sum_{i,j\in[n]} Q_{ij}|F_{ij}| \le 2\zeta_Q. \tag{24}$$

Let $S_+ \coloneqq \{(i, j) \in [n]^2 \mid F_{ij} > 0\}$ and $S_- \coloneqq \{(i, j) \in [n]^2 \mid F_{ij} < 0\}$, and let $S \in \{S_+, S_-\}$. Then

$$\frac{1}{n^2} \sum_{(i,j)\in S} Q_{ij}F_{ij} = \frac{1}{n^2} \sum_{(i,j)\in S} \sum_{k,\ell\in[n]} Q_{ij}Q_{k\ell}(A_{ik} - B_{j\ell})\, y_k$$

is the expression in Eq. (22) for the functions $f, g\colon [n]^2 \to [0,1]$ with $f(i,j) \coloneqq 1$ if $(i,j) \in S$ and $f(i,j) \coloneqq 0$ otherwise, and $g(k,\ell) \coloneqq y_k$. Hence it is at most ζ_Q in absolute value, for $S = S_+$ as well as for $S = S_-$. As $|F_{ij}| = F_{ij}$ on S_+, $|F_{ij}| = -F_{ij}$ on S_-, and $F_{ij} = 0$ otherwise, this proves Eq. (24).

Since the columns of Q sum to one, $(Q^\top \boldsymbol{x})_j = \sum_{i'} Q_{i'j}(F_{i'j} + r_j) = r_j + \sum_{i'} Q_{i'j} F_{i'j}$, and hence

$$x_i - (Q^\top \boldsymbol{x})_j = F_{ij} - \sum_{i' \in [n]} Q_{i'j} F_{i'j} \qquad \text{for all } i,j \in [n].$$

By the triangle inequality and, once more, the fact that the columns of Q sum to one, we have

$$\begin{aligned} \frac{1}{n^2} \sum_{i,j \in [n]} Q_{ij} |x_i - (Q^\top \boldsymbol{x})_j| &\leq \frac{1}{n^2} \sum_{i,j \in [n]} Q_{ij} |F_{ij}| + \frac{1}{n^2} \sum_{j \in [n]} \Big(\sum_{i \in [n]} Q_{ij} \Big) \sum_{i' \in [n]} Q_{i'j} |F_{i'j}| \\ &= \frac{2}{n^2} \sum_{i,j \in [n]} Q_{ij} |F_{ij}| \leq 4\zeta_Q. \end{aligned}$$

◀

It is well known and easy to see from the definition that the cut norm

$$\|K\|_\square = \max_{X,Y \subseteq [n]} \Big| \sum_{i \in X} \sum_{k \in Y} K_{ik} \Big|$$

of a matrix $K \in \mathbb{R}^{n \times n}$ is invariant under transpositions and under permuting rows and columns. That is, $\left\|K^\top\right\|_\square = \|K\|_\square$ and $\left\|PKP^\top\right\|_\square = \|K\|_\square$ for $P \in \mathrm{Perm}(n)$. Furthermore, for $R \in \mathrm{DS}(n)$ we have

$$\|RK\|_\square \leq \|K\|_\square . \tag{25}$$

To see this, let $X, Y \subseteq [n]$ and $\boldsymbol{x} \coloneqq R^\top \mathbb{1}_X$. Then $\boldsymbol{x} \in [0,1]^n$ and

$$\sum_{i \in X} \sum_{k \in Y} (RK)_{ik} = \sum_{j \in [n]} x_j \sum_{k \in Y} K_{jk}.$$

The right-hand side is linear in $\boldsymbol{x}$, so its absolute value over $\boldsymbol{x} \in [0,1]^n$ is maximised at some indicator vector $\mathbb{1}_{X'}$, where it equals $|\sum_{j \in X'} \sum_{k \in Y} K_{jk}| \leq \|K\|_\square$.

With this, we are ready to prove the result of this section.

Theorem 38. *Let $\alpha \coloneqq C_{\mathrm{thin}} \ln(2n) \ln\ln(4n)$, where C_{thin} is the constant of Theorem 35. Then for every $Q \in \mathrm{DS}(n)$ there is a $P \in \mathrm{Perm}(n)$ such that $\zeta_P \leq (1+4\alpha)\zeta_Q$.*

Proof. Let $Q \in \mathrm{DS}(n)$. By Theorem 35, there is a permutation matrix $P \in \mathrm{Perm}(n)$ that is α'-thin with respect to Q for some $\alpha' \leq \alpha$, and since the entries of Q are non-negative, P is then also α-thin with respect to Q. Let π be the permutation of $[n]$ with $P = P_\pi$. We show that

$$\zeta_P \leq (1+4\alpha)\zeta_Q. \tag{26}$$

We first express ζ_P as a cut norm. Since $P_{ij} = 1$ if and only if $j = \pi(i)$, only pairs of the form $(i, \pi(i))$ contribute to the sum in Eq. (21) for ζ_P. Writing $X \coloneqq \{i : (i,\pi(i)) \in S\}$ and $Y \coloneqq \{k : (k,\pi(k)) \in T\}$, the sum equals $\sum_{i \in X} \sum_{k \in Y} (A_{ik} - B_{\pi(i)\pi(k)}) = \sum_{i \in X} \sum_{k \in Y} (A - PBP^\top)_{ik}$, because $(PBP^\top)_{ik} = B_{\pi(i)\pi(k)}$. As X, Y range over all subsets of $[n]$, we obtain

$$\zeta_P = n^{-2} \left\| A - PBP^\top \right\|_\square .$$

Let $R \coloneqq PQ^\top$, which is doubly stochastic and satisfies $(R\boldsymbol{x})_i = (Q^\top \boldsymbol{x})_{\pi(i)}$ for all $\boldsymbol{x} \in \mathbb{R}^n$ and $i \in [n]$. Since $RAR^\top = P(Q^\top AQ)P^\top$, we split

$$A - PBP^\top = (A - RAR^\top) + P(Q^\top AQ - B)P^\top$$

and bound the norm of the two terms of the right-hand side separately. Since the cut norm is invariant under conjugation with P, by the triangle inequality, to prove (26) it suffices to show

$$n^{-2}\left\|Q^\top AQ - B\right\|_\square \le \zeta_Q \qquad \text{and} \qquad n^{-2}\left\|A - RAR^\top\right\|_\square \le 4\alpha\zeta_Q. \tag{27}$$

For the first bound, let $X, Y \subseteq [n]$. Since the columns of Q sum to one, $\sum_{i,k} Q_{ij}Q_{k\ell}B_{j\ell} = B_{j\ell}$, and together with $\sum_{i,k} Q_{ij}Q_{k\ell}A_{ik} = (Q^\top AQ)_{j\ell}$ this gives

$$\frac{1}{n^2}\sum_{j\in X}\sum_{\ell\in Y}((Q^\top AQ)_{j\ell} - B_{j\ell}) = \frac{1}{n^2}\sum_{\substack{(i,j)\in[n]\times X\\(k,\ell)\in[n]\times Y}} Q_{ij}Q_{k\ell}(A_{ik} - B_{j\ell}),$$

which is the expression in Eq. (21) for $S = [n] \times X$ and $T = [n] \times Y$. Hence the left-hand side is at most ζ_Q in absolute value for all $X, Y \subseteq [n]$.

For the second bound in (27), let $E \coloneqq I - R$. Let $Y \subseteq [n]$ and put $\boldsymbol{x} \coloneqq A\mathbb{1}_Y \in \mathbb{R}^n$ and $\boldsymbol{f} \coloneqq EA\mathbb{1}_Y = \boldsymbol{x} - R\boldsymbol{x}$, so that

$$f_i = x_i - (Q^\top \boldsymbol{x})_{\pi(i)} \qquad \text{for all } i \in [n].$$

Applying Lemma 36 to the vectors $\boldsymbol{x}$ and $Q^\top\boldsymbol{x}$, followed by Lemma 37 with $\boldsymbol{y} \coloneqq \mathbb{1}_Y$, gives

$$\frac{1}{n^2}\sum_{i\in[n]}|f_i| \le \frac{\alpha}{n^2}\sum_{i,j\in[n]} Q_{ij}|x_i - (Q^\top\boldsymbol{x})_j| \le 4\alpha\zeta_Q.$$

As the columns of R sum to one, $\sum_i (R\boldsymbol{x})_i = \sum_k x_k \sum_i R_{ik} = \sum_k x_k$. So $\sum_i f_i = \sum_i x_i - \sum_i (R\boldsymbol{x})_i = 0$, and hence the positive and the negative parts of $\boldsymbol{f}$ have the same sum. It follows that $|\sum_{i\in X} f_i| \le \frac{1}{2}\sum_{i\in[n]}|f_i|$ for every $X \subseteq [n]$. Therefore,

$$\frac{1}{n^2}\Big|\sum_{i\in X}\sum_{k\in Y}(EA)_{ik}\Big| = \frac{1}{n^2}\Big|\sum_{i\in X} f_i\Big| \le \frac{1}{2n^2}\sum_{i\in[n]}|f_i| \le 2\alpha\zeta_Q$$

for all $X, Y \subseteq [n]$, that is, $n^{-2}\|EA\|_\square \le 2\alpha\zeta_Q$. Finally, since A is symmetric, we have

$$\begin{aligned} A - RAR^\top &= (A - RA) + (RA - RAR^\top) \\ &= (A - RA) + RA(I - R^\top) \\ &= (A - RA) + R(EA)^\top \\ &= EA + R(EA)^\top, \end{aligned} \tag{28}$$

and Eq. (25) together with the invariance of the cut norm under transposition bounds the cut norms of EA and $R(EA)^\top$ by $\|EA\|_\square$. This gives $n^{-2}\left\|A - RAR^\top\right\|_\square \le 4\alpha\zeta_Q$ and concludes the proof. ◀

E.2.2 Additive Rounding

In this section, we shall prove the following additive bound.

Theorem 39. *Let $d_n \coloneqq \lceil 2\log n \rceil$. For every $Q \in \mathrm{DS}(n)$ there is a permutation matrix $P \in \mathrm{Perm}(n)$ such that*

$$\zeta_P \leq \zeta_Q + 36\sqrt{d_n}\, n^{-1/6}.$$

This is going to be significantly harder than the multiplicative bound of Theorem 38 (which, admittedly, used the thin matchings for free). We start by fixing notations and collecting some tools from geometry and high-dimensional probability.

The Space $\mathcal{L}^2(n)$. We write $\mathcal{L}^2(n)$ for $\mathbb{R}^n$ equipped with the inner product and norm

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle \coloneqq \frac{1}{n} \sum_{i \in [n]} x_i y_i \qquad \text{and} \qquad \|\boldsymbol{x}\| \coloneqq \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle},$$

that is, the L^2-space of the uniform distribution on $[n]$. We may view the elements of $\mathcal{L}^2(n)$ as vectors in $\mathbb{R}^n$ or as functions $[n] \to \mathbb{R}$, whichever is more convenient. In this section, a matrix $M \in \mathbb{R}^{n \times n}$ acts on $\mathcal{L}^2(n)$ as an *averaging operator*, $(M\boldsymbol{x})_i \coloneqq \frac{1}{n}\sum_{j \in [n]} M_{ij} x_j$, and its *operator norm* is

$$\|M\|_{\mathcal{L}^2(n)} \coloneqq \max_{\|\boldsymbol{x}\|=1} \|M\boldsymbol{x}\| = \max_{\|\boldsymbol{x}\|=\|\boldsymbol{y}\|=1} \left|\langle \boldsymbol{x}, M\boldsymbol{y} \rangle\right| = \frac{1}{n} \|M\|_{2 \to 2}\,.$$

For symmetric M, the averaging operator is self-adjoint with respect to $\langle \cdot, \cdot \rangle$, so by the spectral theorem $\mathcal{L}^2(n)$ has an orthonormal basis of eigenvectors of M, and $\|M\|_{\mathcal{L}^2(n)}$ is the largest absolute value of an eigenvalue. Note that the eigenvalues of the averaging operator are those of the matrix M/n.

Kernels and Weighted Cut Norm. For a finite probability space (Ω, μ), a *kernel* is a function $K \colon \Omega \times \Omega \to \mathbb{R}$, i.e. a matrix indexed by Ω, which we always pair with the measure μ as in [Lov12]. Its *weighted cut norm* is

$$\|K\|_{\square,\mu} \coloneqq \max_{S,T \subseteq \Omega} \Big| \sum_{x \in S} \sum_{y \in T} \mu(x)\mu(y) K(x,y) \Big|.$$

Note that if $\Omega = [n]$ and μ is the uniform distribution we have $\|K\|_{\square,\mu} = n^{-2} \|K\|_{\square}$.

Couplings and Discrepancy. For $Q \in \mathrm{DS}(n)$, let $\Omega_Q \coloneqq [n] \times [n]$ carry the probability weights $\mu_Q(i,j) \coloneqq Q_{ij}/n$; thus $\mu_Q \in \mathrm{DS}^*(n)$ is a coupling of the uniform distributions on $[n]$. We view the *discrepancy matrix* $D_{A',B'}$ with entries $D_{A',B'}(i,j,k,\ell) = A'_{ik} - B'_{j\ell}$ (cf. Eq. (6)) as a kernel on Ω_Q. For $A', B' \in \mathbb{R}^{n \times n}$, we let

$$\zeta_Q(A', B') \coloneqq \left\| D_{A',B'} \right\|_{\square,\mu_Q}.$$

Note that for the adjacency matrices A, B of our graphs G, H, we have $\zeta_Q(A,B) = \zeta_Q$ by Eq. (21). We need the following elementary fact about these weighted cut norms.

Lemma 40 (Lifted Kernels). *Let $Q \in \mathrm{DS}(n)$, let $K \in \mathbb{R}^{n \times n}$ with $\|K\|_{\mathcal{L}^2(n)} \leq t$, and let $K^{(1)}$ and $K^{(2)}$ be the kernels on Ω_Q with $K^{(1)}((i,j),(k,\ell)) \coloneqq K_{ik}$ and $K^{(2)}((i,j),(k,\ell)) \coloneqq K_{j\ell}$. Then $\left\|K^{(1)}\right\|_{\square,\mu_Q}, \left\|K^{(2)}\right\|_{\square,\mu_Q} \leq t$.*

Proof. Let $S, T \subseteq \Omega_Q$ and define $\boldsymbol{s}, \boldsymbol{\sigma} \in \mathbb{R}^n$ by $s_i \coloneqq \sum_{j \in [n]} Q_{ij} \mathbb{1}_S(i,j)$ and $\sigma_k \coloneqq \sum_{\ell \in [n]} Q_{k\ell} \mathbb{1}_T(k,\ell)$.

Since the rows of Q sum to one, $\boldsymbol{s}, \boldsymbol{\sigma} \in [0,1]^n$, hence $\|\boldsymbol{s}\|, \|\boldsymbol{\sigma}\| \leq 1$, and

$$\sum_{(i,j)\in S} \sum_{(k,\ell)\in T} \mu_Q(i,j)\, \mu_Q(k,\ell)\, K^{(1)}((i,j),(k,\ell)) = \frac{1}{n^2} \sum_{i,k\in[n]} s_i K_{ik} \sigma_k = \langle \boldsymbol{s}, K\boldsymbol{\sigma} \rangle$$

has absolute value at most $\|\boldsymbol{s}\|\, \|K\|_{\mathcal{L}^2(n)}\, \|\boldsymbol{\sigma}\| \leq t$ by Cauchy–Schwarz. Maximising over S, T gives $\left\|K^{(1)}\right\|_{\square,\mu_Q} \leq t$. For $K^{(2)}$, the same argument applies with $s_j \coloneqq \sum_{i\in[n]} Q_{ij} \mathbb{1}_S(i,j)$ and $\sigma_\ell \coloneqq \sum_{k\in[n]} Q_{k\ell} \mathbb{1}_T(k,\ell)$, which lie in $[0,1]$ since the columns of Q sum to one. ◀

Tools from Geometry. In the remainder of this section, $\mathcal{H} \coloneqq \mathbb{R}^m$ for some $m \in \mathbb{N}_{>0}$, equipped with the standard inner product $\langle \boldsymbol{u}, \boldsymbol{w} \rangle \coloneqq \sum_{r\in[m]} u_r w_r$ and norm $\|\boldsymbol{u}\| \coloneqq \sqrt{\langle \boldsymbol{u}, \boldsymbol{u} \rangle}$; in our application, $m \leq 2n$ is fixed in the proof of Lemma 45. For a linear operator T on $\mathcal{H}$, we write $\|T\|_{2\to 2} \coloneqq \max_{\|\boldsymbol{u}\|=1} \|T\boldsymbol{u}\|$ for its operator norm; if T is diagonal, this is the largest absolute value of a diagonal entry. We use the same symbols $\langle \cdot, \cdot \rangle$ and $\|\cdot\|$ as for $\mathcal{L}^2(n)$; the two spaces only meet in the proof of Lemma 45, where we say which one is meant. For a non-empty compact convex set $Z \subseteq \mathcal{H}$, let

$$h_Z(\boldsymbol{w}) \coloneqq \max_{\boldsymbol{u}\in Z} \langle \boldsymbol{w}, \boldsymbol{u} \rangle \qquad \text{for } \boldsymbol{w} \in \mathcal{H}$$

be its *support function*, and let $B_{\mathcal{H}}$ be the closed unit ball of $\mathcal{H}$.

Lemma 41 (Close Support Functions). *Let $Z, Z' \subseteq \mathcal{H}$ be non-empty compact convex sets contained in the ball of radius M with $|h_Z(\boldsymbol{w}) - h_{Z'}(\boldsymbol{w})| \leq \eta$ for all $\boldsymbol{w} \in B_{\mathcal{H}}$, and let T be a linear operator with $\|T\|_{2\to 2} \leq \tau$. Then*

$$\Big| \max_{\boldsymbol{u},\boldsymbol{u}'\in Z} |\langle \boldsymbol{u}, T\boldsymbol{u}' \rangle| - \max_{\boldsymbol{u},\boldsymbol{u}'\in Z'} |\langle \boldsymbol{u}, T\boldsymbol{u}' \rangle| \Big| \leq 2M\eta\tau.$$

Proof. The hypothesis gives $|h_Z(\boldsymbol{w}) - h_{Z'}(\boldsymbol{w})| \leq \eta \|\boldsymbol{w}\|$ for all $\boldsymbol{w} \in \mathcal{H}$. For non-empty compact convex $Y, Y' \subseteq \mathcal{H}$, let $m_T(Y, Y') \coloneqq \max_{\boldsymbol{u}\in Y, \boldsymbol{u}'\in Y'} \langle \boldsymbol{u}, T\boldsymbol{u}' \rangle$. Maximising over $\boldsymbol{u}$ first gives $m_T(Y, Y') = \max_{\boldsymbol{u}'\in Y'} h_Y(T\boldsymbol{u}')$, and maximising over $\boldsymbol{u}'$ first gives $m_T(Y, Y') = \max_{\boldsymbol{u}\in Y} h_{Y'}(T^*\boldsymbol{u})$ for the adjoint operator T^*, which satisfies $\|T^*\|_{2\to 2} = \|T\|_{2\to 2} \leq \tau$. For $\boldsymbol{u} \in Z$ we have $\|T\boldsymbol{u}\| \leq \tau M$, so $|h_Z(T\boldsymbol{u}) - h_{Z'}(T\boldsymbol{u})| \leq M\eta\tau$, and hence $|m_T(Z, Z) - m_T(Z', Z)| \leq M\eta\tau$. Likewise, $\|T^*\boldsymbol{u}\| \leq \tau M$ for $\boldsymbol{u} \in Z'$ gives $|m_T(Z', Z) - m_T(Z', Z')| \leq M\eta\tau$. Thus $|m_T(Z, Z) - m_T(Z', Z')| \leq 2M\eta\tau$, and the same holds for $-T$. Since $\max_{\boldsymbol{u},\boldsymbol{u}'\in Y} |\langle \boldsymbol{u}, T\boldsymbol{u}' \rangle| = \max\{m_T(Y, Y), m_{-T}(Y, Y)\}$ and $|\max\{a, b\} - \max\{a', b'\}| \leq \max\{|a - a'|, |b - b'|\}$, the claim follows. ◀

Tools from Probability. The following lemma uses a standard argument giving a concentration bound in terms of Rademacher complexity (cf. [vdVW96, Lemma 2.3.1] or [SSBD14, Lemma 26.2]).

Lemma 42. *Let W be a set, let $d_1, \ldots, d_b \colon W \to \mathbb{R}$ be bounded, and let $X_1, \ldots, X_b$ be independent $\{0,1\}$-valued random variables with $\theta_j \coloneqq \mathbb{E}\, X_j$. Then*

$$\mathop{\mathbb{E}}_{X} \sup_{w\in W} \Big| \sum_{j\in[b]} (X_j - \theta_j)\, d_j(w) \Big| \leq 2 \mathop{\mathbb{E}}_{X} \mathop{\mathbb{E}}_{\boldsymbol{\sigma}\sim \mathrm{Unif}\{\pm 1\}^b} \sup_{w\in W} \Big| \sum_{j\in[b]} \sigma_j X_j\, d_j(w) \Big|.$$

Proof. Let X' be an independent copy of X. Since $\theta_j = \mathbb{E}\, X'_j$ and the supremum of absolute values is convex, Jensen's inequality bounds the left-hand side by $\mathbb{E}_{X,X'} \sup_w |\sum_j (X_j - X'_j)\, d_j(w)|$. The differences $X_j - X'_j$ are independent and symmetric, so $(X_j - X'_j)_j$ has the same distribution as $(\sigma_j(X_j - X'_j))_j$ for every fixed $\boldsymbol{\sigma} \in \{\pm 1\}^b$. Averaging over $\boldsymbol{\sigma}$ and applying the triangle inequality,

the left-hand side is at most

$$\mathop{\mathbb{E}}_{X,X'} \mathop{\mathbb{E}}_{\boldsymbol{\sigma}} \sup_{w} \Big| \sum_{j} \sigma_j X_j \, d_j(w) \Big| + \mathop{\mathbb{E}}_{X,X'} \mathop{\mathbb{E}}_{\boldsymbol{\sigma}} \sup_{w} \Big| \sum_{j} \sigma_j X'_j \, d_j(w) \Big|,$$

and both terms equal the expectation on the right-hand side of the lemma. ◀

Next, we need a standard lemma stating that applying a contraction, that is, a 1-Lipschitz function $\gamma\colon \mathbb{R} \to \mathbb{R}$ with $\gamma(0) = 0$, increases the Rademacher complexity by at most a factor of 2 (see [LT91, Theorem 4.12] for a proof; [MRT18, Lemma 5.7] and [SSBD14, Lemma 26.9] give the version without absolute values).

Lemma 43 (Talagrand Contraction Lemma). *Let $S \subseteq \mathbb{R}^b$ be bounded, and let $\gamma_1, \dots, \gamma_b\colon \mathbb{R} \to \mathbb{R}$ be contractions. Then*

$$\mathop{\mathbb{E}}_{\boldsymbol{\sigma} \sim \mathrm{Unif}\{\pm 1\}^b} \sup_{\boldsymbol{x} \in S} \Big| \sum_{j \in [b]} \sigma_j \, \gamma_j(x_j) \Big| \leq 2 \mathop{\mathbb{E}}_{\boldsymbol{\sigma} \sim \mathrm{Unif}\{\pm 1\}^b} \sup_{\boldsymbol{x} \in S} \Big| \sum_{j \in [b]} \sigma_j \, x_j \Big|.$$

The following lemma gives the main estimate we need later. Recall that for $x \in \mathbb{R}$ we let $(x)_+ \coloneqq \max\{x, 0\}$.

Lemma 44. *Let $B_1, \dots, B_b \subseteq [n]$ be pairwise disjoint and non-empty, with $\ell_j = |B_j|$, and let $\boldsymbol{u}_1, \dots, \boldsymbol{u}_n \in \mathcal{H}$ with $\|\boldsymbol{u}_i\| \leq M$. Then*

$$\mathop{\mathbb{E}}_{\boldsymbol{\sigma} \sim \mathrm{Unif}\{\pm 1\}^b} \sup_{\boldsymbol{w} \in B_{\mathcal{H}}} \Big| \sum_{j \in [b]} \sigma_j \sum_{i \in B_j} \big(\langle \boldsymbol{w}, \boldsymbol{u}_i \rangle \big)_+ \Big| \leq 2M \Big(\sum_{j \in [b]} \ell_j^2 \Big)^{1/2}.$$

Proof. Let $I_j \sim \mathrm{Unif}(B_j)$ independently for $j \in [b]$, so that

$$\sum_{i \in B_j} \big(\langle \boldsymbol{w}, \boldsymbol{u}_i \rangle \big)_+ = \mathop{\mathbb{E}}_{I_j} \big(\ell_j \langle \boldsymbol{w}, \boldsymbol{u}_{I_j} \rangle \big)_+.$$

Since the supremum of absolute values is convex, Jensen's inequality bounds the left-hand side of the lemma by the expectation over I of the same expression with $\big(\ell_j \langle \boldsymbol{w}, \boldsymbol{u}_{I_j} \rangle \big)_+$ in place of the block sums. So it suffices to show that, for every fixed $I = (I_1, \dots, I_b)$,

$$\mathop{\mathbb{E}}_{\boldsymbol{\sigma}} \sup_{\boldsymbol{w} \in B_{\mathcal{H}}} \Big| \sum_{j \in [b]} \sigma_j \big(\ell_j \langle \boldsymbol{w}, \boldsymbol{u}_{I_j} \rangle \big)_+ \Big| \leq 2M \Big(\sum_{j \in [b]} \ell_j^2 \Big)^{1/2}. \tag{29}$$

Fix I and let $S \coloneqq \{ (\ell_j \langle \boldsymbol{w}, \boldsymbol{u}_{I_j} \rangle)_{j \in [b]} \mid \boldsymbol{w} \in B_{\mathcal{H}} \} \subseteq \mathbb{R}^b$. This set is bounded, since $|\ell_j \langle \boldsymbol{w}, \boldsymbol{u}_{I_j} \rangle| \leq \ell_j M$ for $\boldsymbol{w} \in B_{\mathcal{H}}$ by Cauchy–Schwarz. The map $s \mapsto (s)_+$ is a contraction, so Lemma 43 with $\gamma_j(s) \coloneqq (s)_+$ for all $j \in [b]$ gives

$$\mathop{\mathbb{E}}_{\boldsymbol{\sigma}} \sup_{\boldsymbol{w} \in B_{\mathcal{H}}} \Big| \sum_{j \in [b]} \sigma_j \big(\ell_j \langle \boldsymbol{w}, \boldsymbol{u}_{I_j} \rangle \big)_+ \Big| \leq 2 \mathop{\mathbb{E}}_{\boldsymbol{\sigma}} \sup_{\boldsymbol{w} \in B_{\mathcal{H}}} \Big| \sum_{j \in [b]} \sigma_j \ell_j \langle \boldsymbol{w}, \boldsymbol{u}_{I_j} \rangle \Big|.$$

For fixed $\boldsymbol{\sigma}$, let $\boldsymbol{g}_{\boldsymbol{\sigma}} \coloneqq \sum_{j \in [b]} \sigma_j \ell_j \boldsymbol{u}_{I_j} \in \mathcal{H}$. By linearity of the inner product, $\sum_{j \in [b]} \sigma_j \ell_j \langle \boldsymbol{w}, \boldsymbol{u}_{I_j} \rangle = \langle \boldsymbol{w}, \boldsymbol{g}_{\boldsymbol{\sigma}} \rangle$, and by Cauchy–Schwarz, $|\langle \boldsymbol{w}, \boldsymbol{g}_{\boldsymbol{\sigma}} \rangle| \leq \|\boldsymbol{g}_{\boldsymbol{\sigma}}\|$ for all $\boldsymbol{w} \in B_{\mathcal{H}}$, with equality for $\boldsymbol{w} = \boldsymbol{g}_{\boldsymbol{\sigma}} / \|\boldsymbol{g}_{\boldsymbol{\sigma}}\|$ if $\boldsymbol{g}_{\boldsymbol{\sigma}} \neq \boldsymbol{0}$. Hence $\sup_{\boldsymbol{w} \in B_{\mathcal{H}}} |\sum_j \sigma_j \ell_j \langle \boldsymbol{w}, \boldsymbol{u}_{I_j} \rangle| = \|\boldsymbol{g}_{\boldsymbol{\sigma}}\|$, and the right-hand side above equals $2 \, \mathbb{E}_{\boldsymbol{\sigma}} \|\boldsymbol{g}_{\boldsymbol{\sigma}}\|$. Since the square root is concave, Jensen's inequality yields $\mathbb{E}_{\boldsymbol{\sigma}} \|\boldsymbol{g}_{\boldsymbol{\sigma}}\| \leq (\mathbb{E}_{\boldsymbol{\sigma}} \|\boldsymbol{g}_{\boldsymbol{\sigma}}\|^2)^{1/2}$. Expanding

the square and using $\mathbb{E}_{\boldsymbol{\sigma}} \sigma_j \sigma_{j'} = 1$ if $j = j'$ and $\mathbb{E}_{\boldsymbol{\sigma}} \sigma_j \sigma_{j'} = 0$ if $j \neq j'$, we obtain

$$\mathop{\mathbb{E}}_{\boldsymbol{\sigma}} \|\boldsymbol{g}_{\boldsymbol{\sigma}}\|^2 = \sum_{j,j' \in [b]} \ell_j \ell_{j'} \langle \boldsymbol{u}_{I_j}, \boldsymbol{u}_{I_{j'}} \rangle \mathop{\mathbb{E}}_{\boldsymbol{\sigma}} \sigma_j \sigma_{j'} = \sum_{j \in [b]} \ell_j^2 \left\| \boldsymbol{u}_{I_j} \right\|^2 \leq M^2 \sum_{j \in [b]} \ell_j^2.$$

Combining the three displays proves Eq. (29), and taking the expectation over I proves the lemma. ◀

Spectral Approximation of the Cut Norm. The following lemma is the spectral heart of the argument. It shows that, up to an additive error $2t$, we may replace the entries $A_{ik} - B_{j\ell}$ of the discrepancy matrix in the definition Eq. (21) of ζ_Q by inner products $\langle \boldsymbol{z}_{ij}, T_t \boldsymbol{z}_{k\ell} \rangle$ of vectors in a Euclidean space, where the vectors are bounded and the operator T_t has norm at most $1/t$. The parameter t thus controls a trade-off between the error made by discarding spectral information and the norm of T_t; it will be chosen at the very end, in the proof of Theorem 39.

Lemma 45. *For every $t > 0$ there are m with $1 \leq m \leq 2n$, vectors $(\boldsymbol{z}_{ij})_{i,j \in [n]}$ in $\mathcal{H} = \mathbb{R}^m$ with $\|\boldsymbol{z}_{ij}\| \leq \sqrt{2}$, and a linear operator T_t on $\mathcal{H}$ with $\|T_t\|_{2 \to 2} \leq 1/t$ such that every $Q \in \mathrm{DS}(n)$ satisfies*

$$\Big| \zeta_Q - \max_{S,T \subseteq [n]^2} \frac{1}{n^2} \Big| \sum_{(i,j) \in S} \sum_{(k,\ell) \in T} Q_{ij} Q_{k\ell} \langle \boldsymbol{z}_{ij}, T_t \boldsymbol{z}_{k\ell} \rangle \Big| \Big| \leq 2t. \tag{30}$$

Proof. By the spectral theorem for the averaging operators A and B on $\mathcal{L}^2(n)$, there are eigenvectors $(\boldsymbol{\phi}_r)_{r \in [n]}$ and $(\boldsymbol{\psi}_s)_{s \in [n]}$, orthonormal in $\mathcal{L}^2(n)$, and eigenvalues $(\lambda_r)_{r \in [n]}$ and $(\nu_s)_{s \in [n]}$ with

$$A_{ik} = \sum_{r \in [n]} \lambda_r \phi_{ri} \phi_{rk} \qquad \text{and} \qquad B_{j\ell} = \sum_{s \in [n]} \nu_s \psi_{sj} \psi_{s\ell}.$$

Let $I_A \coloneqq \{r \in [n] : |\lambda_r| > t\}$ and $I_B \coloneqq \{s \in [n] : |\nu_s| > t\}$, and let

$$(A_t)_{ik} \coloneqq \sum_{r \in I_A} \lambda_r \phi_{ri} \phi_{rk} \qquad \text{and} \qquad (B_t)_{j\ell} \coloneqq \sum_{s \in I_B} \nu_s \psi_{sj} \psi_{s\ell}$$

be the spectral truncations of A and B. The averaging operators $A - A_t$ and $B - B_t$ on $\mathcal{L}^2(n)$ have all their eigenvalues in $[-t, t]$, so $\|A - A_t\|_{\mathcal{L}^2(n)}, \|B - B_t\|_{\mathcal{L}^2(n)} \leq t$. We have

$$D_{A,B} = D_{A_t,B_t} + (A - A_t)^{(1)} - (B - B_t)^{(2)}$$

in the notation of Lemma 40, where the kernels $(A - A_t)^{(1)}$ and $(B - B_t)^{(2)}$ depend only on the first and only on the second coordinates, respectively. Lemma 40 bounds their weighted cut norms $\|\cdot\|_{\square,\mu_Q}$ by t for every $Q \in \mathrm{DS}(n)$. By the triangle inequality for $\|\cdot\|_{\square,\mu_Q}$,

$$\big|\zeta_Q(A, B) - \zeta_Q(A_t, B_t)\big| \leq 2t \qquad \text{for every } Q \in \mathrm{DS}(n). \tag{31}$$

If $I_A = I_B = \emptyset$, then $A_t = B_t = 0$, so $\zeta_Q \leq 2t$ for every $Q \in \mathrm{DS}(n)$ by Eq. (31), and the lemma holds with $m = 1$, $\boldsymbol{z}_{ij} \coloneqq \mathbf{0}$ and $T_t \coloneqq 0$. Otherwise, let $m \coloneqq |I_A| + |I_B| \leq 2n$ and identify $\mathcal{H} = \mathbb{R}^m$ with $\mathbb{R}^{I_A} \oplus \mathbb{R}^{I_B}$, with the standard inner product; let

$$\boldsymbol{z}_{ij} \coloneqq \Big((\lambda_r \phi_{ri})_{r \in I_A}, (\nu_s \psi_{sj})_{s \in I_B} \Big) \qquad \text{and} \qquad T_t \coloneqq \operatorname{diag}\Big((\lambda_r^{-1})_{r \in I_A}, (-\nu_s^{-1})_{s \in I_B} \Big).$$

The i-th row of A, as an element of $\mathcal{L}^2(n)$, equals $\sum_{r \in [n]} \lambda_r \phi_{ri} \boldsymbol{\phi}_r$, so by Parseval's identity in $\mathcal{L}^2(n)$,

the first component of $\boldsymbol{z}_{ij}$ has squared norm $\sum_{r\in I_A}\lambda_r^2\phi_{ri}^2 \leq \frac{1}{n}\sum_{k\in[n]}A_{ik}^2 \leq 1$; similarly for the second component. Hence

$$\|\boldsymbol{z}_{ij}\| \leq \sqrt{2}, \qquad \|T_t\|_{2\to 2} \leq \frac{1}{t}, \qquad \text{and} \qquad \langle \boldsymbol{z}_{ij}, T_t\boldsymbol{z}_{k\ell}\rangle = (A_t)_{ik} - (B_t)_{j\ell}.$$

Now let $Q \in \mathrm{DS}(n)$. By the definition of $\zeta_Q(A_t, B_t)$ as the weighted cut norm of D_{A_t,B_t} on (Ω_Q, μ_Q), and since $\mu_Q(i,j)\mu_Q(k,\ell) = Q_{ij}Q_{k\ell}/n^2$,

$$\begin{aligned}\zeta_Q(A_t,B_t) &= \max_{S,T\subseteq[n]^2}\frac{1}{n^2}\Big|\sum_{(i,j)\in S}\sum_{(k,\ell)\in T} Q_{ij}Q_{k\ell}\big((A_t)_{ik} - (B_t)_{j\ell}\big)\Big| \\ &= \max_{S,T\subseteq[n]^2}\frac{1}{n^2}\Big|\sum_{(i,j)\in S}\sum_{(k,\ell)\in T} Q_{ij}Q_{k\ell}\langle \boldsymbol{z}_{ij}, T_t\boldsymbol{z}_{k\ell}\rangle\Big|.\end{aligned}$$

Together with Eq. (31) and $\zeta_Q(A,B) = \zeta_Q$, this proves Eq. (30). ◀

Zonotopes and a Pseudometric on Doubly Stochastic Matrices. For the remainder of this section, we fix $t > 0$ and, with it, $m \leq 2n$, vectors $(\boldsymbol{z}_{ij})_{i,j\in[n]}$ in $\mathcal{H} = \mathbb{R}^m$ and an operator T_t on $\mathcal{H}$ as in Lemma 45. We let $M \coloneqq \sqrt{2}$, so that $\|\boldsymbol{z}_{ij}\| \leq M$ for all $i,j \in [n]$. For $Q \in \mathrm{DS}(n)$, we let

$$Z_Q \coloneqq \Big\{\frac{1}{n}\sum_{i,j\in[n]} Q_{ij}a_{ij}\boldsymbol{z}_{ij} : a_{ij} \in [0,1] \text{ for } i,j\Big\}. \tag{32}$$

Then Z_Q is a zonotope, the Minkowski sum of the line segments $[0, \frac{1}{n}Q_{ij}\boldsymbol{z}_{ij}]$. Note that Z_Q is a non-empty compact convex set contained in the ball of radius M, because $\frac{1}{n}\sum_{i,j}Q_{ij} = 1$. Its support function $h_Q \coloneqq h_{Z_Q}$ satisfies

$$h_Q(\boldsymbol{w}) = \max_{\boldsymbol{u}\in Z_Q}\langle \boldsymbol{w}, \boldsymbol{u}\rangle = \frac{1}{n}\sum_{i,j\in[n]} Q_{ij}\left(\langle \boldsymbol{w}, \boldsymbol{z}_{ij}\rangle\right)_+ \tag{33}$$

for all $\boldsymbol{w} \in \mathcal{H}$, because the function $a \mapsto \langle \boldsymbol{w}, \frac{1}{n}\sum_{i,j}Q_{ij}a_{ij}\boldsymbol{z}_{ij}\rangle$ on $[0,1]^{n\times n}$ is maximised by $a_{ij} = \mathbb{1}[\langle \boldsymbol{w}, \boldsymbol{z}_{ij}\rangle > 0]$.

For $Q, Q' \in \mathrm{DS}(n)$, let

$$\rho(Q,Q') \coloneqq \sup_{\boldsymbol{w}\in B_{\mathcal{H}}} |h_Q(\boldsymbol{w}) - h_{Q'}(\boldsymbol{w})|. \tag{34}$$

Then ρ is a pseudometric on $\mathrm{DS}(n)$. Furthermore, since by (33), h_Q depends linearly on Q, for all $Q_0, Q_1, Q_0', Q_1' \in \mathrm{DS}(n)$ and $\theta \in [0,1]$ we have

$$\rho((1-\theta)Q_0 + \theta Q_1, (1-\theta)Q_0' + \theta Q_1') \leq (1-\theta)\rho(Q_0,Q_0') + \theta\rho(Q_1,Q_1'). \tag{35}$$

The maximum in Eq. (30) can be written as the maximum of a bilinear form over the zonotope Z_Q: for every $Q \in \mathrm{DS}(n)$,

$$\max_{\boldsymbol{u},\boldsymbol{u}'\in Z_Q}|\langle \boldsymbol{u}, T_t\boldsymbol{u}'\rangle| = \max_{S,T\subseteq[n]^2}\frac{1}{n^2}\Big|\sum_{(i,j)\in S}\sum_{(k,\ell)\in T} Q_{ij}Q_{k\ell}\langle \boldsymbol{z}_{ij}, T_t\boldsymbol{z}_{k\ell}\rangle\Big|, \tag{36}$$

and hence, by Eq. (30),

$$\Big|\zeta_Q - \max_{\boldsymbol{u},\boldsymbol{u}'\in Z_Q}|\langle \boldsymbol{u}, T_t\boldsymbol{u}'\rangle|\Big| \leq 2t. \tag{37}$$

Indeed, for $a, a' \in [0,1]^{n\times n}$, the points $\boldsymbol{u}_a := \frac{1}{n}\sum_{i,j} Q_{ij} a_{ij} \boldsymbol{z}_{ij}$ and $\boldsymbol{u}_{a'} := \frac{1}{n}\sum_{k,\ell} Q_{k\ell} a'_{k\ell} \boldsymbol{z}_{k\ell}$ of Z_Q satisfy

$$\langle \boldsymbol{u}_a, T_t \boldsymbol{u}_{a'} \rangle = \frac{1}{n^2} \sum_{i,j,k,\ell \in [n]} Q_{ij} Q_{k\ell}\, a_{ij}\, a'_{k\ell}\, \langle \boldsymbol{z}_{ij}, T_t \boldsymbol{z}_{k\ell} \rangle$$

by bilinearity of the inner product. The right-hand side is bilinear in (a, a'), so for fixed a' its absolute value is a convex function of a and attains its maximum over the cube $[0,1]^{n\times n}$ at a vertex, that is, at an indicator $a = \mathbb{1}_S$ for some $S \subseteq [n]^2$; likewise for a'. Hence the maximum over Z_Q is attained at indicators $a = \mathbb{1}_S$, $a' = \mathbb{1}_T$, which proves Eq. (36).

Corollary 46. *Let $P, Q \in \mathrm{DS}(n)$. Then*

$$\zeta_P \le \zeta_Q + 4t + \frac{2\sqrt{2}}{t}\,\rho(P,Q).$$

Proof. The zonotopes Z_P and Z_Q are non-empty, compact and convex, and they lie in the ball of radius $M = \sqrt{2}$. By the definition Eq. (34) of ρ, their support functions h_P and h_Q differ by at most $\rho(P,Q)$ on $B_{\mathcal{H}}$. Hence Lemma 41, applied with $M = \sqrt{2}$, $\eta = \rho(P,Q)$ and $\tau = 1/t$, shows that

$$\max_{\boldsymbol{u},\boldsymbol{u}' \in Z_P} \big|\langle \boldsymbol{u}, T_t \boldsymbol{u}' \rangle\big| \le \max_{\boldsymbol{u},\boldsymbol{u}' \in Z_Q} \big|\langle \boldsymbol{u}, T_t \boldsymbol{u}' \rangle\big| + \frac{2\sqrt{2}}{t}\rho(P,Q).$$

Applying Eq. (37) to P and to Q, we obtain

$$\zeta_P \le \max_{\boldsymbol{u},\boldsymbol{u}' \in Z_P} \big|\langle \boldsymbol{u}, T_t \boldsymbol{u}' \rangle\big| + 2t \le \max_{\boldsymbol{u},\boldsymbol{u}' \in Z_Q} \big|\langle \boldsymbol{u}, T_t \boldsymbol{u}' \rangle\big| + \frac{2\sqrt{2}}{t}\rho(P,Q) + 2t \le \zeta_Q + 4t + \frac{2\sqrt{2}}{t}\rho(P,Q). \quad \blacktriangleleft$$

Approximating Doubly Stochastic Matrices by Permutation Matrices. The next lemma is the crucial step of the proof. It shows that, with respect to ρ, every convex combination of two permutation matrices is close to a single permutation matrix. By Corollary 46, the cut norm then cannot increase by much.

Lemma 47 (Merging Two Permutations). *For all permutation matrices P_0, P_1 and $\theta \in [0,1]$, there is a permutation matrix P with*

$$\rho\big(P, (1-\theta)P_0 + \theta P_1\big) \le 20 M n^{-1/3}.$$

Proof. Let $P_0 = P_{\pi_0}$ and $P_1 = P_{\pi_1}$ for permutations π_0, π_1 of $[n]$. The proof below uses no property of the vectors $\boldsymbol{z}_{ij}$ other than $\|\boldsymbol{z}_{ij}\| \le M$, so we may relabel the columns along π_0, that is, replace $\boldsymbol{z}_{ij}$ by $\boldsymbol{z}'_{ij} := \boldsymbol{z}_{i,\pi_0(j)}$ for all $i, j \in [n]$. By Eq. (33), the support function of $Q \in \mathrm{DS}(n)$ with respect to the vectors $\boldsymbol{z}'_{ij}$ equals h_{QP_0}, so the pseudometric ρ' defined by the vectors $\boldsymbol{z}'_{ij}$ satisfies $\rho(Q, Q') = \rho'(QP_0^\top, Q'P_0^\top)$ for all $Q, Q' \in \mathrm{DS}(n)$. Since $Q \mapsto QP_0^\top$ is a linear bijection of $\mathrm{DS}(n)$ that maps $\mathrm{Perm}(n)$ onto itself, P_0 to I and P_1 to $P_1P_0^\top$, it suffices to prove the lemma for the vectors $\boldsymbol{z}'_{ij}$ and the pair $I, P_1P_0^\top$. Hence we may w.l.o.g. assume that $P_0 = I = P_{\mathsf{id}}$ and $P_1 = P_\tau$ for some permutation τ of $[n]$. We merge P_0 and P_1 into a new permutation P by selecting for each row i, either column i or $\tau(i)$. This will generally result in conflicts, so the resulting maps from rows to columns need not be injective. For a map $T\colon [n] \to [n]$, let

$$h_T(\boldsymbol{w}) := \frac{1}{n}\sum_{i\in[n]} \big(\langle \boldsymbol{w}, \boldsymbol{z}_{i,T(i)} \rangle\big)_+ \qquad \text{and} \qquad \Delta(T) := \sup_{\boldsymbol{w}\in B_{\mathcal{H}}} \big|h_T(\boldsymbol{w}) - (1-\theta)h_{\mathrm{id}}(\boldsymbol{w}) - \theta h_\tau(\boldsymbol{w})\big|,$$

so that $h_{P_\pi} = h_\pi$ for permutations π, and $\Delta(\pi) = \rho(P_\pi, (1-\theta)I + \theta P_\tau)$ by linearity of $Q \mapsto h_Q$.

Our goal is to construct a map T that is "close" to being bijective while $\Delta(T)$ remains small. We can then "repair" T to obtain a bijection (permutation matrix) while only increasing the distance under ρ slightly.

Let $L \coloneqq \lceil n^{1/3} \rceil$, so that $n^{1/3} \le L \le 2n^{1/3}$. We partition $[n]$ into blocks $B_1, \dots, B_b$ of consecutive elements $i, \tau(i), \tau^2(i), \dots$ of the cycles of τ. Cycles of length at most L form a block, whereas cycles of length $\ell > L$ are cut into $\lfloor \ell/L \rfloor$ blocks. The resulting blocks are all of length at most L, with (possibly) the exception of the last block, which has length less than $2L$. The following properties hold:

(i) every block has length less than $2L$, so $\sum_{a\in[b]} |B_a|^2 \le 2L \sum_{a\in[b]} |B_a| = 2Ln$;

(ii) if i and $\tau(i)$ lie in different blocks, then i is the last element of a block of length at least L, so there are at most n/L such i.

For $i \in [n]$, let $a(i)$ denote the index of the block containing i.

For independent $X_1, \dots, X_b \sim \mathrm{Ber}(\theta)$, let every block either stay in place or follow τ, that is,

$$T_X(i) \coloneqq \begin{cases} i, & X_{a(i)} = 0, \\ \tau(i), & X_{a(i)} = 1, \end{cases}$$

for all $i \in [n]$. Switching block a from the identity to τ changes the quantity $n \cdot h_{T_X}(\boldsymbol{w})$ by

$$d_a(\boldsymbol{w}) \coloneqq \sum_{i\in B_a} \Big[\big(\langle \boldsymbol{w}, \boldsymbol{z}_{i,\tau(i)} \rangle\big)_+ - \big(\langle \boldsymbol{w}, \boldsymbol{z}_{i,i} \rangle\big)_+ \Big].$$

Hence $h_{T_X} = h_{\mathrm{id}} + \frac{1}{n}\sum_a X_a d_a$ and $h_\tau = h_{\mathrm{id}} + \frac{1}{n}\sum_a d_a$, and therefore

$$n \cdot \big(h_{T_X} - (1-\theta)h_{\mathrm{id}} - \theta h_\tau\big) = \sum_{a\in[b]} (X_a - \theta)\, d_a.$$

By Lemma 42 with $W = B_{\mathcal{H}}$ and $\theta_j = \theta$ for all $j \in [b]$,

$$n \mathop{\mathbb{E}}_X \Delta(T_X) \le 2 \mathop{\mathbb{E}}_X \mathop{\mathbb{E}}_{\boldsymbol{\sigma}} \sup_{\boldsymbol{w}\in B_{\mathcal{H}}} \Big| \sum_{a: X_a = 1} \sigma_a d_a(\boldsymbol{w}) \Big|.$$

Splitting d_a into its two sums and applying Lemma 44 to the blocks $\{B_a \mid a \in [b], X_a = 1\}$, once with $\boldsymbol{u}_i = \boldsymbol{z}_{i,\tau(i)}$ and once with $\boldsymbol{u}_i = \boldsymbol{z}_{i,i}$, bounds the right-hand side by

$$2 \cdot (2M + 2M) \Big(\sum_{a\in[b]} |B_a|^2 \Big)^{1/2} \le 8M\sqrt{2Ln},$$

by (i). In particular, there is an $X \in \{0,1\}^b$ with

$$\Delta(T_X) \le 8M \sqrt{\frac{2L}{n}} \le 16 M n^{-1/3}. \tag{38}$$

It remains to transform T_X into a permutation matrix. By (ii), at most $\kappa \le n/L$ columns have two preimages. Removing the row $\tau^{-1}(c)$ for each of them leaves an injective map on $n - \kappa$ rows. Matching the κ removed rows to the κ unused columns arbitrarily then gives a permutation π that differs from T_X in at most n/L rows. Since $s \mapsto s_+$ is 1-Lipschitz, changing the column of one row i

from j to j' changes $h_T(\boldsymbol{w})$ for $\boldsymbol{w} \in B_{\mathcal{H}}$ by at most

$$\frac{1}{n}\Big|\big(\langle \boldsymbol{w}, \boldsymbol{z}_{ij}\rangle\big)_+ - \big(\langle \boldsymbol{w}, \boldsymbol{z}_{ij'}\rangle\big)_+\Big| \le \frac{1}{n}\,\|\boldsymbol{z}_{ij} - \boldsymbol{z}_{ij'}\| \le \frac{2M}{n}.$$

Hence

$$\sup_{\boldsymbol{w}\in B_{\mathcal{H}}} |h_\pi(\boldsymbol{w}) - h_{T_X}(\boldsymbol{w})| \le \frac{n}{L}\cdot\frac{2M}{n} \le 2Mn^{-1/3},$$

and together with Eq. (38) the triangle inequality gives

$$\rho(P_\pi, (1-\theta)I + \theta P_\tau) = \Delta(\pi) \le 18Mn^{-1/3},$$

concluding the proof. ◀

We now apply Lemma 47 to the permutations in the support of Q. To keep the error small we merge permutations along a binary tree. In doing so the error only scales logarithmically with the number of permutations.

Lemma 48. *Recall that $n \ge 2$ and $d_n = \lceil 2\log n\rceil = \lceil \log(n^2)\rceil$. For every $Q \in \mathrm{DS}(n)$ there is a permutation matrix P with*

$$\rho(P, Q) \le 20Md_n n^{-1/3}.$$

Proof. Let $\eta_0 \coloneqq 20Mn^{-1/3}$ be the bound of Lemma 47. By Eq. (3) and Carathéodory's theorem [Sch86], we can write $Q = \sum_{s\in[N]} \lambda_s P_s$ as a convex combination of $N \le n^2$ permutation matrices. Place the terms at the leaves of a binary tree of height $\lceil \log N\rceil \le d_n$, padding with leaves of weight zero. For a node ν with leaf weight $\Lambda_\nu > 0$, let

$$Q_\nu \coloneqq \frac{1}{\Lambda_\nu}\sum_{s\in \mathrm{leaves}(\nu)} \lambda_s P_s,$$

so that $Q_{\mathrm{root}} = Q$. If both children ν_0, ν_1 of ν have positive weight, then $Q_\nu = (1-\theta_\nu)Q_{\nu_0} + \theta_\nu Q_{\nu_1}$ with $\theta_\nu = \Lambda_{\nu_1}/\Lambda_\nu$. Bottom-up, give each leaf its P_s, let a node with one child of positive weight copy that child's matrix, and otherwise let P_ν be the matrix provided by Lemma 47 for P_{ν_0}, P_{ν_1} and θ_ν. By induction on the height $k(\nu)$, the triangle inequality and Eq. (35), we have

$$\begin{aligned}\rho(P_\nu, Q_\nu) &\le \eta_0 + (1-\theta_\nu)\rho(P_{\nu_0}, Q_{\nu_0}) + \theta_\nu \rho(P_{\nu_1}, Q_{\nu_1})\\ &\le \eta_0 + (k(\nu)-1)\eta_0\\ &= k(\nu)\eta_0.\end{aligned}$$

Since the root has height at most d_n, the matrix $P = P_{\mathrm{root}}$ satisfies $\rho(P, Q) \le d_n\eta_0$. ◀

We can now prove Theorem 39 by choosing t so as to balance the truncation error $4t$ against the transfer error $2\sqrt{2}\,\rho(P,Q)/t$ in Corollary 46.

Proof of Theorem 39. Recall that $M = \sqrt{2}$ and that $t > 0$ was arbitrary, so we may choose it below. By Lemma 48, there is a permutation matrix P with $\rho(P, Q) \le 20\sqrt{2}\,d_n n^{-1/3}$. Corollary 46 gives

$$\zeta_P \le \zeta_Q + 4t + \frac{2\sqrt{2}\cdot 20\sqrt{2}\,d_n n^{-1/3}}{t} = \zeta_Q + 4t + \frac{80\,d_n n^{-1/3}}{t}.$$

For $t = \sqrt{20d_n}\,n^{-1/6}$ both error terms equal $4t$, and $8t = 8\sqrt{20}\sqrt{d_n}\,n^{-1/6} < 36\sqrt{d_n}\,n^{-1/6}$. ◀

E.2.3 Proof of Theorem 11

Proof. The lower bound holds because permutation matrices are doubly stochastic. Let $\varepsilon := \delta_{\square}^{\mathsf{OT}}(G,H)$ and let $Q \in \mathrm{DS}(n)$ be optimal, so that $\zeta_Q = \varepsilon$. Since $\ln(2n)\ln\ln(4n) \geq \ln 4 \cdot \ln\ln 8 > 1$ for all $n \geq 2$, Theorem 38 gives, with $C_1 := 1 + 4C_{\mathrm{thin}}$,

$$\delta_{\square}(G,H) \leq C_1 \ln(2n)\ln\ln(4n)\,\varepsilon. \tag{39}$$

This settles $\varepsilon = 0$. If $\varepsilon \geq 1/2$, the claim holds for $C \geq 2$, since $\delta_{\square}(G,H) \leq 1$ and both logarithmic factors are at least one. So let $0 < \varepsilon < 1/2$ and set $L_\varepsilon := \ln(e/\varepsilon) = 1 + \ln(1/\varepsilon)$.

We distinguish two cases: If $n \leq \varepsilon^{-12}$, then

$$\ln(2n) \leq \ln 2 + 12\ln(1/\varepsilon) \leq 13L_\varepsilon \qquad \text{and} \qquad \ln\ln(4n) \leq \ln(14L_\varepsilon) \leq 3\ln(eL_\varepsilon),$$

where the last step uses $\ln 14 < 3$. Hence Eq. (39) gives the claim with $C = 39C_1$.

Now suppose that $n > \varepsilon^{-12}$. Then $n > 2^{12}$ and

$$d_n \leq 2\log n + 1 \leq 3\log n = \frac{3}{\ln 2}\ln n.$$

The function $x \mapsto x^{-1/6}\sqrt{\ln x}$ is decreasing for $x \geq e^3$, so the error term of Theorem 39 satisfies

$$36\sqrt{d_n}\,n^{-1/6} \leq 36\sqrt{\tfrac{3}{\ln 2}}\,(\varepsilon^{-12})^{-1/6}\sqrt{12\ln(1/\varepsilon)} = \frac{216}{\sqrt{\ln 2}}\,\varepsilon\cdot\varepsilon\sqrt{\ln(1/\varepsilon)} \leq \frac{216}{\sqrt{2e\ln 2}}\,\varepsilon < 112\,\varepsilon,$$

using $\varepsilon\sqrt{\ln(1/\varepsilon)} \leq (2e)^{-1/2}$. So $\delta_{\square}(G,H) \leq 113\varepsilon$, and the claim holds with $C = \max\{2, 39C_1, 113\}$.

Since $\varepsilon^{1-\alpha}\ln(e/\varepsilon)\ln(e\ln(e/\varepsilon))$ is bounded on $(0,1]$ for every $\alpha < 1$, it follows that

$$\delta_{\square}(G,H) \leq C_\alpha\,\delta_{\square}^{\mathsf{OT}}(G,H)^\alpha$$

for a constant $C_\alpha > 0$ depending only on α. This concludes the proof of Theorem 11. ◀

F The Integrality Gap of the Spectral Distance

Before considering the integrality gap of δ_2, we first prove the relation between δ_2 and $\delta_{\square}$.

Proof of Proposition 12. Let V, W be finite sets, $m := |V|$ and $n := |W|$. We show that for all real matrices $A \in \mathbb{R}^{V\times W}$, it holds that

$$\frac{1}{\sqrt{mn}}\,\|A\|_{\square} \leq \|A\|_2 \leq 2\sqrt{\|A\|_{(\infty)}\,\|A\|_{\square}}$$

where $\|A\|_{(\infty)} := \max_{v\in V, w\in W}|A_{vw}|$. This implies the proposition by applying it to the matrices $A_G^\pi - A_H$, which have entrywise maximum norm at most one, and observing the different normalisation factors of $\delta_{\square}$ and δ_2.

Let $A \in \mathbb{R}^{V\times W}$. For every set $U \subseteq V$, we denote by $\mathbb{1}_U \in \mathbb{R}^V$ the characteristic vector of U, that is, $\mathbb{1}_U(i) = 1$ if $i \in U$ and $\mathbb{1}_U(i) = 0$ otherwise and similarly for subsets of W. Observe that $\|\mathbb{1}_U\|_2 \leq \sqrt{|U|}$.

For the first inequality, we have

$$\|A\|_{\square} = \max_{S\subseteq V,\, T\subseteq W}\left|\sum_{s\in S,\, t\in T} A_{st}\right|$$

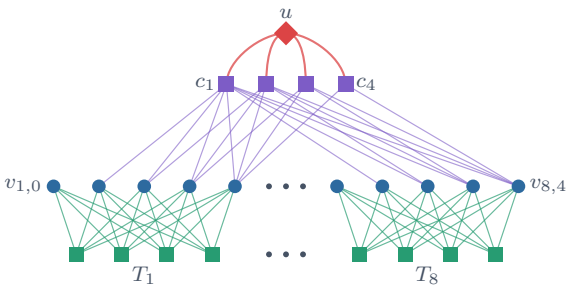


Figure 7 The graph H_4 of Proposition 13, only the first and last branch are drawn. The vertices $v_{i,0}, \dots, v_{i,4}$ of a branch i are arranged horizontally and fully connected to its corresponding tag set T_i. Every vertex $v_{i,k}$ is adjacent to the clock markers $c_1, \dots, c_k$. The graph G_4 is obtained by removing the red edges incident to u.

$$
\begin{aligned}
&= \max_{S \subseteq V, T \subseteq W} \left| \langle \mathbb{1}_S, A \mathbb{1}_T \rangle \right| \\
&\leq \max_{S \subseteq V, T \subseteq W} \|\mathbb{1}_S\|_2 \cdot \|A \mathbb{1}_T\|_2 && \text{(Cauchy–Schwarz)} \\
&\leq \sqrt{mn} \max_{T \subseteq W, T \neq \emptyset} \frac{\|A \mathbb{1}_T\|_2}{\|\mathbb{1}_T\|_2} \\
&\leq \sqrt{mn}\, \|A\|_2.
\end{aligned}
$$

The second inequality was shown by Nikiforov [Nik09, Theorem 3]. ◀

F.1 The Lower Bound

The construction is a simplified version of the one in Appendix D.1, without the bits and without the need for additional vertices to fix markers. Similarly to before, bijections incur a large cost compared to fractional alignments, since the latter can spread error across many branches, reducing the spectral norm of the resulting matrix.

Fix an integer $r \geq 1$. The graphs G_r and H_r have the common vertex set $V \coloneqq X \cup M$ with the *core vertices* and *markers*

$$
X \coloneqq \{u\} \cup \{v_{i,k} \mid i \in [2r], 0 \leq k \leq r\} \qquad \text{and} \qquad M \coloneqq C \cup T_1 \cup \cdots \cup T_{2r},
$$

where $C \coloneqq \{c_1, \dots, c_r\}$ are the *clock markers* and $T_i \coloneqq \{t_{i,1}, \dots, t_{i,r}\}$ are the *tags* of branch $i \in [2r]$. Thus $|X| = 2r(r+1)+1$, $|M| = r + 2r^2$, and let $n \coloneqq 4r^2 + 3r + 1$. In both graphs, $N(v_{i,k}) = T_i \cup \{c_1, \dots, c_k\}$. In G_r, the vertex u is isolated, while in H_r it is adjacent to all clock markers. There are no further edges, so X and M are independent sets in both graphs, and G_r and H_r are bipartite. Along branch i, the tags T_i are present throughout, while the clock markers are switched on one by one (Figure 7). Note that u is isolated in G_r and H_r has minimum degree r.

It is easy to see that $\mathsf{d}_2(G_r, H_r) = \sqrt{r}$. Every bijection matches the all-zeros row of u with a row with at least r non-zero entries since $\deg_{H_r}(\pi(u)) \geq r$, so this row has Euclidean norm at least $\sqrt{r}$, and as the spectral norm of a matrix is at least the Euclidean norm of each of its rows, hence $\|A_G^{\pi} - A_H\|_2 \geq \sqrt{r}$. Taking the identity alignment, $A_G - A_H$ is the adjacency matrix of a star with r leaves (up to entries being negative) whose spectral norm is $\sqrt{r}$.

Lemma 49. $\frac{r}{\sqrt{n}} \leq \mathsf{d}_2^{\mathsf{OT}}(G_r, H_r) \leq 1$.

Proof. For the upper bound, for $i \in [2r]$, let σ_i be the bijection from G_r to H_r that shifts the core vertices of the i-th branch along one step, that is $\sigma_i(u) = v_{i,0}$, $\sigma_i(v_{i,k}) = v_{i,k+1}$ for $k \in \{0, \dots, r-1\}$ and $\sigma_i(v_{i,r}) = u$. Then define

$$Q \coloneqq \frac{1}{2r}\sum_{i=1}^{2r} P_{\sigma_i}.$$

Similarly to the proof of Lemma 27, Q fixes all markers and spreads u uniformly across the first vertices of the $2r$ branches, moves mass $\frac{1}{2r}$ one step along each branch, and moves mass $\frac{1}{2r}$ from the end of each branch to u again. For a pair $(x,y) \in \operatorname{supp}(Q)$ of core vertices, let $\boldsymbol{d}_{xy} \coloneqq \mathbb{1}_{N_{G_r}(x)} - \mathbb{1}_{N_{H_r}(y)} \in \{-1,0,1\}^M$. The nonzero vectors of this kind are

$$\boldsymbol{d}_{uv_{i,0}} = -\mathbb{1}_{T_i}, \qquad \boldsymbol{d}_{v_{i,k}v_{i,k+1}} = -\mathbb{1}_{\{c_{k+1}\}} \text{ for } 0 \le k < r, \qquad \text{and} \qquad \boldsymbol{d}_{v_{i,r}u} = \mathbb{1}_{T_i},$$

and for each such pair (x,y), we have $Q(x,y) = \frac{1}{2r}$. As X and M are independent sets in both graphs, $D \coloneqq D_{G_r,H_r}$ is zero for pairs of atoms that both consist of core vertices or both consist of markers. Moreover, $D((x,y),(z,z)) = \boldsymbol{d}_{xy}(z)$ for a pair $(x,y) \in \operatorname{supp}(Q)$ of core vertices and a marker $z \in M$. Hence the matrix $B_{2\to 2}(G_r,H_r,Q)$ has the block form $\left(\begin{smallmatrix} 0 & K \\ K^\top & 0 \end{smallmatrix}\right)$, where K has rows $\sqrt{Q(x,y)}\boldsymbol{d}_{xy}^\top$, and therefore

$$\mathcal{C}_2(Q)^2 = \|K\|_2^2 = \left\|K^\top K\right\|_2 = \Big\| \sum_{(x,y)\in\operatorname{supp}(Q)} Q(x,y)\, \boldsymbol{d}_{xy}\boldsymbol{d}_{xy}^\top \Big\|_2 = \Big\| \frac{1}{r}\sum_{i=1}^{2r} \mathbb{1}_{T_i}\mathbb{1}_{T_i}^\top + \sum_{j=1}^{r} \mathbb{1}_{\{c_j\}}\mathbb{1}_{\{c_j\}}^\top \Big\|_2.$$

Here every tag set contributes twice with mass $\frac{1}{2r}$, and every clock marker contributes once on each of the $2r$ branches with mass $\frac{1}{2r}$. The sets $C, T_1, \dots, T_{2r}$ are disjoint, so the matrix on the right-hand side is block-diagonal with blocks $\frac{1}{r}\mathbb{1}_{T_i}\mathbb{1}_{T_i}^\top$ of norm $\frac{|T_i|}{r} = 1$ and an identity block for C. Thus $\mathcal{C}_2(Q) = 1$ and $\mathsf{d}_2^{\mathsf{OT}}(G_r,H_r) \le 1$.

For the lower bound, let $R \in \mathrm{DS}(V,W)$ be arbitrary and let $\mathbf{1}_R \in \mathbb{R}^{V\times W}$ be the unit vector with $\mathbf{1}_R(x,y) \coloneqq \sqrt{R(x,y)/n}$ for $(x,y) \in \operatorname{supp}(R)$ and zero otherwise. Then, writing $B \coloneqq B_{2\to 2}(G_r,H_r,R)$, we have

$$\begin{aligned}(B\mathbf{1}_R)(x,y) &= \sqrt{R(x,y)} \sum_{(x',y')\in\operatorname{supp}(R)} D((x,y),(x',y'))\frac{R(x',y')}{\sqrt{n}} \\ &= \mathbf{1}_R(x,y)\big(\deg_{G_r}(x) - \deg_{H_r}(y)\big).\end{aligned}$$

for every $(x,y) \in V \times W$ since R has row and column sums one.

As u is isolated in G_r and H_r has minimum degree r, we have $|\deg_{G_r}(u) - \deg_{H_r}(y)| \ge r$ for all pairs $(u,y) \in \operatorname{supp}(R)$, and these pairs satisfy $\sum_y \mathbf{1}_R(u,y)^2 = \frac{1}{n}\sum_y R(u,y) = \frac{1}{n}$. Hence

$$\mathcal{C}_2(R) \ge \|B\mathbf{1}_R\|_2 \ge \Big(\sum_{(u,y)\in\operatorname{supp}(R)} \big(B\mathbf{1}_R(u,y)\big)^2\Big)^{1/2} \ge \Big(r^2 \sum_{(u,y)\in\operatorname{supp}(R)} \mathbf{1}_R(u,y)^2\Big)^{1/2} = \frac{r}{\sqrt{n}},$$

and thus $\mathsf{d}_2^{\mathsf{OT}}(G_r,H_r) \ge \frac{r}{\sqrt{n}}$. ◀

Proof of Proposition 13. With Lemma 49 together with the fact that $\mathsf{d}_2(G_r,H_r) = \sqrt{r}$, it remains to observe that $\frac{r}{\sqrt{n}} \ge \frac{1}{2\sqrt{2}}$ since $8r^2 \ge 4r^2 + 3r + 1 = n$. ◀

F.2 The Upper Bound

Throughout this section, G and H are graphs of the same order n. For simplicity, we assume that $V(G) = V(H) = [n]$. The adjacency matrices of G, H are $A, B \in \{0,1\}^{n\times n}$. As in Appendix E.2.1,

matrices act on vectors by ordinary matrix multiplication. We recall that $\|K\|_2 \le \|K\|_F$ and $\|K\|_F^2 = \sum_{r\in[n]} \|K\boldsymbol{\phi}_r\|_2^2$ for every orthonormal basis $(\boldsymbol{\phi}_r)_{r\in[n]}$ of $\mathbb{R}^n$.

We show that the spectral distance is bounded by the square root of its optimal transport relaxation, that is, we prove Theorem 14. The proof is mostly an adaptation of the multiplicative rounding for the cut distance (Theorem 38), however a crucial step based on the transport via thin matchings breaks down here and needs to be repaired. We do this by means of spectral truncation similar to the one that was used in the additive cut estimate (compare Lemma 45). We therefore establish the necessary results for arbitrary matrices A' and B', however they will be chosen to be "close" to A and B later in the proof.

We first translate Eq. (22) to the setting of the spectral distance. There, the test functions f, g represented cuts in the cut norm. Here, similar functions are used to encode the vectors in the optimisation within the spectral norm. Let $Q \in \mathrm{DS}(n)$. For functions $f, g\colon [n]^2 \to \mathbb{R}$, we define the *$Q$-weighted inner product* and the *Q-weighted norm* by

$$\langle f, g\rangle_Q := \frac{1}{n} \sum_{i,j\in[n]} Q_{ij} f(i,j) g(i,j) \qquad \text{and} \qquad \|f\|_Q := \sqrt{\langle f, f\rangle_Q}.$$

Strictly speaking, $\langle\cdot,\cdot\rangle_Q$ is only a semi-inner product, as $\|f\|_Q = 0$ whenever f vanishes on the support of Q, however, this does not affect any of our arguments. Then, for $A', B' \in \mathbb{R}^{n\times n}$, we let

$$\gamma_Q(A', B') := \max_{f,g} \frac{1}{n^2} \Big| \sum_{i,j,k,\ell\in[n]} f(i,j)\, Q_{ij} Q_{k\ell} (A'_{ik} - B'_{j\ell})\, g(k,\ell) \Big| \qquad \text{and} \qquad \gamma_Q := \gamma_Q(A, B), \quad (40)$$

where the maximum is over all $f, g\colon [n]^2 \to \mathbb{R}$ with $\|f\|_Q, \|g\|_Q = 1$. Note that

$$\delta_2^{\mathsf{OT}}(G, H) = \min_{Q\in\mathrm{DS}(n)} \gamma_Q \qquad \text{and} \qquad \delta_2(G, H) = \min_{P\in\mathrm{Perm}(n)} \gamma_P.$$

We recall the proof of Theorem 38. For a given optimal $Q \in \mathrm{DS}(n)$ and a permutation matrix $P = P_\pi$, we let $R := PQ^\top$ so that $(R\boldsymbol{x})_i = (Q^\top \boldsymbol{x})_{\pi(i)}$ for all $\boldsymbol{x} \in \mathbb{R}^n$. Using $RAR^\top = P(Q^\top A Q)P^\top$, we split

$$A - PBP^\top = (A - RAR^\top) + P(Q^\top A Q - B)P^\top \quad (41)$$

and bound the norm of the two terms of the right-hand side separately as for Eq. (27).

The second term can be bounded by γ_Q using the following lemma and the fact that the spectral norm is invariant under conjugation with P.

Lemma 50. *Let $Q \in \mathrm{DS}(n)$ and $A', B' \in \mathbb{R}^{n\times n}$. Then $n^{-1} \left\|Q^\top A' Q - B'\right\|_2 \le \gamma_Q(A', B')$.*

The proof is essentially the same as for the first bound in Eq. (27): testing $\gamma_Q(A', B')$ with $f(i,j) := \sqrt{n}\, x_j$ and $g(k,\ell) := \sqrt{n}\, y_\ell$ for unit vectors $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ yields $n^{-1}|\boldsymbol{x}^\top (Q^\top A' Q - B')\boldsymbol{y}|$, and since the columns of Q sum to one, these lifts satisfy the norm constraint in Eq. (40), just as the indicator of $[n] \times X$ is an admissible test set for the cut norm. Then, maximising over $\boldsymbol{x}, \boldsymbol{y}$ and applying Lemma 19 proves the claim.

Expanding $A - RAR^\top = EA + R(EA)^\top$ with $E := I - R$ (see Eq. (28)), our goal is to show that $\|EA\|_2$ is small. This suffices because $\|RK\|_2 \le \|R\|_2 \|K\|_2 \le \|K\|_2$ for $R \in \mathrm{DS}(n)$, as R is a convex combination of permutation matrices, which have spectral norm one. We bound $\|EA'\|_2$ where A' is a suitable approximation of A. For a test vector $\boldsymbol{y}$ and $\boldsymbol{x} := A'\boldsymbol{y}$ it holds that $(EA'\boldsymbol{y})_i = \boldsymbol{x}_i - (Q^\top \boldsymbol{x})_{\pi(i)}$, so we need the value of $\boldsymbol{x}$ at each i to be close to its Q-weighted average

over the vertices coupled with $\pi(i)$.

We do this in two steps. The first is a local agreement property similar to the one that was shown for the cut distance: We show that $x_i \approx (Q^\top x)_j$ on average over the pairs (i,j) weighted by Q.

Lemma 51 (Local Agreement). *Let $Q \in \mathrm{DS}(n)$, $A', B' \in \mathbb{R}^{n\times n}$ and $\boldsymbol{y} \in \mathbb{R}^n$, and let $\boldsymbol{x} \coloneqq A'\boldsymbol{y} \in \mathbb{R}^n$. Then*

$$\frac{1}{n^2} \sum_{i,j\in[n]} Q_{ij}\big(x_i - (Q^\top \boldsymbol{x})_j\big)^2 \leq \gamma_Q(A',B')^2 \, \|\boldsymbol{y}\|_2^2 \,.$$

Proof. As in the proof of Lemma 37, for $i,j \in [n]$, let

$$F_{ij} \coloneqq \sum_{k,\ell\in[n]} Q_{k\ell}(A'_{ik} - B'_{j\ell})\, y_k = x_i - r_j, \qquad \text{where } \boldsymbol{r} \coloneqq B'Q^\top \boldsymbol{y},$$

and the identity uses that the rows of Q sum to one. We first show that

$$\frac{1}{n^2} \sum_{i,j\in[n]} Q_{ij} F_{ij}^2 \leq \gamma_Q(A',B')^2 \, \|\boldsymbol{y}\|_2^2 \,. \tag{42}$$

To this end, define $S \coloneqq \big(\sum_{i,j\in[n]} Q_{ij}F_{ij}^2\big)^{1/2}$. Then

$$\frac{S^2}{n^2} = \frac{1}{n^2} \sum_{i,j\in[n]} F_{ij}\, Q_{ij} F_{ij} = \frac{1}{n^2} \sum_{i,j,k,\ell\in[n]} F_{ij}\, Q_{ij} Q_{k\ell}(A'_{ik} - B'_{j\ell})\, y_k$$

is the expression in Eq. (40) for the functions $f, g\colon [n]^2 \to \mathbb{R}$ with $f(i,j) \coloneqq F_{ij}$ and $g(k,\ell) \coloneqq y_k$. As this expression is bilinear in f and g, it is at most $\gamma_Q(A',B') \, \|f\|_Q \, \|g\|_Q$, where $\|f\|_Q = \frac{1}{\sqrt{n}} S$ and, since the rows of Q sum to one, $\|g\|_Q = \frac{1}{\sqrt{n}} \|\boldsymbol{y}\|_2$. Hence $S^2 \leq n\,\gamma_Q(A',B') \, \|\boldsymbol{y}\|_2 \, S$, which proves Eq. (42), the case $S = 0$ being trivial.

Since the columns of Q sum to one, we have $(Q^\top \boldsymbol{x})_j = \sum_{i'} Q_{i'j}(F_{i'j} + r_j) = r_j + m_j$ with $m_j \coloneqq \sum_{i'\in[n]} Q_{i'j} F_{i'j}$, and hence

$$x_i - (Q^\top \boldsymbol{x})_j = F_{ij} - m_j \qquad \text{for all } i,j \in [n].$$

Expanding the square and using once more that the columns of Q sum to one, we have

$$\frac{1}{n^2} \sum_{i,j\in[n]} Q_{ij}\big(x_i - (Q^\top \boldsymbol{x})_j\big)^2 = \frac{1}{n^2} \sum_{i,j\in[n]} Q_{ij} F_{ij}^2 - \frac{1}{n^2} \sum_{j\in[n]} m_j^2 \leq \gamma_Q(A',B')^2 \, \|\boldsymbol{y}\|_2^2 \,. \qquad \blacktriangleleft$$

Compared to Lemma 37, the factors of two disappear since we can test with F itself rather than with its positive and negative parts.

In a second step, we now want to transfer this guarantee from the pairs (i,j) weighted by Q to the pairs $(i,\pi(i))$. However, here we cannot rely on a thinness argument as for the cut distance (Lemma 36). Indeed, the graphs we constructed in Appendix F.1 show that the loss of an ℓ_2-version of such a transfer can, in general, grow with the square root of the order of the graphs.

We therefore follow a different strategy, again rounding to a permutation in a Birkhoff decomposition of Q by making a random choice, see Eq. (3).

Lemma 52. *Let $Q \in \mathrm{DS}(n)$, let $\boldsymbol{P} = P_{\boldsymbol{\pi}}$ be a random permutation matrix with $\mathbb{E}[\boldsymbol{P}] = Q$, and let*

$\boldsymbol{R} \coloneqq \boldsymbol{P}Q^\top$. *Then for every* $\boldsymbol{x} \in \mathbb{R}^n$,

$$\mathbb{E}\,\|(I - \boldsymbol{R})\boldsymbol{x}\|_2^2 = \sum_{i,j\in[n]} Q_{ij}\big(x_i - (Q^\top \boldsymbol{x})_j\big)^2.$$

Proof. We have $((I - \boldsymbol{R})\boldsymbol{x})_i = x_i - (Q^\top \boldsymbol{x})_{\boldsymbol{\pi}(i)}$ and $\Pr(\boldsymbol{\pi}(i) = j) = \mathbb{E}[\boldsymbol{P}_{ij}] = Q_{ij}$ for all $i, j \in [n]$. Linearity of expectation proves the claim. ◀

Thus, a random permutation satisfies this "ℓ_2-version" of Lemma 36 with constant one on average for each fixed test vector. Of course, the permutation we finally choose has to be good for all test vectors simultaneously. This is the central obstacle in the following proof.

Proof of Theorem 14. Let $Q \in \mathrm{DS}(n)$ be optimal, so that $\gamma_Q = \delta_2^{\mathsf{OT}}(G, H)$. We start by "centering" the matrices A and B, which allows us to control parts of the rounding error via the Frobenius norm later in the proof. Let $p_G \coloneqq 2|E(G)|/n^2$ and $p_H \coloneqq 2|E(H)|/n^2$, and define $h \coloneqq \min\{\sqrt{p_G(1-p_G)}, \sqrt{p_H(1-p_H)}\}$. By symmetry, we may assume that $h = \sqrt{p_G(1-p_G)}$. Letting J be the all-ones matrix, we define $A' \coloneqq A - p_G J$ and $B' \coloneqq B - p_G J$. Then $D_{A',B'} = D_{A,B}$, so $\gamma_Q(A', B') = \gamma_Q$, and $A' - PB'P^\top = A - PBP^\top$ for every $P \in \mathrm{Perm}(n)$, as $PJP^\top = J$. Since $A_{ik}^2 = A_{ik}$ and $\sum_{i,k\in[n]} A_{ik} = n^2 p_G$, we have $\|A'\|_F^2 = n^2(p_G - p_G^2) = n^2h^2$.

Since A' is symmetric, by the spectral theorem, we can write $A' = \sum_{r\in[n]} \lambda_r \boldsymbol{\phi}_r \boldsymbol{\phi}_r^\top$ where $(\boldsymbol{\phi}_r)_{r\in[n]}$ is an orthonormal basis. For $t > 0$, let

$$A'_t \coloneqq \sum_{r\in I_t} \lambda_r \boldsymbol{\phi}_r \boldsymbol{\phi}_r^\top \qquad \text{with} \qquad I_t \coloneqq \{r \in [n] : |\lambda_r| > nt\}$$

be its spectral truncation. (Note the similarity to the spectral truncation used in the additive cut estimate, compare Lemma 45.) This truncation reduces the size of the orthonormal basis, however the resulting simplified representation is lossy. This yields two different sources of error which we bound in the following. For the first kind of error, as $A' - A'_t$ has all its eigenvalues in $[-nt, nt]$, we have $\|A' - A'_t\|_2 \le nt$.

Turning to the second kind of error, we first show that $|I_t|$ can be bounded by a function of t. Recalling that $n^2h^2 = \|A'\|_F^2$, we have

$$n^2h^2 = \|A'\|_F^2 = \sum_{r\in[n]} \lambda_r^2 \ge \sum_{r\in I_t} \lambda_r^2 \ge \sum_{r\in I_t} (nt)^2 = n^2t^2|I_t|,$$

so $|I_t| \le h^2/t^2$. Our goal is to now show that there is a permutation P within the support of Q such that its spectral norm restricted to optimisation over vectors in the span of the eigenvectors of eigenvalues in I_t is bounded by a quantity depending on γ_Q. Intuitively, we consider the expected sum of costs for all eigenvectors in I_t where the expectation is over the choice of $\boldsymbol{P}$. Hence if the total cost is small, there must also exist some choice of P with small cost.

Since the maximum does not commute with expectation, we first bound the spectral norm by the Frobenius norm and then use linearity of expectation to bound the spectral norm for each of the $|I_t|$ eigenvectors individually.

Let $\boldsymbol{P}$ and $\boldsymbol{R} \coloneqq \boldsymbol{P}Q^\top$ as in Lemma 52, and define $\boldsymbol{E} \coloneqq I - \boldsymbol{R}$. Then, we have

$$\mathbb{E}\,\|\boldsymbol{E}A'_t\|_2^2 \le \mathbb{E}\,\|\boldsymbol{E}A'_t\|_F^2 = \sum_{r\in I_t} \mathbb{E}\,\|\boldsymbol{E}A'\boldsymbol{\phi}_r\|_2^2 \le n^2\gamma_Q^2|I_t| \le \frac{n^2\gamma_Q^2h^2}{t^2}. \tag{43}$$

where the first inequality holds since $\|\cdot\|_2 \le \|\cdot\|_F$. For the equality, let Φ be the orthogonal

matrix with columns $\boldsymbol{\phi}_r$. Since the Frobenius norm is invariant under multiplication with orthogonal matrices, every matrix $M \in \mathbb{R}^{n\times n}$ satisfies $\|M\|_F^2 = \|M\Phi\|_F^2 = \sum_{r\in[n]} \|M\boldsymbol{\phi}_r\|_2^2$. We apply this to $M := \boldsymbol{E}A'_t$ and use that $A'_t\boldsymbol{\phi}_r = \lambda_r\boldsymbol{\phi}_r = A'\boldsymbol{\phi}_r$ for $r \in I_t$ and $A'_t\boldsymbol{\phi}_r = \mathbf{0}$ for $r \notin I_t$; linearity of expectation then gives the equality. The next inequality follows by chaining Lemma 52 and Lemma 51 with $\boldsymbol{x} := A'\boldsymbol{\phi}_r$ and $\boldsymbol{y} := \boldsymbol{\phi}_r$, respectively, using that $\|\boldsymbol{\phi}_r\|_2 = 1$. The last inequality then follows from the size bound on I_t.

Taking the square root of both sides in Eq. (43) therefore shows that there is a value P of $\boldsymbol{P}$ that satisfies $\|EA'_t\|_2 \le n\gamma_Q h/t$ for $R := PQ^\top$ and $E := I - R$.

Splitting $A = (A - A'_t) + A'_t$, we have

$$\begin{aligned}
\left\|A' - RA'R^\top\right\|_2 &= \left\|(A' - A'_t) - R(A' - A'_t)R^\top + A'_t - RA'_tR^\top\right\|_2 \\
&\le \left\|(A' - A'_t) - R(A' - A'_t)R^\top\right\|_2 + \left\|A'_t - RA'_tR^\top\right\|_2 \\
&\le \left\|(A' - A'_t) - R(A' - A'_t)R^\top\right\|_2 + 2\,\|EA'_t\|_2 \qquad \text{(a)} \\
&\le \left\|(A' - A'_t) - R(A' - A'_t)R^\top\right\|_2 + \frac{2n\gamma_Q h}{t} \\
&\le \|A' - A'_t\|_2 + \left\|R(A' - A'_t)R^\top\right\|_2 + \frac{2n\gamma_Q h}{t} \\
&\le 2nt + \frac{2n\gamma_Q h}{t}
\end{aligned}$$

where we used Eq. (28) in step (a) with A'_t instead of A and the fact that $\|R\|_2 \le 1$ for the last inequality. Hence, together with Eq. (41) for A', B' and Lemma 50, we get

$$\delta_2(G,H) \le \gamma_P = \frac{1}{n}\left\|A' - PB'P^\top\right\|_2 \le \gamma_Q + 2t + \frac{2\gamma_Q h}{t}.$$

If $\gamma_Q h > 0$, we choose $t := \sqrt{\gamma_Q h}$ to balance the two error terms optimally and obtain $\delta_2(G,H) \le \gamma_Q + 4\sqrt{\gamma_Q h}$. Otherwise, letting $t \to 0$ shows that $\delta_2(G,H) \le \gamma_Q$ as the second error term is always zero. Finally, we always have $h \le \frac{1}{2}$, since $x(1-x) \le \frac{1}{4}$ for all $x \in \mathbb{R}$, so $4\sqrt{h\gamma_Q} \le 2\sqrt{2\gamma_Q}$. ◀

G Proofs Omitted from Section 4

Proof of Theorem 15. We give a reduction from the following NP-hard variant of the Three-Partition problem [GJ79]. The input consists of integers A and $a_1, \dots, a_{3m}$ in unary representation satisfying $\sum_{i=1}^{3m} a_i = mA$ and $A/4 < a_i < A/2$ for $1 \le i \le 3m$. The question then is to decide whether $a_1, \dots, a_{3m}$ can be partitioned into m triples such that the sum of each triple is exactly A. Given an instance $(A, a_1, \dots, a_{3m})$, we construct a forest F_1 which is the disjoint union of paths on a_i vertices for each $i \in [3m]$. The forest F_2 is the disjoint union of m paths on A vertices. Moreover, we set $t := 4m$. Denote the vertex sets of F_1 and F_2 by V and W, respectively.

We claim that the resulting instance (F_1, F_2, t) is a YES-instance of $\textsc{GSim}(\mathsf{d}_{\mathsf{ed}}^{\mathsf{OT}})$ if and only if $(A, a_1, \dots, a_{3m})$ is a YES-instance of Three-Partition. By [Gro25, Theorem 3.10], $\mathsf{d}_{\mathsf{ed}}^{\mathsf{OT}}(F_1, F_2) = \mathsf{d}_{\mathsf{ed}}^{\bullet}(F_1, F_2)$, so we may argue with blowups. In the following, we assume that $a_i \ge 4$ for all $i \in [3m]$. It is easy to see that we can always add 3 to every a_i and 9 to A to ensure that this is the case.

Given a YES-instance of Three-Partition, let $\sigma : [3m] \to [m]$ be the assignment of the a_i to the sums they occur in, i.e. $|\sigma^{-1}(\{\ell\})| = 3$ for every $\ell \in [m]$. Clearly, we can map the paths of F_1 to the paths of F_2 according to σ, incurring 2 mismatches per triple, let $\rho \in \mathrm{Bij}(V, W)$ be the corresponding mapping. Note that ρ maps every edge of F_1 to an edge of F_2, so the mismatches are exactly the $2m$ edges of F_2 that are not hit by ρ. For any $k \ge 1$, consider the bijection $\rho^{\bullet k} \in \mathrm{Bij}(V \times [k], W \times [k])$

between $F_1^{\bullet k}$ and $F_2^{\bullet k}$ defined by $\rho^{\bullet k}((v,j)) := (\rho(v), j)$ for $v \in V$ and $j \in [k]$. In words, the clouds of the blowups get mapped to each other as if they were the same vertex. The number of mismatches in the k-blowup increases by a factor of k^2 compared to the underlying graphs, so there are $2mk^2$ mismatches, and as $\|\cdot\|_{(1)}$ counts every mismatched edge twice we obtain $\mathsf{d}_{\mathsf{ed}}(F_1^{\bullet k}, F_2^{\bullet k}) \leq 4mk^2$. Since this holds for all $k \geq 1$ it follows that $\mathsf{d}_{\mathsf{ed}}^{\bullet}(F_1, F_2) \leq t$.

Conversely, suppose that $\mathsf{d}_{\mathsf{ed}}^{\bullet}(F_1, F_2) \leq t$, and set $\gamma := \frac{1}{2mA}$. Then, there exist $k \geq 1$ and an alignment π of the k-blowups $F_1' := F_1^{\bullet k}$ and $F_2' := F_2^{\bullet k}$ such that $\left\|A^{\pi}_{F_1'} - A_{F_2'}\right\|_{(1)} \leq (4m + 4\gamma)k^2$. Let b be the number of edges of F_1' that π maps to non-edges of F_2'. Then exactly $|E(F_2')| - |E(F_1')| + b$ edges of F_2' are not hit. Since each edge discrepancy contributes twice, the edge counts give

$$\left\|A^{\pi}_{F_1'} - A_{F_2'}\right\|_{(1)} = 4mk^2 + 4b.$$

Thus $b \leq \gamma k^2$. For $v \in V$ and $j \in [m]$, let $n_j(v)$ be the number of vertices in the cloud of v that π maps into the blowup of the j-th path of F_2. In particular, $\sum_j n_j(v) = k$. If $vv' \in E(F_1)$, an edge between their clouds can be preserved only if both endpoints are mapped into the same path of F_2. Hence at least

$$k^2 - \sum_j n_j(v) n_j(v') \geq k(k - \max_j n_j(v))$$

of the edges between these clouds are not preserved. These edges are counted by b, so $\max_j n_j(v) \geq (1-\gamma)k$. Every vertex in V has a neighbour, so this condition holds for every $v \in V$. As $\gamma = \frac{1}{2mA} \leq 1/4$, each $v \in V$ has a unique index $j(v)$ satisfying $n_{j(v)}(v) \geq (1-\gamma)k$.

We claim that $j(v) = j(v')$ whenever $vv' \in E(F_1)$. Otherwise, writing $a := j(v)$, we have $n_a(v') \leq \gamma k$, because $j(v') \neq a$. Also, $\sum_{j \neq a} n_j(v) = k - n_a(v) \leq \gamma k$. Consequently,

$$\sum_j n_j(v) n_j(v') \leq k\, n_a(v') + k \sum_{j \neq a} n_j(v) \leq 2\gamma k^2.$$

At least $(1-2\gamma)k^2$ edges between these clouds are therefore not preserved. This implies $b \geq (1-2\gamma)k^2 > \gamma k^2$, since $\gamma \leq 1/4$, contradicting the bound on b. Thus $j(\cdot)$ is constant on every path P_i of F_1. We denote this common value by $j(i)$ for each $i \in [3m]$.

Finally, let $j \in [m]$ and $S_j := \sum_{i\,:\,j(i)=j} a_i$. As π is a bijection, $\sum_v n_j(v) = Ak$, where the S_j vertices of paths P_i with $j(i) = j$ contribute more than $(1-\gamma)k$ each and all other vertices less than γk each. Thus $|A - S_j| < \gamma m A = \frac{1}{2}$, so $S_j = A$, and as $A/4 < a_i < A/2$, exactly three paths satisfy $j(i) = j$. Hence we obtain a valid solution for the Three-Partition-instance. ◀

Proof of Theorem 16. We again reduce from Three-Partition. This time F_2 is the collection of m disjoint cycles on A vertices each. As in the proof of Theorem 15, we may assume that $a_i \geq 3$ for all $i \in [3m]$: adding 3 to every a_i and 9 to A yields an equivalent instance that still satisfies $A/4 < a_i < A/2$. In particular every path of F_1 has at least three vertices. We show that $\mathsf{d}_1^{\bullet}(F_1, F_2) \leq 1$ if $\mathcal{I} = (A, a_1, \ldots, a_{3m})$ is a YES-instance, and $\mathsf{d}_1^{\bullet}(F_1, F_2) \geq 2$ otherwise.

A YES-instance of Three-Partition gives a mapping of the triples into the cycles such that the vertices at the endpoints of each path in F_1 get an additional neighbour in F_2. Let $\pi \in \mathrm{Sym}([mA])$ be the corresponding permutation. Then for each $k \geq 1$, define $\pi^{\bullet k} \in \mathrm{Sym}([mA] \times [k])$ by letting $\pi^{\bullet k}(v, j) := (\pi(v), j)$, that is $\pi^{\bullet k}$ maps clouds in $F_1^{\bullet k}$ to clouds in $F_2^{\bullet k}$. Clearly, the vertices in the clouds corresponding to endpoints of paths of F_1 gain k new neighbours under $\pi^{\bullet k}$ whereas (non-)adjacency is preserved for all other vertices of $F_1^{\bullet k}$. Hence $\mathsf{d}_1(F_1^{\bullet k}, F_2^{\bullet k}) \leq k$ which proves $\mathsf{d}_1^{\bullet}(F_1, F_2) \leq 1$.

For the other direction, assume that $(A, a_1, \dots, a_{3m})$ is a NO-instance, let $k \geq 1$ and let $\pi \in \mathrm{Sym}([mA] \times [k])$ be an arbitrary alignment of $F_1^{\bullet k}$ and $F_2^{\bullet k}$. We show that some vertex incurs a mismatch of at least $2k$ under π. For $i \in [3m]$, we write $P_i^{\bullet k}$ for the k-blowup of the path P_i of F_1. Similarly, for each $j \in [m]$, we write $C_j^{\bullet k}$ for the blowup of the cycle C_j in F_2.

Note that it cannot hold that for each $i \in [3m]$ there is a $j \in [m]$ such that

$$\pi\left(V(P_i^{\bullet k})\right) \subseteq V(C_j^{\bullet k}). \tag{44}$$

Indeed, otherwise, since π is a bijection, and since $|C_j^{\bullet k}| = Ak$ for each $j \in [m]$, and $|P_i^{\bullet k}| = a_i k$ and $A/4 < a_i < A/2$ for each $i \in [3m]$, it follows that exactly three paths $P_i^{\bullet k}$ are mapped to each $C_j^{\bullet k}$. Hence, the corresponding partition mapping $\sigma : [3m] \to [m]$ yields a solution for the original instance $(A, a_1, \dots, a_{3m})$, contradicting that $\mathcal{I}$ is a NO-instance.

Hence there is a path P whose blowup is mapped to multiple blown-up cycles. There are two cases to distinguish.

Case 1. π maps all vertices of each cloud of $P^{\bullet k}$ to the same cycle. Since π does not map all vertices to the same cycle, there are vertices $v, w \in V(P)$ with $vw \in E(P)$ such that $\{v\} \times [k]$ is mapped to a different cycle (blowup) in $F_2^{\bullet k}$ than $\{w\} \times [k]$ under π. Since each path has at least three vertices, at least one of v and w has degree two in P, w.l.o.g. we assume this to be v. Then clearly $\pi(v, j)\pi(w, j') \notin E(F_2^{\bullet k})$ for any choice $j, j' \in [k]$, so the vertices (v, j) are missing k adjacencies from $F_1^{\bullet k}$. Moreover since $\deg_{F_2^{\bullet k}}(\pi(v, j)) = 2k$ there are at least k vertices that $\pi(v, j)$ is adjacent to in $F_2^{\bullet k}$ whose preimages under π are not adjacent to (v, j) in $F_1^{\bullet k}$. Hence the mismatch count for (v, j) is at least $2k$.

Case 2. A cloud is split across multiple cycles. In particular, there is $v \in V(P)$ and $j, j' \in [k]$ such that $N_{F_2^{\bullet k}}(\pi(v, j)) \cap N_{F_2^{\bullet k}}(\pi(v, j')) = \emptyset$ and $|N_{F_2^{\bullet k}}(\pi(v, j)) \cup N_{F_2^{\bullet k}}(\pi(v, j'))| = 4k$. Letting $d := \deg_{F_1}(v)$, the set $B := N_{F_1^{\bullet k}}(v, j) = N_{F_1^{\bullet k}}(v, j')$ has size dk. Let ℓ, ℓ' be the number of neighbours of $\pi(v, j)$ (respectively $\pi(v, j')$) in $F_2^{\bullet k}$ that are also neighbours of (v, j) in $F_1^{\bullet k}$ (respectively (v, j')). Clearly, $\ell + \ell' \leq dk$. Then, since (v, j) only keeps ℓ adjacencies, equivalently it loses $dk - \ell$ adjacencies. However since $\deg_{F_2^{\bullet k}}(\pi(v, j)) = 2k$ it also gains adjacency to $2k - \ell$ other vertices whose preimage under π is not adjacent to (v, j). The same holds for (v, j'). Thus, the total number of mismatches of (v, j) under π is

$$(2k - \ell) + (dk - \ell) = 2k + (dk - 2\ell).$$

Since $\ell + \ell' \leq dk$, either $dk - 2\ell \geq 0$ or $dk - 2\ell' \geq 0$ which implies that the mismatch count of at least one of (v, j) and (v, j') under π is at least $2k$.

In either case, it follows that there is a vertex incurring a mismatch of at least $2k$ under π. Since $\left\| A^{\pi}_{F_1^{\bullet k}} - A_{F_2^{\bullet k}} \right\|_1$ is the maximum column sum, i.e. the maximum mismatch count of a vertex, and π was arbitrary, we obtain $\mathsf{d}_1(F_1^{\bullet k}, F_2^{\bullet k}) \geq 2k$ for every $k \geq 1$, and hence $\mathsf{d}_1^{\bullet}(F_1, F_2) \geq 2$.

By Theorem 1, $\mathsf{d}_1^{\mathsf{OT}} = \mathsf{d}_1^{\bullet}$, so YES-instances satisfy $\mathsf{d}_1^{\mathsf{OT}}(F_1, F_2) \leq 1$ and NO-instances satisfy $\mathsf{d}_1^{\mathsf{OT}}(F_1, F_2) \geq 2$. A polynomial-time algorithm approximating $\mathsf{d}_1^{\mathsf{OT}}$ (equivalently, δ_1^{OT}) within a factor of $2 - \varepsilon$ would distinguish the two cases and thus decide Three-Partition. In particular, $\mathrm{GSim}(\mathsf{d}_1^{\mathsf{OT}})$ is NP-hard. This proves correctness of our reduction and concludes the proof. ◀

Our main tool for proving Theorem 17 is a lower bound that reduces the analysis of $\|B_p(G, H, Q)\|_p$ to that of a smaller matrix which averages the discrepancies according to Q. For $Q \in \mathrm{DS}(V, W)$,

define the $V \times V$ matrix $A^Q_{G,H} \coloneqq A_G - QA_HQ^\top$ with entries

$$A^Q_{G,H}(v,v') \coloneqq \sum_{w,w' \in W} Q(v,w)\, D_{G,H}(v,w,v',w')\, Q(v',w') \tag{45}$$

for all $v, v' \in V$. The following lemma allows us to derive a lower bound from $A^Q_{G,H}$.

Lemma 53. *For all graphs G, H on the same number of nodes, all $Q \in \mathrm{DS}(V,W)$, and all $1 \leq p \leq \infty$ it holds that*

$$\left\| A^Q_{G,H} \right\|_p \leq \left\| B_p(G,H,Q) \right\|_p .$$

Proof of Lemma 53. By Lemma 19, let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^V$ with $\|\boldsymbol{x}\|_p = 1$ and $\|\boldsymbol{y}\|_{p^*} = 1$ such that $\left\| A^Q_{G,H} \right\|_p = \left| \boldsymbol{y}^\top A^Q_{G,H} \boldsymbol{x} \right|$. We define vectors $\tilde{\boldsymbol{x}}, \tilde{\boldsymbol{y}} \in \mathbb{R}^{V \times W}$ by letting

$$\tilde{\boldsymbol{y}}(v,w) \coloneqq Q(v,w)^{1/p^*} \boldsymbol{y}(v) = Q(v,w)^{1-1/p} \boldsymbol{y}(v), \quad \text{and}$$
$$\tilde{\boldsymbol{x}}(v,w) \coloneqq Q(v,w)^{1/p} \boldsymbol{x}(v)$$

for all $v \in V$, $w \in W$. We claim that $\|\tilde{\boldsymbol{y}}\|_{p^*} = 1$ and $\|\tilde{\boldsymbol{x}}\|_p = 1$. Let $(\boldsymbol{z}, r) \in \{(\boldsymbol{x}, p), (\boldsymbol{y}, p^*)\}$. If $r < \infty$, then, as $\sum_{w \in W} Q(v,w) = 1$ for all $v \in V$,

$$\|\tilde{\boldsymbol{z}}\|_r = \left(\sum_{(v,w) \in V \times W} Q(v,w)\, |\boldsymbol{z}(v)|^r \right)^{1/r} = \left(\sum_{v \in V} |\boldsymbol{z}(v)|^r \sum_{w \in W} Q(v,w) \right)^{1/r} = \|\boldsymbol{z}\|_r = 1.$$

If $r = \infty$, then by the convention $0^0 = 0$ we have $\tilde{\boldsymbol{z}}(v,w) = \boldsymbol{z}(v)$ if $Q(v,w) > 0$ and $\tilde{\boldsymbol{z}}(v,w) = 0$ otherwise; since every row of Q has a positive entry, $\|\tilde{\boldsymbol{z}}\|_\infty = \|\boldsymbol{z}\|_\infty = 1$.

Moreover, $Q(v,w) = Q(v,w)^{1/p} \cdot Q(v,w)^{1/p^*}$ and $Q(v',w') = Q(v',w')^{1-1/p} \cdot Q(v',w')^{1/p}$ for all entries (if $Q(v,w) = 0$, both sides vanish by the convention $0^0 = 0$). Letting $B \coloneqq B_p(G,H,Q)$, we hence obtain

$$\begin{aligned}
&\left| \boldsymbol{y}^\top A^Q_{G,H} \boldsymbol{x} \right| \\
&= \left| \sum_{v,v' \in V} \boldsymbol{y}(v) \left(\sum_{w,w' \in W} D(v,w,v',w') Q(v,w) Q(v',w') \right) \boldsymbol{x}(v') \right| \\
&= \left| \sum_{\substack{v,v' \in V \\ w,w' \in W}} D(v,w,v',w') Q(v,w) \boldsymbol{y}(v) Q(v',w') \boldsymbol{x}(v') \right| \\
&= \left| \sum_{\substack{v,v' \in V \\ w,w' \in W}} D(v,w,v',w') Q(v,w)^{1/p} \underbrace{Q(v,w)^{1-1/p} \boldsymbol{y}(v)}_{\tilde{\boldsymbol{y}}(v,w)} Q(v',w')^{1-1/p} \underbrace{Q(v',w')^{1/p} \boldsymbol{x}(v')}_{\tilde{\boldsymbol{x}}(v',w')} \right| \\
&= \left| \sum_{(v,w),(v',w') \in V \times W} \tilde{\boldsymbol{y}}(v,w) \underbrace{D(v,w,v',w') Q(v,w)^{1/p} Q(v',w')^{1-1/p}}_{= B((v,w),(v',w'))} \tilde{\boldsymbol{x}}(v',w') \right| \\
&\leq \max_{\|\tilde{\boldsymbol{y}}\|_{p^*}=1, \|\tilde{\boldsymbol{x}}\|_p=1} \left| \tilde{\boldsymbol{y}}^\top B \tilde{\boldsymbol{x}} \right| = \|B\|_p
\end{aligned}$$

which completes the proof. ◀

The above lemma has the following simple consequence.

Corollary 54. *Let $1 \leq p \leq \infty$. Suppose the matrix $A^Q_{G,H}$ has a column with at least k entries of absolute value at least ε. Then $\left\|A^Q_{G,H}\right\|_p \geq k^{b(p)}\varepsilon$. In particular, if $k > (c/\varepsilon)^{1/b(p)}$ for some $c > 0$, then $\|B_p(G,H,Q)\|_p > c$.*

Proof. Let $\mathbf{e}_v \in \mathbb{R}^V$ denote the unit vector with $\mathbf{e}_v(v) = 1$ and $\mathbf{e}_v(u) = 0$ for $u \neq v$, so that $\|\mathbf{e}_v\|_s = 1$ for all $1 \leq s \leq \infty$. As $A^Q_{G,H}\mathbf{e}_v$ is the column $A^Q_{G,H}(\cdot, v)$, we get

$$\left\|A^Q_{G,H}\right\|_p \geq \left\|A^Q_{G,H}\mathbf{e}_v\right\|_p \geq (k \cdot \varepsilon^p)^{1/p} = k^{1/p}\varepsilon,$$

where for $p = \infty$ the right-hand side equals ε. Moreover, $A^Q_{G,H}$ is symmetric (exchange (v,w) and (v',w') in Eq. (45)), and $\|A\|_p = \left\|A^\top\right\|_{p^*}$ for every matrix A (see Appendix A). Hence, analogously,

$$\left\|A^Q_{G,H}\right\|_p = \left\|A^Q_{G,H}\right\|_{p^*} \geq \left\|A^Q_{G,H}\mathbf{e}_v\right\|_{p^*} \geq k^{1/p^*}\varepsilon = k^{1-1/p}\varepsilon.$$

Together, $\left\|A^Q_{G,H}\right\|_p \geq k^{b(p)}\varepsilon$ with $b(p) = \max\{1/p, 1-1/p\}$. If $k > (c/\varepsilon)^{1/b(p)}$, then $k^{b(p)}\varepsilon > c$, and Lemma 53 yields $\|B_p(G,H,Q)\|_p \geq \left\|A^Q_{G,H}\right\|_p > c$ for this fixed Q. ◀

Proof of Theorem 17. The proof is by reduction from Three-Partition. Specifically, for a given instance $\mathcal{I} = (A, a_1, \ldots, a_{3m})$, the graph G' consists of $3m$ paths where the i-th path, denoted P_i has a_i vertices. We assume w.l.o.g. that $a_i \geq 3$ for all $i \in [3m]$. Consequently, $A > 2a_i \geq 6$, so $A \geq 7$. The graph H' consists of m disjoint cycles on A vertices each, which are denoted C_j. The graphs G and H are obtained from G' and H' respectively, by attaching at each vertex u a total of M leaves. The value of M will be specified later in the proof. We refer to the vertex set of G' and H' by V and W, respectively. The vertex sets of G and H are referred to as X and Y and we have $V \subseteq X$ and $W \subseteq Y$. For each graph, we partition the vertex set into the base vertex sets $\mathcal{B}_G, \mathcal{B}_H$, which correspond to the vertices in G and H that were already present in G' and H', and the leaf sets $\mathcal{L}_G, \mathcal{L}_H$ containing the vertices that were added in the augmentation. For a base vertex $v \in \mathcal{B}_G$, let $\mathcal{L}(v) \coloneqq N_G(v) \cap \mathcal{L}_G$ denote the set of leaves attached to v. We use the same notation for vertices in H.

For a fixed $Q \in \mathrm{DS}(X,Y)$, we allow both arguments to Q to also be sets, letting $Q(S,T) \coloneqq \sum_{s\in S, t\in T} Q(s,t)$ for $S \subseteq X, T \subseteq Y$. For simplicity we omit the set notation for input sets that are singletons. If $S = \emptyset$ or $T = \emptyset$, then $Q(S,T) = 0$.

Finally we make the following useful observation that will be used throughout the proof. If $xx' \in E(G)$, then

$$A^Q_{G,H}(x,x') = \sum_{yy' \notin E(H)} Q(x,y)Q(x',y').$$

It remains to show correctness of the proposed reduction. Fix $0 < \varepsilon^* < \sqrt{2} - 1$. We show that YES-instances satisfy $\mathsf{d}_p^{\mathsf{OT}}(G,H) \leq 1$, whereas NO-instances satisfy $\mathsf{d}_p^{\mathsf{OT}}(G,H) > 2^{b(p)} - \varepsilon^* > 1$, where the last inequality holds as $b(p) \geq 1/2$. This proves both NP-hardness and the claimed inapproximability (a $(2^{b(p)} - \varepsilon^*)$-approximation would return a value of at most $2^{b(p)} - \varepsilon^*$ on YES-instances and of more than $2^{b(p)} - \varepsilon^*$ on NO-instances).

If $\mathcal{I}$ is a YES-instance, then the witnessing partition function $\sigma : [3m] \to [m]$ yields a natural way of linking up three paths into a cycle by inserting three edges, yielding a bijection $\tau : \mathcal{B}_G \to \mathcal{B}_H$. We extend τ to a bijection $\pi : X \to Y$ by choosing an arbitrary bijection $\mu_v \colon \mathcal{L}(v) \to \mathcal{L}(\tau(v))$ for each $v \in \mathcal{B}_G$ and setting $\pi \coloneqq \tau \cup \bigcup_{v \in \mathcal{B}_G} \mu_v$. Since $A_H - A_G^\pi$ corresponds to the adjacency matrix of a graph consisting of $3m$ disjoint edges and isolated vertices, we have $\|A_G^\pi - A_H\|_p = 1$.

In the following, let $\mathcal{I}$ be a NO-instance. Fix an arbitrary $Q \in \mathrm{DS}(X, Y)$. We start with two claims that allow us to conclude that any admissible Q is approximately a bijection between $\mathcal{B}_G$ and $\mathcal{B}_H$.

Claim 55. *Let $v \in \mathcal{B}_G$ and suppose that $Q(v, \mathcal{L}_H) \geq \eta$ for some $\eta > 0$. If $M > 2\left(\frac{4}{\eta}\right)^2 + 1$, then $\|B_p(G, H, Q)\|_p > 2^{b(p)}$.*

Proof. Let $N \coloneqq N_G(v) \subseteq X$ and note that $|N| \geq M + 1$. Let $L \coloneqq \{y \in \mathcal{L}_H \mid Q(v, y) > 0\}$ and denote by z_y the unique neighbour of y in H and let $Z \coloneqq \{z_y \mid y \in L\} \subseteq \mathcal{B}_H$. We distinguish the following two cases.

Case 1. For all $w \in \mathcal{B}_H$, $Q(v, \mathcal{L}(w)) \leq \eta/2$.
Then, for all $x \in N$, we have

$$\begin{aligned} A^Q_{G,H}(x, v) &\geq \sum_{y \in L} Q(v, y) \sum_{y' \in Y, yy' \notin E(H)} Q(x, y') \\ &\geq \sum_{y \in L} Q(v, y) \sum_{y' \in Y \setminus \{z_y\}} Q(x, y') \\ &= \sum_{y' \in Y} Q(x, y') \sum_{y \in L : z_y \neq y'} Q(v, y) \\ &\geq \frac{\eta}{2} \sum_{y' \in Y} Q(x, y') \\ &\geq \frac{\eta}{2}. \end{aligned}$$

Case 2. There is $w \in \mathcal{B}_H$ with $Q(v, \mathcal{L}(w)) > \eta/2$.
For any $0 < \delta < 1$, at most $\lfloor 1/\delta \rfloor$ vertices x can satisfy $Q(x, w) > \delta$. Conversely, at least $M+1-\lfloor 1/\delta \rfloor$ vertices $x \in N$ have $Q(x, w) \leq \delta$, or equivalently $Q(x, Y \setminus \{w\}) \geq 1 - \delta$. For any such x it follows that

$$A^Q_{G,H}(x, v) \geq \left(\sum_{y \in L : z_y = w} Q(v, y) \right) \left(\sum_{y' \in Y \setminus \{w\}} Q(x, y') \right) \geq \frac{\eta}{2}(1 - \delta).$$

Choosing $\delta \coloneqq 1/2$, in either case we find at least $M - 1$ entries in column v of $A^Q_{G,H}$ that are at least $\eta/4$ each. Hence, if $M - 1 > 2\left(\frac{4}{\eta}\right)^2$, then $\|B_p(G, H, Q)\|_p > 2^{b(p)}$ by Corollary 54. ◁

Claim 56. *Let $v \in \mathcal{B}_G$ and $w, w' \in \mathcal{B}_H$ be distinct with $Q(v, w), Q(v, w') \geq \varepsilon$ for some $\varepsilon > 0$. If $M > 2\left(\frac{2}{\varepsilon}\right)^2 + 1$, then $\|B_p(G, H, Q)\|_p > 2^{b(p)}$.*

Proof. Let $I \coloneqq N_H(w) \cap N_H(w')$ and observe that $|I| \leq 1$ since $|V(C_j)| = A \geq 5$. Let $N \coloneqq N_G(v)$. Since $|N| \geq M + 1$, a total mass of at least M of the vertices in N is mapped such that it is adjacent to at most one of w and w'. For $0 < \delta < 1$, there are at most $\lfloor 1/\delta \rfloor$ vertices $x \in N$ with $Q(x, I) > \delta$, so for at least $|N| - \lfloor 1/\delta \rfloor$ vertices we have that $Q(x, Y \setminus I) \geq 1 - \delta$. In particular for each $y \in Y \setminus I$

there is a $w_y \in \{w, w'\}$ such that $yw_y \notin E(H)$. Thus for at least $M + 1 - \lfloor 1/\delta \rfloor$ vertices $x \in N$, it holds that

$$\begin{aligned} A^Q_{G,H}(x,v) &\geq \sum_{y \in Y \setminus I} Q(x,y) \left(\sum_{y' \in Y, yy' \notin E(H)} Q(v,y') \right) \\ &\geq \sum_{y \in Y \setminus I} Q(x,y) Q(v,w_y) \\ &\geq \varepsilon \cdot \sum_{y \in Y \setminus I} Q(x,y) \\ &\geq \varepsilon(1-\delta). \end{aligned}$$

Choosing $\delta \coloneqq 1/2$ it follows that the v-th column of $A^Q_{G,H}$ has at least $M-1$ entries that are at least $\frac{\varepsilon}{2}$ each. Then $\|B_p(G,H,Q)\|_p > 2^{b(p)}$ by Corollary 54. ◁

Let $\eta \coloneqq \frac{\varepsilon^*}{8}$, $\varepsilon \coloneqq \frac{\varepsilon^*}{8}$ and set $\varepsilon' \coloneqq \frac{\varepsilon}{mA-1}$ and choose

$$M > \max\left\{ 2\left(\frac{4}{\eta}\right)^2 + 1,\ 2\left(\frac{2}{\varepsilon'}\right)^2 + 1 \right\}. \tag{46}$$

Since ε^* is a fixed constant, M can be chosen as $O((mA)^2)$, independently of p. If the precondition of Claim 55 or Claim 56 holds for some $v \in \mathcal{B}_G$, then $\|B_p(G,H,Q)\|_p > 2^{b(p)} > 2^{b(p)} - \varepsilon^*$ and we are done. So assume otherwise. Invoking Claim 56 with ε', for every $v \in \mathcal{B}_G$, there are no two distinct vertices $w, w' \in \mathcal{B}_H$ with $Q(v,w), Q(v,w') \geq \varepsilon'$. Since $|\mathcal{B}_H| = mA$, in particular, there is a $w \in \mathcal{B}_H$ with $Q(v, \mathcal{L}_H \cup \{w\}) \geq 1 - (mA-1)\varepsilon' = 1 - \varepsilon$. Additionally, by Claim 55, $Q(v, \mathcal{B}_H) \geq 1 - \eta$ or equivalently $Q(v, \mathcal{L}_H) \leq \eta$ which implies that $Q(v,w) \geq 1 - \varepsilon - \eta$.

As $1 - \varepsilon - \eta = 1 - \varepsilon^*/4 > 1/2$, Q induces a bijection $\tau : \mathcal{B}_G \to \mathcal{B}_H$, by letting $\tau(v) \coloneqq \arg\max_{w \in \mathcal{B}_H} Q(v,w)$.

Now since $\mathcal{I}$ is a NO-instance, for all partition functions $\sigma : [3m] \to [m]$ with $|\sigma^{-1}(j)| = 3$ for each $j \in [m]$, there is a $j \in [m]$ with $\sum_{i \in \sigma^{-1}(j)} a_i \geq A + 1$. We claim that there exist $i \in [3m]$ and $v, v' \in V(P_i)$ with $vv' \in E(P_i)$ such that $\tau(v)$ and $\tau(v')$ lie in distinct cycles of H'. Otherwise, by connectivity of the paths, for each i there is $j(i)$ with $\tau(V(P_i)) \subseteq V(C_{j(i)})$. Since τ is a bijection $\mathcal{B}_G \to \mathcal{B}_H$, each cycle C_j receives entire paths of total size $\sum_{i : j(i) = j} a_i = A$, and the assumption $A/4 < a_i < A/2$ forces exactly three paths to be mapped into each cycle. Hence, this yields a valid 3-partition, contradicting that $\mathcal{I}$ is a NO-instance. Specifically, $Q(v, \tau(v)), Q(v', \tau(v')) \geq 1 - \varepsilon - \eta$ and $\tau(v)\tau(v') \notin E(H)$. This implies that $A^Q_{G,H}(v', v) \geq (1 - \varepsilon - \eta)^2$.

As $a_i \geq 3$ and hence $|V(P_i)| \geq 3$, we may assume that $\deg_{P_i}(v) = 2$ and denote by v'' the second neighbour of v in P_i. If $\tau(v'') \notin N_H(\tau(v))$, then similarly to before $A^Q_{G,H}(v'', v) \geq (1 - \varepsilon - \eta)^2$. So suppose that $\tau(v'') \in N_H(\tau(v))$. Since $\deg_{H[\mathcal{B}_H]}(\tau(v)) = 2$, there exists $w^* \in (N_H(\tau(v)) \cap \mathcal{B}_H) \setminus \{\tau(v'')\}$ with $\tau^{-1}(w^*) \notin N_G(v)$. It follows that $A^Q_{G,H}(\tau^{-1}(w^*), v) \leq -(1 - \varepsilon - \eta)^2$. Intuitively, we have shown that v either loses both its neighbours in P_i, or loses adjacency to one neighbour and gains adjacency to at least one non-neighbour under τ.

In either case, $A^Q_{G,H}(\cdot, v)$ contains at least two entries that are at least $(1 - \varepsilon - \eta)^2$ in absolute value, and therefore Corollary 54 (with $k = 2$) and Lemma 53 yield

$$\|B_p(G,H,Q)\|_p \geq \left\| A^Q_{G,H} \right\|_p \geq 2^{b(p)}(1 - \varepsilon - \eta)^2.$$

By the choice $\varepsilon = \eta = \varepsilon^*/8$, we have $(1-\varepsilon-\eta)^2 > 1-\varepsilon^*/2$, and hence, using $b(p) \le 1$,

$$\|B_p(G,H,Q)\|_p > 2^{b(p)}(1-\varepsilon^*/2) \ge 2^{b(p)} - \varepsilon^*.$$

Since Q was arbitrary, we obtain $\mathsf{d}_p^{\mathsf{OT}}(G,H) > 2^{b(p)} - \varepsilon^*$. This shows correctness of our reduction, thus concluding the proof. ◀

Proposition 57. *Let G, H be graphs of orders n, m and let $\Diamond \in \{1, \mathsf{ed}\}$.*

1. $\textsc{GSim}(\delta_1^{\mathsf{OT}})$ *and* $\textsc{GSim}(\delta_{\mathsf{ed}}^{\mathsf{OT}})$ *are* NP*-complete.*
2. $\delta_1^{\mathsf{OT}}(G,H)$ *can be computed in time* $2^{nm} \cdot \mathsf{poly}(n+m)$.
3. *There is* $k \in \mathbb{N}_{>0}$ *with* $\log k \le \mathsf{poly}(n+m)$ *such that* $\delta_\Diamond^{\mathsf{OT}}(G,H) = \delta_\Diamond(G^{\bullet mk}, H^{\bullet nk})$.

Proof. NP-hardness follows from Theorems 15 and 16 since $\mathsf{d}_\Diamond^{\mathsf{OT}}$ and $\delta_\Diamond^{\mathsf{OT}}$ differ by a factor only depending on the order of the graphs. By Eq. (7), $\delta_1^{\mathsf{OT}}(G,H)$ is the optimal value of the mixed integer linear program

$$\begin{aligned}
\min \quad & z \\
\text{s.t.} \quad & \sum_{(v,w)\in V\times W} \left|D(v,w,v',w')\right| x_{v,w} \le z + 1 - y_{v',w'} \quad \forall\,(v',w') \in V \times W, \\
& \boldsymbol{x} \le \boldsymbol{y}, \\
& \boldsymbol{x} \in \mathrm{DS}^*(V,W), \\
& \boldsymbol{y} \in \{0,1\}^{V\times W}, \\
& z \ge 0.
\end{aligned}$$

Indeed, $x_{v',w'} > 0$ forces $y_{v',w'} = 1$, and for $y_{v',w'} = 0$ the first constraint is void as column sums are at most $\sum_{v,w} x_{v,w} = 1$; so an optimal choice is $y = \mathbb{1}_{\mathrm{supp}(x)}$. For fixed $\boldsymbol{y}$, this is a linear program of polynomial size, solvable in polynomial time with an optimal vertex solution of polynomial encoding length [Sch86, Theorem 10.2]. Hence guessing $\boldsymbol{y}$ shows $\textsc{GSim}(\delta_1^{\mathsf{OT}}) \in \mathsf{NP}$, enumerating all 2^{nm} choices of $\boldsymbol{y}$ proves the second statement, and δ_1^{OT} has an optimal Q^* of polynomial encoding length. Similarly, $\delta_{\mathsf{ed}}^{\mathsf{OT}}(G,H) = \min_{\boldsymbol{x}\in\mathrm{DS}^*(V,W)} \sum |D(v,w,v',w')|\, x_{v,w}x_{v',w'}$ is a quadratic program, which by [Vav90] has an optimal solution Q^* of polynomial encoding length; it serves as the certificate.

For the third statement, let k be a common denominator of the entries of Q^* and $N \coloneqq nmk$. For an alignment π of $G^{\bullet mk}$ and $H^{\bullet nk}$, let $NQ_\pi(v,w)$ be the number of vertices of $G^{\bullet mk}$ mapped from the cloud of v to the cloud of w; then $Q_\pi \in \mathrm{DS}^*(V,W)$. Counting mismatches, $\left\|A^\pi_{G^{\bullet mk}} - A_{H^{\bullet nk}}\right\|_\Diamond = N^{\kappa_\Diamond} \left\|B_\Diamond(G,H,Q_\pi)\right\|_\Diamond$ where $\kappa_1 = 1$ and $\kappa_{\mathsf{ed}} = 2$, so $\delta_\Diamond(G^{\bullet mk}, H^{\bullet nk}) \ge \delta_\Diamond^{\mathsf{OT}}(G,H)$, with equality for π satisfying $Q_\pi = Q^*$, which exists as NQ^* is integral. ◀

Proof of Theorem 18. We follow the proof of [GG22, Theorem 23], in which they showed the hardness of $\textsc{GSim}(\mathsf{d}_\square)$. Given a graph F on n vertices and m edges, let G', H' be the graphs constructed in their proof (before they additionally introduce leaves attached to each vertex), which satisfy $\mathsf{d}_\square(G',H') = \mathcal{C}_\square(\pi) = 2 \cdot \mathsf{maxcut}(F)$ where $\mathsf{maxcut}(F)$ denotes the size of a maximum cut in F and π is the canonical bijection between the two graphs. Let $n' \coloneqq 2m + n$ denote the order of G' and H' and let $v_1, \dots, v_{n'}$ and $w_1, \dots, w_{n'}$ be enumerations of their respective vertex sets. Recall that in the final step of their construction, $i \cdot (n^2+n)$ leaves are attached to the i-th vertex in each graph. Our graphs G, H are obtained by adding $i \cdot (n^2+n) \cdot k$ leaves to the i-th vertex in both graphs, where k will be chosen later. Note that each non-leaf vertex has at least $M \coloneqq k(n^2+n)$

adjacent leaves. Let N denote the order of the resulting graphs. We write π for the canonical bijection $v_i \mapsto w_i$ between the base vertices (both carry the same number ℓ_i of leaves). We claim that

$$\frac{33}{17} \cdot \mathsf{maxcut}(F) \leq \mathsf{d}_{\square}^{\mathsf{OT}}(G,H) \leq 2 \cdot \mathsf{maxcut}(F). \tag{47}$$

The second inequality follows from $\mathsf{d}_{\square}^{\mathsf{OT}}(G,H) \leq \mathsf{d}_{\square}(G,H)$ for all graphs G, H of the same order and the original proof in [GG22]. Hence, we only need to show the first inequality. Note that Eq. (47) implies that exact computation of $\mathsf{d}_{\square}^{\mathsf{OT}}$ is NP-hard, because for any $\varepsilon > 0$, unless $\mathsf{P} = \mathsf{NP}$, MaxCut does not admit a polynomial-time approximation algorithm with multiplicative guarantee $\frac{16}{17} + \varepsilon$ [Hås01]. We fix ε in the previous equation to be $\varepsilon \coloneqq \frac{1}{34}$.

The reduction of [GG22] leverages the quantities of leaves attached at each vertex to force the optimal bijection between the graphs G and H to be the identity.

We employ a similar strategy, however we first have to rule out fractional mappings that split base vertices, that is, the vertices that are not added as leaves, too much. Let $\mathcal{B}_G, \mathcal{B}_H$ denote the base vertices of the construction and $\mathcal{L}_G, \mathcal{L}_H$ the leaf vertices. For a vertex $v \in \mathcal{B}_G$, we define $\mathcal{L}(v) \coloneqq \mathcal{L}_G \cap N_G(v)$, the definition for vertices in H is analogous. Note that this notation suppresses the subscript indicating the graph, however it will be clear which graph we refer to by vertex naming. In the remainder of the proof, $V \coloneqq V(G)$ and $W \coloneqq V(H)$ denote the vertex sets of the graphs. Fix an arbitrary $Q \in \mathrm{DS}(V,W)$ and write $B \coloneqq B(G,H,Q)$. We show that $\|B\|_{\square} \geq \frac{33}{17}\mathsf{maxcut}(F)$; as Q is arbitrary, this proves the first inequality of Eq. (47).

Claim 58. *Let $v \in \mathcal{B}_G$ and suppose that $Q(v, \mathcal{L}_H) \geq \eta$ for some $\eta > 0$. If $k \geq \frac{4}{\eta}$, then $\|B(G,H,Q)\|_{\square} \geq 2 \cdot \mathsf{maxcut}(F)$.*

Proof. Let $U \coloneqq \mathcal{L}(v) \subseteq V$ and note that $|U| \geq M$. Let $L \coloneqq \{y \in \mathcal{L}_H \mid Q(v,y) > 0\}$ and denote by z_y the unique neighbour of y in H and let $Z \coloneqq \{z_y \mid y \in L\} \subseteq \mathcal{B}_H$.

Write $B \coloneqq B(G,H,Q)$. We distinguish two cases.

Case 1. For all $w \in \mathcal{B}_H$, $Q(v, \mathcal{L}(w)) \leq \eta/2$.

Set $S' \coloneqq \{v\} \times L$ and $T' \coloneqq U \times W$. Then

$$\begin{aligned}
B(S',T') &= \sum_{\substack{v' \in U \\ w \in L, w' \in W}} \Big(\underbrace{A_G(v,v')}_{=1} - A_H(w,w') \Big) Q(v,w) Q(v',w') \\
&= \sum_{\substack{v' \in U \\ w \in L, w' \in W \setminus \{z_w\}}} Q(v,w) Q(v',w') \\
&= \sum_{v' \in U, w' \in W} Q(v',w') \underbrace{\sum_{w \in L : z_w \neq w'} Q(v,w)}_{\geq \frac{\eta}{2}} \\
&\geq \frac{\eta}{2} \sum_{v' \in U, w' \in W} Q(v',w') \\
&\geq M \cdot \frac{\eta}{2}
\end{aligned}$$

and thus $\|B\|_{\square} \geq \frac{1}{2}\eta M$.

Case 2. There is $w^* \in \mathcal{B}_H$ with $Q(v, \mathcal{L}(w^*)) > \eta/2$.

For any $0 < \delta < 1$, at most $\lfloor 1/\delta \rfloor$ vertices x can satisfy $Q(x, w^*) > \delta$. Conversely, at least $M - \lfloor 1/\delta \rfloor$ vertices $x \in U$ have $Q(x, w^*) \leq \delta$, or equivalently $Q(x, W \setminus \{w^*\}) \geq 1 - \delta$. Let $U' \coloneqq \{x \in U \mid Q(x, W \setminus \{w^*\}) \geq 1 - \delta\}$. By the previous observation $|U'| \geq M - \lfloor 1/\delta \rfloor$. Choose

$S' \coloneqq \{v\} \times \mathcal{L}(w^*)$ and $T' \coloneqq U' \times (W \setminus \{w^*\})$.

For $w \in \mathcal{L}(w^*)$ and $w' \neq w^*$, we have $A_H(w, w') = 0$, for each $v' \in U'$ we have $A_G(v, v') = 1$. Therefore

$$\begin{aligned} B(S', T') &= \sum_{\substack{v' \in U' \\ w \in \mathcal{L}(w^*), w' \in W \setminus \{w^*\}}} \Big(\underbrace{A_G(v, v')}_{=1} - \underbrace{A_H(w, w')}_{=0} \Big) Q(v, w) Q(v', w') \\ &= \sum_{\substack{v' \in U' \\ w \in \mathcal{L}(w^*), w' \in W \setminus \{w^*\}}} Q(v, w) Q(v', w') \\ &= \underbrace{Q(v, \mathcal{L}(w^*))}_{> \frac{\eta}{2}} \sum_{v' \in U'} \underbrace{Q(v', W \setminus \{w^*\})}_{\geq 1 - \delta} \\ &\geq \frac{\eta}{2}(1 - \delta) \left(M - \left\lfloor \frac{1}{\delta} \right\rfloor \right) \end{aligned}$$

Choosing $\delta \coloneqq \frac{1}{2}$ now guarantees that $B(S', T') \geq \frac{1}{8}\eta M$ given that $M \geq 4$ which holds if $n \geq 2$ or $k \geq 4$.

In either case we have shown that $\|B\|_\square \geq \frac{1}{8}\eta M$. Note that $\mathsf{maxcut}(F) \leq \frac{n^2}{4}$ for any graph F. Hence the claim follows if $\frac{1}{8}\eta M \geq \frac{n^2}{2}$. Since $M \geq kn^2$, this holds given that $kn^2 \geq \frac{4n^2}{\eta}$ which is equivalent to $k \geq \frac{4}{\eta}$. ◁

Claim 59. *Let $v \in \mathcal{B}_G$ and $w, w' \in \mathcal{B}_H$ be distinct with $Q(v, w), Q(v, w') \geq \tau$ for some $\tau > 0$. If $k \geq \max\left\{\frac{2}{\tau}, 4(n-1)\right\}$, then $\|B(G, H, Q)\|_\square \geq 2 \cdot \mathsf{maxcut}(F)$.*

Proof. Let $U \coloneqq N_G(v)$, $I \coloneqq N_H(w) \cap N_H(w')$ and observe that $|U| \geq M$ and $|I| \leq n - 1$. To see the latter, let A denote the $2m \times n$ matrix constructed by Alon and Naor from the input graph F (compare [AN06, Proposition 3.2]). In this matrix, every edge of F contributes two rows, each with exactly one entry $+1$ and one entry -1. Hence, for every vertex u of F, the column of u contains exactly $\deg_F(u) \leq n - 1$ entries $+1$ and $\deg_F(u)$ entries -1. Since G' contains the edges of the positive entries and H' those of the negative entries, column vertices have degree at most $n - 1$ and row vertices have degree 1 in both G' and H'. Thus G' and H' have maximum degree at most $n - 1$. Leaves are adjacent to only one vertex, so they never lie in I, and hence $|I| \leq n - 1$. For $0 < \delta < 1$ there are at most $\lfloor 1/\delta \rfloor \cdot |I|$ vertices $x \in U$ with $Q(x, I) > \delta$, so for at least $|U| - \lfloor 1/\delta \rfloor \cdot |I|$ vertices, we have that $Q(x, W \setminus I) \geq 1 - \delta$. Let $U' \coloneqq \{x \in U \mid Q(x, W \setminus I) \geq 1 - \delta\}$. For all $y \in W \setminus I$, there is a $w_y \in \{w, w'\}$ such that $yw_y \notin E(H)$. Let $\overline{w_y}$ denote the unique element in $\{w, w'\} \setminus \{w_y\}$. Now set $S' \coloneqq \{(v, w), (v, w')\}$ and $T' \coloneqq U' \times (W \setminus I)$. Then

$$\begin{aligned} B(S', T') &= \sum_{z \in \{w, w'\}} \sum_{v' \in U', z' \in W \setminus I} \Big(\underbrace{A_G(v, v')}_{=1} - A_H(z, z') \Big) Q(v, z) Q(v', z') \\ &\geq \sum_{v' \in U'} \sum_{z' \in W \setminus I} \underbrace{Q(v, w_{z'})}_{\geq \tau} Q(v', z') \\ &\geq \tau \sum_{v' \in U'} Q(v', W \setminus I) \\ &\geq \tau(1 - \delta) \left(M - \left\lfloor \frac{1}{\delta} \right\rfloor \cdot |I| \right) \end{aligned}$$

where the first inequality holds by choosing for each z', the $z \in \{w, w'\}$ such that $z = w_{z'}$. The summand with $\overline{w_{z'}}$ is non-negative, so omitting it yields a valid lower bound. Choosing $\delta \coloneqq \frac{1}{2}$, we

have $B(S', T') \geq \frac{1}{4}\tau M$ given that $M \geq 4(n-1) \geq 4|I|$, since in this case $M - \lfloor 1/\delta \rfloor \cdot |I| \geq M/2$. This implies that $\|B\|_\square \geq \frac{1}{4}\tau M$. Now the claim follows, since $M \geq kn^2 \geq \frac{2n^2}{\tau}$ implies that $\|B\|_\square \geq \frac{n^2}{2} \geq 2 \cdot \mathsf{maxcut}(F)$. ◁

Let $\eta, \varepsilon > 0$ be sufficiently small. Invoking Claim 59 with $\tau \coloneqq \frac{\varepsilon}{n'-1}$, for every $v \in \mathcal{B}_G$, there are no two distinct vertices $w, w' \in \mathcal{B}_H$ with $Q(v, w), Q(v, w') \geq \tau$. Since $|\mathcal{B}_H| = n'$, it follows that for each $v \in \mathcal{B}_G$ there is a $w \in \mathcal{B}_H$ with $Q(v, \mathcal{L}_H \cup \{w\}) \geq 1 - (n'-1)\tau = 1 - \varepsilon$. Moreover, by Claim 58, $Q(v, \mathcal{L}_H) \leq \eta$ which implies that $Q(v, w) \geq 1 - \rho$ where $\rho \coloneqq \varepsilon + \eta$.

Recall the enumerations $v_1, \dots, v_{n'}$ and $w_1, \dots, w_{n'}$ where $\pi(v_i) = w_i$ for all $i \in [n']$. The next claim shows that if Q does not "respect" this mapping approximately, then the cut norm is already large.

Claim 60. *Let $i, j \in [n']$ be distinct with $Q(v_i, w_j) \geq 1 - \rho$. Then, if $k \geq \frac{1}{2(1-\rho)}$, it holds that $\|B(G, H, Q)\|_\square \geq 2 \cdot \mathsf{maxcut}(F)$.*

Proof. We show the claim for the case $i < j$; the other case is identical after reversing the roles of the two graphs. Let ℓ_i, ℓ_j denote the number of leaves attached to vertices v_i, v_j (w_i, w_j), respectively. Recall that for each $i \in [n']$ we have $\ell_i = i(n^2 + n)k$. Define $R \coloneqq V(G) \setminus N_G(v_i)$ and choose $S' \coloneqq \{(v_i, w_j)\}$ and $T' \coloneqq R \times \mathcal{L}(w_j)$. Then

$$\begin{aligned} B(S', T') &= \sum_{x \in R, y \in \mathcal{L}(w_j)} \Big(\underbrace{A_G(v_i, x) - A_H(w_j, y)}_{=-1} \Big) Q(v_i, w_j) Q(x, y) \\ &\leq -(1-\rho) \sum_{x \in R} Q(x, \mathcal{L}(w_j)) \end{aligned}$$

Note that $|R| \geq N - (\ell_i + n)$ since v_i has ℓ_i neighbours that are leaves and at most $n-1$ neighbours within $\mathcal{B}_G$ due to the properties of the matrix of Alon and Naor. Moreover, $|V(H) \setminus \mathcal{L}(w_j)| = N - \ell_j$. Intuitively, there is more mass in R than what can possibly fit into $V(H) \setminus \mathcal{L}(w_j)$. In particular this implies that the mass transported from R to $\mathcal{L}(w_j)$ is at least $\ell_j - (\ell_i + n)$, so

$$\sum_{x \in R} Q(x, \mathcal{L}(w_j)) \geq \ell_j - (\ell_i + n)$$

Combined with the previous inequality, this gives

$$\begin{aligned} |B(S', T')| &\geq (1-\rho)(\ell_j - \ell_i - n) \\ &\geq (1-\rho)((n^2+n)k - n) \\ &\geq (1-\rho) n^2 k \end{aligned}$$

Thus choosing k sufficiently large such that

$$2 \cdot \mathsf{maxcut}(F) \leq \frac{n^2}{2} \leq (1-\rho) n^2 k$$

concludes the proof. This is equivalent to $k \geq \frac{1}{2(1-\rho)}$. ◁

To conclude the proof, we now show that if Q is approximately the identity bijection (on the base vertices), then the cut norm of $B(G, H, Q)$ cannot be significantly smaller than $2 \cdot \mathsf{maxcut}(F)$. This is the last case we need to consider and thus proves Eq. (47). The basic idea is that we take

the maximiser (S, T) under the bijection π realising the value $2 \cdot \mathsf{maxcut}(F)$ and show that the same pair yields a sufficiently close value under the assumption that Q is "close" to being the bijection π.

Claim 61. *Let $\delta \in (0,1)$ and $0 \leq \alpha \leq \frac{\delta}{(2n')^2}$. If $Q(v_i, w_i) \geq 1 - \alpha$ for all $i \in [n']$, then*

$$\|B(G, H, Q)\|_{\square} \geq 2(1-\delta) \cdot \mathsf{maxcut}(F).$$

Proof. Write $B \coloneqq B(G, H, Q)$. Let $S, T \subseteq \mathcal{B}_G$ with

$$R \coloneqq \left| \left(A_G - A_H^{\pi^{-1}} \right)(S, T) \right| = 2 \cdot \mathsf{maxcut}(F).$$

For $s \in S, t \in T$, let $A(s,t) \coloneqq A_G(s,t) - A_H(\pi(s), \pi(t))$. Moreover, let $\sigma \in \{-1, +1\}$ be the sign such that $\sigma A(S,T) = R$. We choose $S' \coloneqq S \times W$ and $T' \coloneqq T \times W$. For $s \in S \cup T$, we write $q_s \coloneqq Q(s, \pi(s))$.

We decompose $B(S', T')$ into two parts: The first part (M) is the contribution of terms corresponding to the identity alignment π, the second part (E) is the error term accounting for the contribution of all other terms. Hence, to bound $B(S', T')$, we bound M and E individually.

Since $|A(s,t)| \leq 1$, we have

$$\begin{aligned}
\sigma M &= \sum_{s \in S, t \in T} \sigma A(s,t) q_s q_t \\
&= \sum_{s \in S, t \in T} \sigma A(s,t) - \sum_{s \in S, t \in T} \sigma A(s,t)(1 - q_s q_t) \\
&\geq R - \sum_{s \in S, t \in T} (1 - q_s q_t) \\
&\geq R - 2\alpha |S||T|.
\end{aligned}$$

For the last inequality, we used

$$1 - q_s q_t \leq 1 - (1-\alpha)^2 = 2\alpha - \alpha^2 \leq 2\alpha.$$

Next, we estimate the contribution of terms that do not correspond to mapping according to π. For each pair (s,t), the total mass under Q of all non-identity aligned pairs $(w, w') \neq (\pi(s), \pi(t))$ is at most $1 - q_s q_t \leq 2\alpha$. Since every adjacency difference is bounded by one in absolute value, summing this over all pairs (s,t) yields $|E| \leq 2\alpha |S||T|$.

Put together, we obtain

$$\begin{aligned}
|B(S', T')| &\geq \sigma B(S', T') \\
&= \sigma M + \sigma E \\
&\geq R - 2\alpha |S||T| - |E| \\
&\geq R - 4\alpha |S||T|.
\end{aligned}$$

Now since $S, T \subseteq \mathcal{B}_G$, it holds that $|S||T| \leq n'^2$ and hence

$$\|B\|_{\square} \geq |B(S', T')| \geq R - 4\alpha n'^2 \geq 2(1-\delta) \cdot \mathsf{maxcut}(F). \qquad \triangleleft$$

Fix a constant c with $1 \leq c < \frac{17}{16}$ and choose a constant $0 < \delta < 1 - \frac{16c}{17}$. Observe that Claims 58 to 61 exhaustively cover the space of possible matrices Q. It remains to choose k sufficiently large as to ensure that the preconditions of these claims are all satisfied. For simplicity, in the following

we do not optimize the values chosen. Since $n' = 2m + n \leq n^2$, we may choose $\alpha \coloneqq \frac{\delta}{200n^4} \leq \frac{\delta}{2n'^2}$, $\eta \coloneqq \frac{\alpha}{2} = \frac{\delta}{400n^4}$, $\varepsilon \coloneqq \frac{\alpha}{2} = \frac{\delta}{400n^4}$ and hence $\tau = \frac{\alpha}{2(n'-1)}$ and $\rho = \alpha = \frac{\delta}{200n^4}$. For convenience, we instead choose $\tau \coloneqq \frac{\alpha}{4n^2} \leq \frac{\alpha}{2(n'-1)}$, so that $\tau = \frac{\delta}{800n^6}$.

Then, it suffices to choose k such that

$$k \geq \max\left\{\frac{4}{\eta}, \frac{2}{\tau}, 4(n-1), \frac{1}{2(1-\rho)}\right\}$$

to ensure that the preconditions of Claims 58 to 61 are satisfied. Since $\delta < 1$ and $n \geq 1$, we have $\rho = \alpha \leq \frac{1}{200} \leq \frac{1}{2}$ and hence $\frac{1}{2(1-\rho)} \leq 1$. Plugging in the values, we get

$$k \geq \max\left\{\frac{1600n^4}{\delta}, \frac{1600n^6}{\delta}, 4(n-1), 1\right\},$$

and we choose $k \coloneqq \left\lceil \frac{1600n^6}{\delta} \right\rceil$. This means that the resulting graphs are of order at most

$$\begin{aligned}
\sum_{i=1}^{n'}(1+\ell_i) &\leq \sum_{i=1}^{n'}\left(1 + i(n^2+n)\left\lceil\frac{1600n^6}{\delta}\right\rceil\right) \\
&\leq n' + \frac{n'(n'+1)}{2} \cdot 2n^2 \cdot \left\lceil\frac{1600n^6}{\delta}\right\rceil \\
&\leq n^2 + 2n^6\left(\frac{1600n^6}{\delta} + 1\right) = O\left(\frac{n^{12}}{\delta}\right),
\end{aligned}$$

which is polynomial in the order of the input graph F, since δ is fixed.

By the preceding claims and the upper bound established earlier,

$$2(1-\delta)\mathsf{maxcut}(F) \leq \mathsf{d}_{\square}^{\mathsf{OT}}(G,H) \leq 2\mathsf{maxcut}(F).$$

Hence, a polynomial-time c-approximation algorithm for $\mathsf{d}_{\square}^{\mathsf{OT}}$ would approximate the value $\mathsf{maxcut}(F)$ with multiplicative error at most

$$\frac{c}{1-\delta} < \frac{17}{16},$$

contradicting the inapproximability result for MaxCut, assuming that $\mathsf{P} \neq \mathsf{NP}$. Since $c < \frac{17}{16}$ was arbitrary, this concludes the proof. ◀

Remark 62 (Hardness for Other Operator $p \to q$-norms). Lemma 53 and Corollary 54 carry over to $\|\cdot\|_{p\to q}$ with $b(p)$ replaced by $r \coloneqq \max\{1/q, 1-1/p\}$. For $p \leq q$, the value of YES-instances under the canonical bijection remains 1, so the proof of Theorem 17 yields inapproximability within $2^r - \varepsilon$ whenever $(p,q) \neq (1,\infty)$. For $p > q$, the value of YES-instances grows like $m^{1/q-1/p}$, so the single-column bound of Corollary 54 no longer yields a gap. In the proof of Theorem 18, leaves force every "good" Q to be close to the canonical bijection. Similarly to our proof above, one can show that this gives $(1 - \frac{1}{\mathsf{poly}(n)})\,\|D\|_{p\to q} \leq \mathsf{d}_{p\to q}^{\mathsf{OT}}(G,H) \leq \|D\|_{p\to q}$ for any symmetric $D \in \{-1,0,1\}^{n'\times n'}$ with zero diagonal. Hence, hardness reduces to approximating $\|D\|_{p\to q}$. For $p = \infty > q > 1$, NP-hardness then follows from MaxCut, as the oriented incidence matrix M of F satisfies $\|M\|_{\infty\to q} = 2 \cdot \mathsf{maxcut}(F)^{1/q}$. For $1 < q < p < \infty$ the complexity remains open.

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